go-ledger

THE LEDGER BOOK

Double-entry accounting with invariants impossible to violate.
Built in the open over twelve weeks.

every debit has an equal credit

Sohag Hasan

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A Note Before You Begin

A while back a friend paid for dinner before the rest of us could reach for our phones. On the walk home I opened a payment app, sent him my share, and watched my balance drop before I reached the bus stop. Somewhere a system had just moved money between two accounts, atomically, in a way that both of us, the company, and some future auditor could trust completely. I have spent most of my career building on top of systems like that. Using a ledger taught me to operate one. It never taught me to defend one. The gap between those two is this book.

I am Sohag Hasan. For the better part of a decade I have shipped software for fintech and payments teams, most recently as a tech lead, on things like cross-border money transfers and integrations with banks. My day to day has mostly been PHP and Laravel. But the ledger, the part that actually decides whether money is real, was always someone else's library or someone else's table. I wanted to build that part myself, and I wanted to do it in Go. I needed concurrency I could trust: thousands of postings racing on a single account, a race detector that made the failure real, tests that actually caught it. Go fought me on the sharp edges of shared state, which was exactly the lesson I came for.

So I built it in the open, one week at a time. No single week was the hard one. Each had a wall I did not see coming, and understanding arrived slowly, until one day the whole system fit in my head and I could defend any piece of it. That is the bar I was after.

Every week I shipped a piece of the ledger, wrote a post explaining the decisions, and then sat myself down with a list of hard interview questions about what I had just built. I have spent years on both sides of the interview table, and the questions that expose shallow understanding are the same ones I now ask myself. More than once, a question I could not answer sent me straight back to the code. The posts are the story. The questions are the stress test. Reviewing my own work this way changed how I review everyone else's: the bar is no longer "does it pass" but "can you defend it", with the rejected alternatives written down.

I wrote this for the engineer who reaches for a payments API and never thinks about what is underneath, the way I once didn't. If you take one idea from these pages, take this: your balance is not a number someone saved. It is the

sum of every posting that ever touched your account. The history is the truth, and everything else is a cache.

Everything here is real. Every line of code, every schema, every trade-off lives in the open at github.com/sohag-pro/go-ledger, running, tested, and deployed. I have been writing and building in public for years, at notes.sohag.pro, from Dhaka, Bangladesh. This book is part of that habit, not a departure from it. If any of this sounds familiar, you are in the right place.

How to Read This Book

The book is organized the way it was built: one week at a time. Each week has two chapters.

The first chapter of each week is the essay. It tells the story of what I built that week and, more importantly, why: the decisions, the trade-offs, the things I deliberately left out. Read it the way you would read a build log from someone sitting next to you.

The second chapter is the interview. It is a set of questions a senior engineer would face if this project were on their resume, each with the key points a strong answer covers. Use it two ways: to check whether the essay actually landed, and to rehearse defending the design out loud. If you are preparing for backend or systems interviews, this is the part to drill.

At the back you will find the Architecture Decision Records. These are the formal, dated decisions made during the build, kept in the repository and reproduced here. When a chapter says “see ADR-003”, this is where it points. They are terse on purpose.

A few conventions. Code, identifiers, and file paths are set in a monospaced font. Everything in these pages is real and lives in the open at github.com/sohag-pro/go-ledger. You can read the source, run it, or follow the live service at go.sohag.pro.

You can read this front to back, or jump to the week that solves the problem you have in front of you right now. Both work.

go-ledger

WEEK 1

CHAPTER 1

Why I’m Building a Production Go Payment Ledger

Last night I had dinner with two friends, and one of them paid the whole bill, 1,800 taka, before the rest of us could reach for our phones. So on the way home I opened my payment app, typed 600 taka, and sent him my share. My balance dropped before I reached the bus stop. Then I stopped and thought about what just happened. Somewhere, a system recorded that money left my account and entered his, atomically, in milliseconds, in a way that both of us, the payment company, and possibly an auditor three years from now can trust completely.

That system is called a ledger, and I’m going to build one. In public, open source, over twelve weeks, with a blog post every week. This is post 1 of 12.

1.1 Why a ledger, of all things?

Most side projects die for one of two reasons: they’re too ambitious to finish, or too trivial to matter. A todo app teaches you nothing new. A “build your own database” project takes two years and ships never.

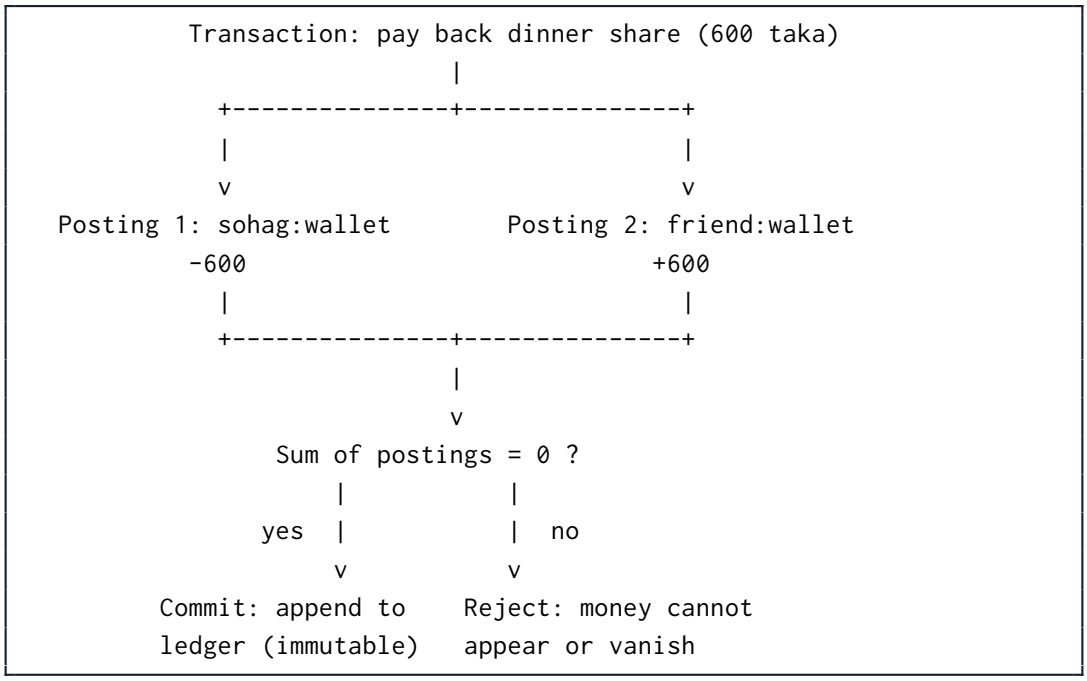
A payment ledger sits in the sweet spot. It’s small enough to build in twelve weeks, and deep enough that doing it *right* forces you through almost everything that makes backend engineering hard: data integrity, concurrency, idempotency, observability, and deployment.

Every fintech company has a ledger at its core, and most are built on an idea older than the printing press: **double-entry bookkeeping**. The rule is simple. Money never appears or disappears. It only *moves*. Every transaction is recorded as two or more postings that must sum to zero. Your balance isn’t a number someone updated; it’s the sum of every posting that ever touched your account. The history *is* the state.

Back to my dinner. Here’s what paying my friend back his 600 taka looks like as a double-entry transaction:

Posting	Account	Amount
1	sohag:wallet	-600
2	friend:wallet	+600
	Sum	**0**

Two postings, summing to zero. If a bug ever produces a transaction where the sum isn't zero, money was created or destroyed out of thin air, and the system must reject it before it persists:



That one constraint is why this project is interesting to build:

- **Correctness is non-negotiable.** A bug in a CRUD app shows a stale page. A bug in a ledger creates or destroys money. The invariant must hold under any failure, any concurrency, any retry.
- **The hard problems are real ones.** What happens when two requests post to the same account simultaneously? When a client retries a payment because the network dropped the response? When the database dies mid-transaction? These aren't interview puzzles; they're Tuesday.
- **It's bounded.** The domain fits in your head. Accounts, transactions, postings, balances. No ML, no infinite feature surface. The depth is vertical, not horizontal.

1.2 Why Go?

I want a language where the production story is boring. Go gives me a single static binary, a stellar standard library (`net/http`, `log/slog`, `database/sql` and friends), first-class concurrency for the stress tests I have planned, and an ecosystem where the "right" choice for most infrastructure questions is well-trodden.

The stack is locked upfront, because relitigating decisions mid-project is how twelve weeks becomes thirty:

- **Router:** chi (idiomatic, minimal)
- **Database:** Postgres with pgx + sqlc for type-safe queries, goose for migrations
- **API:** REST first, gRPC later (with buf for proto management)
- **Testing:** standard library + testcontainers-go, k6 for load, property-based tests for the balance invariant
- **Observability:** OpenTelemetry traces, slog structured logs, Prometheus metrics
- **Deployment:** Docker image, deployed to a VPS via GitHub Actions CI/CD

No Kubernetes, no Kafka, no event sourcing, no multi-currency. Not because they're bad, but because scope discipline is the difference between shipping and abandoning. Single currency, one region, no admin UI. Everything cut from v1 becomes a follow-up post if the project earns it.

1.3 The 12-week plan

1. **Scaffolding**: repo, module layout, tooling, hello ledger *(this post)*
2. **Domain model**: double-entry accounting in Go types; invariants the compiler helps enforce
3. **Postgres schema + repository layer**: append-only postings, integration tests against real Postgres
4. **Transaction posting service**: atomicity, SERIALIZABLE isolation, correctness under 10,000 concurrent posts
5. **REST API**: chi, validation, RFC 7807 errors, OpenAPI
6. **Idempotency keys + immutable audit log**: the two patterns that make fintech APIs boring (in a good way)
7. **gRPC layer**: shared domain services behind both protocols

8. **OpenTelemetry**: one trace from HTTP handler to SQL query
9. **Testing**: unit, integration, property-based, load, and a little chaos
10. **Docker + infrastructure**: multi-stage builds, distroless, tiny images
11. **Live deployment + CI/CD**: push to main, live in minutes
12. **Launch + retrospective**

1.4 Week 1: the unglamorous part

This week was scaffolding, deliberately light. The goal: anyone can clone the repo, run `make run`, and hit a health check.

The layout follows the standard Go convention:

```
go-ledger/├──
 cmd/server/        # main.go, wiring only, no logic├──
 internal/          # domain code lives here, unimportable from outside
    ├──
 pkg/               # public packages, if any earn the spot├──
 docs/adr/          # architecture decision records├──
 Makefile           # run, build, test, lint, dev├──
 Dockerfile         # multi-stage skeleton, distroless base├──
 .golangci.yml      # lint config└──
 .air.toml          # hot reload
```

The server itself is about seventy lines of standard library: an `http.ServeMux` (the 1.22+ pattern-matching routes mean I don't even need chi yet), `slog` JSON logging from day one, and graceful shutdown on SIGTERM, because writing it now is free and bolting it on later never happens.

```
$ curl localhost:8080/healthz
{"status":"ok"}
```

Not impressive. But it builds clean, lints clean (`golangci-lint` with gosec and friends enabled from commit one), and ships with the project's first architecture decision record: **`ADR-001, why double-entry`**. Writing ADRs for a solo project feels ceremonial until you're in week 9 trying to remember why you rejected event sourcing. Future me will thank present me.

1.5 My AI companion

Full transparency: I'm not building this alone. I have an AI pair, **`Claude`**, running in my terminal via Claude Code.

This week it handled the unglamorous parts: installed the Go toolchain, scaffolded the project layout, wrote the Makefile and lint config, drafted the first ADR, and caught its own lint failures before I saw them. That's the deal I've settled on: Claude accelerates the mechanical work, and I own every decision that matters. The architecture, the invariants, the trade-offs in the ADRs, and the understanding stay mine. A ledger where I can't explain *why* every line exists would defeat the entire point of this project.

I'll be honest about the split as the series goes. When Claude writes something substantial, I'll say so. When I overrule it, that's probably worth a paragraph too; those disagreements tend to be where the interesting engineering lives. If you're curious how an AI-assisted workflow holds up across a twelve-week production build, that's now a quiet subplot of this series.

1.6 Why you should build one too

Because "I read about idempotency keys" and "I watched 100 goroutines try to corrupt my ledger and fail" are different kinds of knowledge. The second kind is what production engineering actually is, and you can't get it from blog posts, including this one. You get it by building the thing, breaking it, and fixing it.

The repo is open source at **`github.com/sohag-pro/go-ledger`**. Follow along, steal the plan, or build your own. Next week: modeling double-entry accounting in Go's type system, and making the balance invariant impossible to violate.

CHAPTER 2

Week 1 Interview Q&A

Practice set for defending go-ledger in a senior engineering interview. Everything here is answerable from this week's actual deliverables: `cmd/server/main.go`, the project layout, the Makefile, the Dockerfile, `.golangci.yml`, and `docs/adr/001-why-double-entry.md`.

2.1 Theme 1: Domain design

2.1.1 1. Warm-up: Walk me through why you chose double-entry accounting for this ledger instead of a simple balances table.

What a strong answer covers:

- A mutable balance row tells you what the balance is, never how it got there; the audit trail and the state drift apart.
- Double-entry makes every transaction a set of postings that must sum to zero, so money structurally cannot appear or vanish.
- Balances become derived values (sum of postings), so the history is the source of truth.
- ADR-001 records this reasoning, including the rejected alternatives.

2.1.2 2. Your ADR says postings are append-only and balances are never stored as mutable primary state. What does that buy you, concretely?

What a strong answer covers:

- Append-only writes sidestep read-modify-write races on a balance row entirely.

- Every balance is explainable: it is an aggregation over immutable facts.
- A corrupted cache can be rebuilt from postings; the reverse is not true.
- The trade-off is that reads become aggregations, accepted in ADR-001 with a rebuildable cache as the escape hatch.

2.1.3 3. Challenge a locked decision: Double-entry is centuries old and adds friction. Clients now have to submit balanced postings instead of "add 600 to account X". Why is that complexity worth it for a v1?

What a strong answer covers:

- The friction is the feature: an unbalanced request is a bug surfaced at the API boundary instead of corrupted money discovered months later.
- "Add 600 to X" has no answer to "from where?", which is exactly the question auditors and reconciliation ask.
- The domain is fintech; correctness is the product. A todo app can take the simpler model.
- The model matches what payment partners and accountants already expect, reducing integration friction later.

2.1.4 4. Hard follow-up: ADR-001 claims append-only postings reduce concurrency control to "serializing transaction inserts". Two requests race to post against the same account at the same moment. What can still go wrong, and what is your plan?

What a strong answer covers:

- Inserts of postings themselves do not conflict, but any read-then-decide logic does (e.g. "reject if balance would go negative" reads a balance that another in-flight transaction is changing).
- This is a write skew shaped problem; append-only does not make it disappear.
- The plan (Week 4) is a DB transaction at SERIALIZABLE isolation plus row locks on involved accounts, with the invariant also enforced by a Postgres CHECK constraint as a backstop.

- Knowing the failure mode before writing the code is why the ADR and plan exist.

2.2 Theme 2: Project structure and scaffolding

2.2.1 5. Warm-up: Explain your repo layout. Why `cmd/server/`, `internal/`, and `pkg/`?

What a strong answer covers:

- `cmd/server/main.go` is wiring only: config, server start, shutdown. No business logic, so the entry point stays trivially readable.
- `internal/` is compiler-enforced privacy; no external module can import the domain code, which keeps the public API surface deliberate.
- `pkg/` exists but is empty; packages must earn public status, not default to it.
- This mirrors the standard Go project convention, so any Go developer can navigate it immediately.

2.2.2 6. Why did you write an ADR for a solo side project? Who is it for?

What a strong answer covers:

- Future self in week 9, deciding whether to relitigate event sourcing, reads why it was rejected in week 1.
- Interviews and open source contributors get the reasoning, not just the result.
- ADRs force the alternatives to be written down at decision time, when the trade-offs are freshest.
- Cheap to write now, expensive to reconstruct later.

2.2.3 7. You wrote graceful shutdown handling in `main.go` in week 1, before there is any real traffic. Premature?

What a strong answer covers:

- The shutdown path (signal.NotifyContext on SIGINT/SIGTERM, `srv.Shutdown` with a 10s timeout) is about 15 lines now and free to write.
- Retrofitting it later competes with feature work and never happens; meanwhile every deploy hard-kills in-flight requests.
- For a ledger specifically, killing a request mid-transaction is the exact class of bug the project exists to prevent.
- It also makes the service well-behaved under Docker and CI from the first deploy, since both deliver SIGTERM.

2.3 Theme 3: HTTP server decisions

2.3.1 8. The stack locks in chi as the router, but `main.go` uses the standard library `http.ServeMux`. Contradiction?

What a strong answer covers:

- Go 1.22+ ServeMux supports method-prefixed patterns (`GET /healthz`), enough for two routes.
- chi earns its place when the middleware chain arrives (logging, recovery, request ID) in Week 5.
- Adding a dependency before its value exists is scope creep at the dependency level.
- The handlers will not change when chi lands; only the wiring in main.go does.

2.3.2 9. Challenge a locked decision: Why chi at all then? The 1.22 ServeMux handles routing, and you clearly know it. Defend the dependency.

What a strong answer covers:

- chi adds composable middleware, route groups, and URL param handling that ServeMux still lacks, all on stdlib types (`http.Handler` everywhere).
- Because chi is stdlib-compatible, the exit cost is near zero; handlers never know chi exists.

- The alternative is hand-rolling middleware chaining and mounting, which is rebuilding chi badly.
- Locked decisions exist to prevent mid-project relitigating; the ADR-style reasoning happened once, upfront.

2.3.3 10. Why does your `http.Server` set `ReadHeaderTimeout`, and what would happen without it?

What a strong answer covers:

- Without it, a client can open a connection and trickle header bytes forever (slowloris), pinning server resources.
- gosec flags the omission; the lint config enforces it from commit one.
- 5 seconds bounds the attack window without affecting legitimate clients.
- Related timeouts (read, write, idle) will matter once real handlers exist; this is the minimum safe default for a health-check server.

2.3.4 11. Hard follow-up: Your server goroutine sends errors into a channel, and main selects on that channel and a signal context. Why this shape instead of just calling `ListenAndServe` and handling the error?

What a strong answer covers:

- `ListenAndServe` blocks, so it must run in a goroutine if main also waits on signals.
- The buffered error channel (size 1) lets the goroutine exit even if main has already moved on, avoiding a goroutine leak.
- The select races startup failure (port in use) against shutdown signal; both paths produce a clean exit with the error logged.
- `http.ErrServerClosed` is filtered because it is the expected result of `Shutdown`, not a failure.

2.4 Theme 4: Tooling and build

2.4.1 12. Your Makefile builds with `CGO_ENABLED=0` and tests with `-race`. Those flags look contradictory. Explain.

What a strong answer covers:

- `CGO_ENABLED=0` for builds produces a fully static binary, which is what lets the Dockerfile use a distroless static base image.
- The race detector requires cgo, so `make test` simply does not set the variable; build and test have different goals.
- Race detection from day one is cheap insurance for a project whose week 4 milestone is 10,000 concurrent posts.
- Knowing why each flag exists beats cargo-culting a Makefile.

2.4.2 13. Why distroless for the final Docker image instead of alpine or scratch?

What a strong answer covers:

- Distroless static ships CA certificates, tzdata, and a nonroot user, which scratch lacks and the app will need (TLS to Postgres, HTTPS egress).
- No shell and no package manager shrinks the attack surface compared to alpine.
- The nonroot tag means the container does not run as root by default.
- Image size target (<20MB) supports fast pulls and deploys in the Week 11 CI/CD pipeline.

2.4.3 14. Your lint config enables gofumpt, gosec, revive, and friends from the first commit. Why front-load lint strictness instead of adding it when the codebase grows?

What a strong answer covers:

- Lint debt compounds; enabling gosec on 10,000 existing lines produces a wall of findings nobody fixes.
- It already paid off in week 1: errcheck caught unchecked `Fprint` returns and revive demanded the package comment.

- Style arguments (gofumpt) are settled by the tool, not by future code review back-and-forth.
- CI (Week 9) will reuse the exact same config, so local clean means CI clean.

2.5 Theme 5: Trade-offs and scope

2.5.1 15. Hard follow-up: Your v1 cuts multi-currency, and ADR-001 punts money representation to ADR-002. An interviewer pushes: "You are building a ledger and have not decided how to represent money. Is that not the first decision?"

What a strong answer covers:

- The balance invariant (postings sum to zero) is representation-independent; it holds for int64 cents and for decimals alike, so the domain model is not blocked.
- Deferring with a written deadline (ADR-002, Week 2) is different from not deciding; the decision lands before the first `Money` type is written.
- Single currency in v1 removes the hardest representation pressure (FX, mixed-currency transactions) deliberately, recorded in the scope discipline list.
- Sequencing decisions to land exactly when code needs them is the discipline that keeps a 12-week plan at 12 weeks.

WEEK 2

CHAPTER 3

Modeling Double-Entry Accounting in Go

A few years ago I shipped a bug that moved money and forgot to take it from anywhere. The code added a credit to one account and, because of a botched early return, never wrote the matching debit. Nothing crashed. No test failed. The balance just quietly grew, like a bathtub with the drain unplugged, until someone in finance noticed the numbers didn't tie out and the hunt began.

That bug is the reason this week exists. Last week I explained *why* I'm building a payment ledger and got a "hello ledger" health check responding. This week is the part that would have caught that bug at compile time, or close to it: the domain model. No database yet, no HTTP. Just Go types and the one rule they exist to protect. This is post 2 of 12.

3.1 The one rule

Double-entry accounting has a single invariant that everything else hangs off:

> Every transaction is two or more postings, and those postings must sum to zero.

Money never appears or vanishes. It moves. If I pay a friend back 600 taka, one account goes down by 600 and another goes up by 600, and the two cancel out. The sum is zero. If the sum is ever anything but zero, money was created or destroyed, and that transaction is a bug wearing a transaction's clothes.

My whole goal for the week: make a transaction that doesn't balance hard to even build, and impossible to validate. If the type system and one small method stand guard, the bathtub bug from my past can't happen here.

3.2 First decision: how do you even store money?

Before modeling a transaction I had to answer a deceptively boring question: what *is* an amount? This got its own architecture decision record (ADR-002), because getting it wrong is expensive and quiet.

The wrong answer is `float64`. Floating point can't represent most decimal fractions exactly. In Go, `0.1 + 0.2` is `0.30000000000000004`. That rounding error is invisible on one transaction and catastrophic across a million of them. Money and floats do not mix.

The answer I picked: store every amount as a signed `int64` count of the currency's smallest unit. For US dollars that's cents. So $10.50 is the integer `1050`. Exact, fast, no allocation, and it maps straight onto a Postgres `BIGINT` when the database arrives next week. The tradeoff is that I have to guard against overflow myself, which turns out to be a few lines.

Here's the type. The fields are unexported so a `Money` can only be built through a constructor that validates the currency:

```
type Money struct {
    amount   int64    // signed minor units: cents for USD
    currency Currency
}

func NewMoney(amount int64, currency Currency) (Money, error) {
    if err := currency.Validate(); err != nil {
        return Money{}, err
    }
    return Money{amount: amount, currency: currency}, nil
}
```

The interesting part is arithmetic. Adding two amounts can overflow `int64`, and an overflow that silently wraps a balance from a huge positive to a huge negative is exactly the kind of horror this project is meant to avoid. So `Add` checks and returns an error instead of wrapping:

```
func (m Money) Add(other Money) (Money, error) {
    if m.currency != other.currency {
        return Money{}, ErrCurrencyMismatch
    }
    sum := m.amount + other.amount
    // Overflow happened if both operands share a sign but the result
       flipped.
    if (m.amount > 0 && other.amount > 0 && sum < 0) ||
```

```
        (m.amount < 0 && other.amount < 0 && sum > 0) {
        return Money{}, ErrOverflow
    }
    return Money{amount: sum, currency: m.currency}, nil
}
```

Two things fall out of this for free. Different currencies can't be added, so a multi-currency mistake (a thing explicitly out of scope for v1) can't silently corrupt a sum. And overflow is a typed error you can test, not undefined behavior you discover in production.

3.3 Modeling the transaction

A **posting** is a single signed entry against one account. The sign carries the direction: a positive amount is a debit, a negative amount is a credit. One field, not two, because the sign already tells you everything.

```
type Posting struct {
    AccountID string
    Amount    Money
}

type Transaction struct {
    ID       string
    Postings []Posting
}
```

Back to paying my friend 600 taka. As postings:

Posting	Account	Amount
1	sohag:wallet	-600
2	friend:wallet	+600
	Sum	**0**

Now the chokepoint. `Transaction.Validate()` is the single place in the domain where the invariant lives. It needs at least two postings, every posting has to name an account, every leg has to be the same currency, and the signed amounts have to sum to exactly zero:

```go
func (t Transaction) Validate() error {
    if len(t.Postings) < 2 {
        return ErrTooFewPostings
    }
    for _, p := range t.Postings {
        if err := p.Validate(); err != nil {
            return err
        }
    }
    sum := t.Postings[0].Amount
    for _, p := range t.Postings[1:] {
        next, err := sum.Add(p.Amount) // surfaces currency mismatch and
            overflow
        if err != nil {
            return err
        }
        sum = next
    }
    if !sum.IsZero() {
        return ErrUnbalanced
    }
    return nil
}
```

Notice how the currency check and the overflow check come for free. I don't write them in `Validate` at all. They happen because the running sum is built with `Money.Add`, and `Add` already refuses to mix currencies or wrap. The small type did the work so the big method didn't have to.

Here's the same flow as a picture:

```
        Transaction: pay back dinner share
                        |
        +---------------+--------------+
        |               |              |
        v               v              v
   Posting 1       Posting 2      Posting N
  sohag -600      friend +600        ...
        |               |              |
        +---------------+--------------+
                        |
            fold with Money.Add  ->  same currency? overflow?
                        |
                        v
                  Sum == 0 ?
                  /        \
              yes /          \ no
                 v            v
```

```
        Validate passes   ErrUnbalanced
        (safe to persist)  (rejected)
```

Every arrow that could go wrong has a named error waiting for it: `ErrTooFewPostings`, `ErrInvalidPosting`, `ErrCurrencyMismatch`, `ErrOverflow`, `ErrUnbalanced`. When something fails, the caller knows exactly which rule it broke.

3.4 Proving it holds, hundreds of times

Table-driven unit tests cover the cases I can think of: a balanced two-leg transaction, a balanced three-leg one, an unbalanced one, a single posting, a currency mismatch, an empty account ID. Good, but those are the cases *I* imagined. The bug from my past was one I hadn't imagined. So the balance invariant gets a second kind of test.

Property-based testing flips the script. Instead of asserting "this specific input gives this specific output," you assert a property that must hold for *all* inputs, then let the library throw hundreds of randomized cases at it and try to find one that breaks the rule. I'm using `pgregory.net/rapid`.

The generator builds a transaction that is balanced by construction: pick a random number of legs, give each a random amount, then make the last leg the exact negation of everything before it. The sum is zero on purpose. The property: any transaction built this way must pass `Validate`.

```
func TestProp_BalancedAlwaysValid(t *testing.T) {
    rapid.Check(t, func(t *rapid.T) {
        tx := Transaction{ID: "tx", Postings: genBalancedPostings(t)}
        if err := tx.Validate(); err != nil {
            t.Fatalf("balanced transaction rejected: %v", err)
        }
    })
}
```

Then the mirror image: take a balanced transaction, bump exactly one leg by a non-zero amount, and assert it *always* fails. A balanced ledger you can nudge into imbalance and it still passes would be a disaster.

```
func TestProp_PerturbedAlwaysInvalid(t *testing.T) {
    rapid.Check(t, func(t *rapid.T) {
        postings := genBalancedPostings(t)
        idx := rapid.IntRange(0, len(postings)-1).Draw(t, "idx")
```

```
        delta := rapid.Int64Range(1, 1_000_000).Draw(t, "delta")
        bumped, _ := NewMoney(postings[idx].Amount.Amount()+delta, "USD")
        postings[idx].Amount = bumped
        tx := Transaction{ID: "tx", Postings: postings}
        if err := tx.Validate(); err == nil {
            t.Fatal("perturbed transaction passed Validate, expected
                failure")
        }
    })
}
```

If `rapid` ever finds a counterexample, it shrinks it down to the smallest input that still breaks and hands it to me. That's the part I love: the test doesn't just say "something's wrong," it says "here is the simplest case that's wrong." For an invariant this important, I want a tireless adversary trying to break it on every run, not just the handful of examples I was clever enough to write down.

Everything passes, the domain package sits at 98% coverage, and the linter (gosec included) is clean.

3.5 What I deliberately left out

No `Account` balance field. Anywhere. A balance is never stored as mutable state; it's the sum of every posting that ever touched the account, derived when you ask. That's the whole point of double-entry, and storing a balance you also derive is how the two drift apart and lie to you. The `Account` type this week is identity and classification only: an ID, a name, a type (asset, liability, equity, income, expense), and a currency.

No persistence, no concurrency control, no money math beyond add, subtract, and negate. Those have their own weeks. This week was about getting the shapes right, because every layer above this one (the repository, the service, the API) is going to lean on these types being correct.

3.6 The honest split

Same as last week: I own the decisions, Claude does the mechanical lifting. The choice of `int64` minor units over a decimal library, the signed-posting

convention, the idea of making `Validate` the single chokepoint, those are mine and they're written down in ADR-002 so future me can't forget the reasoning.

What Claude did, working from a detailed plan I reviewed first, was the test-driven grind: write the failing test, watch it fail, write the minimal code, watch it pass, commit, repeat. It also caught a real edge case I'd waved off. My `Money.String` formatter negated the amount to print a minus sign, which breaks for the single most-negative `int64` value, whose negation doesn't fit in an `int64`. An absurd amount for a real ledger, but "absurd inputs shouldn't produce nonsense" is exactly the standard this project holds itself to, so we fixed it with an unsigned-magnitude trick and added a test pinning the behavior. Small thing. But small things quietly wrong is the whole genre of bug I'm trying to engineer out of existence.

3.7 Next week

The types are solid and proven. Next week they meet reality: a Postgres schema with append-only postings, a repository layer with sqlc, and the first integration test that creates accounts, posts a real transaction, and reads a balance back, all against an actual database spun up in a container.

The repo is open at **`github.com/sohag-pro/go-ledger`**. The domain model from this post lives in `internal/domain`, and **`ADR-002`** explains the money decision in full.

CHAPTER 4

Week 2 Interview Q&A

Interview-style questions for the Week 2 work: the pure `internal/domain` package (Money, Currency, Account, Posting, Transaction) and ADR-002. Every question is answerable from this repo's code, ADRs, and tests.

4.1 Domain design

4.1.1 1. Walk me through how you modeled a double-entry transaction in Go. What are the core types and how do they relate?

What a strong answer covers:

- `Money` (signed `int64` minor units plus `Currency`), `Account` (identity and classification), `Posting` (one signed `Money` entry against an account ID), `Transaction` (an ID plus a slice of `Posting`).
- A `Transaction` is two or more `Postings`; the sign on `Posting.Amount` carries direction (positive = debit, negative = credit).
- Balance is never a stored field; it is derived by summing postings (see ADR-001).
- The invariant (postings sum to zero) is enforced in one place: `Transaction.Validate()`.

4.1.2 2. Why is `Posting.Amount` a single signed value instead of an explicit Debit/Credit direction field plus a positive amount?

What a strong answer covers:

- One signed field makes the invariant a plain sum: `Validate` folds the postings with `Money.Add` and checks the result is zero.
- An explicit direction enum would force `Validate` to split sums by side and reconcile them, more code and more room for error.
- The sign convention is documented on the `Posting` type so it is not folklore.
- Tradeoff acknowledged: a direction enum reads more explicitly to an accountant; the signed model was chosen for a simpler, harder-to-break invariant check.

4.1.3 3. Why are Money's fields unexported, and why does construction go through NewMoney?

What a strong answer covers:

- Unexported `amount` and `currency` mean a `Money` can only be built through `NewMoney`, which validates the currency shape.
- It keeps `Money` an immutable value type: operations return new values, nothing mutates in place.
- Prevents an invalid `Money` (bad currency) from ever existing in the first place, a "make illegal states unrepresentable" move.
- The accessors `Amount()`, `Currency()`, `IsZero()` give read access without exposing mutation.

4.1.4 4. The zero value of AccountType is invalid by design. Explain that choice.

What a strong answer covers:

- The constants start at `iota + 1`, so `Asset` is 1, not 0.
- An uninitialized or zero `AccountType` fails `Validate()` with `ErrInvalidAccountType` instead of silently passing as a real type.
- Defends against the Go footgun where a struct field defaults to a meaningful enum value nobody set on purpose.
- `IsDebitNormal()` and `String()` also treat the zero/undefined value safely (not debit-normal, "unknown").

4.2 Data integrity and the invariant

4.2.1 5. Where exactly is the balance invariant enforced, and why concentrate it in one place?

What a strong answer covers:

- `Transaction.Validate()` is the single chokepoint: at least two postings, every posting valid, single currency, signed amounts sum to exactly zero.
- One enforcement point means one place to audit, test, and reason about; no scattered checks that can drift.
- It returns the first rule broken via named sentinel errors (`ErrTooFewPostings`, `ErrInvalidPosting`, `ErrCurrencyMismatch`, `ErrOverflow`, `ErrUnbalanced`).
- The domain check is layer one; ADR-001 notes a Postgres CHECK constraint will enforce it again at the database in a later week (defense in depth).

4.2.2 6. `Validate` never explicitly checks for currency mismatch or overflow, yet it catches both. How?

What a strong answer covers:

- The running sum is built with `Money.Add`, which already returns `ErrCurrencyMismatch` on differing currencies and `ErrOverflow` on int64 overflow.
- `Validate` just propagates the first error from the fold, so the small type does the work and the method stays short.
- This is composition: correctness pushed down into `Money` so every consumer inherits it.
- Demonstrates the value of error-returning arithmetic over panicking or silently wrapping.

4.2.3 7. Why does the ledger derive balances by summing postings instead of storing a balance column you update?

What a strong answer covers:

- A stored mutable balance can drift from the postings that are supposed to explain it; the history would stop being the source of truth.
- Append-only postings give a built-in audit trail: every balance is reconstructable.
- A single-sided "add 10 to A" update has no structural guarantee the money came from anywhere; double-entry makes that impossible (ADR-001).
- Tradeoff: balance reads become aggregations, not single-row lookups; acceptable for v1, and a rebuildable cache can come later if it gets hot.

4.3 Money representation (ADR-002)

4.3.1 8. Why `int64` minor units for money instead of a float or a decimal library?

What a strong answer covers:

- Floats can't represent most decimal fractions exactly (`0.1 + 0.2 != 0.3`); rounding error accumulates across many postings, unacceptable for money.
- `int64` minor units (cents) are exact, allocation-free, and map cleanly onto Postgres `BIGINT` with no conversion.
- A decimal library (for example shopspring/decimal) is exact too but allocates a big.Int per op and maps to `NUMERIC`; more machinery than single-currency v1 needs.
- The cost of int64 is overflow risk, handled explicitly in the arithmetic.

4.3.2 9. What is the downside of int64 minor units, and how did you handle it?

What a strong answer covers:

- int64 caps the representable magnitude near 9.2e16 minor units, far beyond any real balance, but arithmetic can still overflow.
- `Add`, `Sub`, `Neg` are overflow-guarded and return `ErrOverflow` rather than wrapping or panicking.

- `Add` detects overflow with sign logic: overflow happened if both operands share a sign but the result's sign flipped.
- `Neg` special-cases `math.MinInt64`, whose negation does not fit in an int64.

4.3.3 10. Your `Money` carries a `Currency` even though v1 is single-currency. Why pay that cost now?

What a strong answer covers:

- Cross-currency arithmetic returns `ErrCurrencyMismatch`, so a future multi-currency mistake cannot silently corrupt a sum.
- It is cheap insurance: the field and the check cost little and make the invariant currency-aware from day one.
- Multi-currency is explicitly out of scope (ADR-002, build plan), so there is no per-currency scaling logic yet, just the guard.
- Shows scope discipline: build the cheap guardrail, defer the expensive feature.

4.4 Testing

4.4.1 11. You wrote both table-driven tests and property-based tests for the same invariant. Why both, and what does each catch?

What a strong answer covers:

- Table-driven tests pin specific known cases: balanced two-leg and multi-leg, unbalanced, too-few postings, currency mismatch, empty account ID.
- Property tests (`pgregory.net/rapid`) assert a rule for hundreds of randomized inputs, catching cases the author never imagined.
- Two properties: any balanced transaction passes `Validate`; any one-leg perturbation of a balanced transaction fails.
- rapid shrinks a failure to the minimal counterexample, so a regression points straight at the smallest breaking input.

4.4.2 12. Explain the `genBalancedPostings` generator. How do you know it produces genuinely balanced transactions, and what keeps it from triggering false overflow?

What a strong answer covers:

- It draws n-1 random legs, tracks the running int64 sum, then adds a final leg equal to the negation of that sum, so the total is zero by construction.
- A separate property (`TestProp_SumIsZero`) verifies the generator's own output sums to zero.
- Amounts are bounded to roughly +/-1e9 over at most 8 legs, so the running sum stays well within int64 and the fold never hits a spurious `ErrOverflow`.
- The perturbation delta is always nonzero and within bounds, so a perturbed transaction is genuinely unbalanced, not accidentally rebalanced.

4.4.3 13. The domain package sits around 98% coverage. What is the uncovered slice, and why is that acceptable?

What a strong answer covers:

- The small gap is an error path in `Sub` reached via `Neg` (the MinInt64 underflow), already exercised indirectly by the overflow test cases.
- Coverage is a signal, not a goal; the invariant itself is covered by both example and property tests.
- Chasing the last percent on an implicitly-tested error branch is low value versus the property coverage already in place.
- Honest about what is and is not exercised rather than gaming the number.

4.5 Trade-offs and defending decisions

4.5.1 14. Defend the double-entry model itself. Why not a simpler balances table with debit/credit helper functions?

What a strong answer covers:

- A mutable balances table has no structural audit trail and no guarantee money came from somewhere; bugs create or destroy money silently (ADR-001 context).
- Double-entry makes an unbalanced transaction rejectable before it persists; the invariant is the schema, not a convention.
- Append-only postings sidestep read-modify-write races on a balance row.
- It matches what accountants, auditors, and payment partners already expect, lowering integration friction.

4.5.2 15. You pulled in `pgregory.net/rapid` in Week 2 even though the plan first names it in Week 9. Justify adding a dependency early.

What a strong answer covers:

- The balance invariant is the single most important property in the system; it deserves adversarial randomized testing from the moment it exists, not seven weeks later.
- rapid is a test-only dependency, so it adds zero weight to the production binary.
- Introducing it once and using it consistently beats hand-rolling generators now and swapping later.
- Defensible reversal of a plan detail with a clear reason, which is the point of the question.

4.5.3 16. A `Money.String` edge case came up during review: formatting `math.MinInt64`. What was the bug and how did you fix it without changing the method signature?

What a strong answer covers:

- The original `String` negated the amount to print a minus sign, but negating `math.MinInt64` overflows int64 and produces nonsense.
- `String` must satisfy `fmt.Stringer`, so it cannot start returning an error; the fix had to stay signature-compatible.
- The fix computes the absolute value as a `uint64` (unsigned negation yields the correct magnitude even for MinInt64), with a `//nolint:gosec` and a comment explaining the intentional signed-to-unsigned reinterpretation.
- A test pins the MinInt64 output so the behavior cannot silently regress; the broader lesson is that absurd inputs still must not produce garbage.

WEEK 3

CHAPTER 5

Postgres Schema for a Multi-Tenant Ledger

Open your banking app and look at the balance. It feels like a saved number, a single field in a row somewhere with your name on it, that goes up when money arrives and down when it leaves. That mental model is wrong, and the gap between that model and how a real ledger works is most of what Week 3 was about.

In a proper ledger your balance is not stored anywhere. It is computed, every time, by adding up every entry that ever touched your account. The history is the truth. The balance is just a question you ask of that history. Last week I built the double-entry domain model in Go types. This week I had to put it on disk, in Postgres, in a way that keeps that property honest. This is post 3 of 12.

5.1 The one decision that shapes the rest

Here is the fork in the road. When someone posts a transaction, you can do one of two things with their balance.

Option A: keep a balance column. Each account row has a `balance` field. Every posting reads it, adds the amount, and writes it back. Reads are trivial, one column lookup.

Option B: keep only the postings. Postings are append-only rows. The balance is `SUM(amount)` over an account. Nothing is ever updated in place.

Option A is faster to read and it is the one most people reach for first. It is also the exact bug double-entry exists to prevent. The moment you store a balance, you have two copies of the truth: the number in the column, and the sum of the postings. The day those two disagree, and they will, after a crash mid-write, a missed update, a racy retry, you have no way to tell which one is right. Money is now ambiguous, which in a ledger is the worst thing money can be.

So go-ledger uses Option B. Postings are append-only. Balances are derived. There is exactly one source of truth, and it cannot drift from itself.

Picture my dinner debt from post 1, the 600 taka I owed a friend, now sitting in the actual tables:

```
postings table (the only truth)
+------------+--------------+---------+
| account    | txn          | amount  |
+------------+--------------+---------+
| sohag      | dinner-debt  |  -600   |
| friend     | dinner-debt  |  +600   |
+------------+--------------+---------+
          |                      |
          | SUM(amount)          | SUM(amount)
          | WHERE account=sohag  | WHERE account=friend
          v                      v
     balance: -600          balance: +600

No balance column anywhere. Ask the question, get the answer.
```

The trade is real and worth saying out loud: reads are now an aggregate, not a lookup. At v1 scale that is a non-issue. If it ever bites, the fix is a cached rollup that is *rebuilt from the postings*, never a mutable primary balance. The history stays sacred.

5.2 Three tables, not five

My build plan listed five tables for this week: `accounts`, `transactions`, `postings`, `idempotency_keys`, and `audit_log`. I built three.

The last two belong to Week 6. I do not know their columns yet, because I have not written the code that uses them. Designing a table before its consumer exists just means I get to `ALTER` it later anyway, so there is no real saving in rushing it. Migrations are cheap and ordered. Each table arrives with the feature that needs it. That is the whole argument, and it is the kind of small discipline that keeps a twelve-week project from quietly turning into a thirty-week one.

So `0001_initial.sql` creates `accounts`, `transactions`, and `postings`. That is it.

5.3 Multi-tenant from the first migration

The blog title says multi-tenant, and that is the part most schema posts skip. A real ledger does not hold one person's accounts. It holds many independent tenants' books in the same database, and tenant A must never, under any bug, see tenant B's money.

There is no login or auth in go-ledger yet. So why is there a `tenant_id` on every single table already? Because adding it later is the migration nobody wants to write. Retrofitting tenancy onto a populated ledger means a new column, a backfill across every row, and rebuilt foreign keys, all on live data. Adding the column now, while the tables are empty, costs nothing. The column goes in on day one even though the code that resolves a tenant comes much later.

But a column is just a label. The interesting part is making the database itself refuse to mix tenants. Here is the trick:

```
accounts                        transactions
+-----------------------+       +-----------------------+
| id          (uuid)    |       | id          (uuid)    |
| tenant_id   (uuid)    |       | tenant_id   (uuid)    |
| ...                   |       | ...                   |
| UNIQUE (tenant_id,id) |       | UNIQUE (tenant_id,id) |
+-----------------------+       +-----------------------+
          ^                                 ^
          |  FK (tenant_id, account_id)     |  FK (tenant_id,
              transaction_id)
          |                                 |
      +-------------------------------------+
      | postings                            |
      |   tenant_id, account_id, transaction_id|
      +-------------------------------------+

A posting must match its account's tenant AND its transaction's tenant.
A cross-tenant posting is not "discouraged". It is impossible to insert
    .
```

Instead of a plain foreign key on `id`, accounts and transactions carry a `UNIQUE (tenant_id, id)`, and postings reference the pair. Now Postgres will reject a posting that points at an account in a different tenant. Tenant isolation stops being a thing every query has to remember and becomes a guarantee the database enforces. The query that forgets its `WHERE tenant_id = ?` is still a bug, but the catastrophic version, a posting that silently links two tenants' books, simply cannot be written.

5.4 The small choices that add up

A few smaller decisions, each with a fork worth a sentence:

UUIDv7 keys, generated in Go. Keys are UUIDs, not auto-incrementing integers, and the app generates them before insert using UUIDv7. Version 7 is time-ordered, so it keeps the nice index locality that makes serial keys fast, while staying globally unique and not coupled to the database. Generating them in Go means the caller knows the ID without a round-trip, and tests are deterministic. The cost is 16 bytes instead of 8. Worth it.

Currency on the transaction, not the posting. The domain already guarantees every posting in a transaction shares one currency. So I store the currency once, on the transaction, and a posting stays a single signed integer amount. Less duplication, and the single-currency rule is visible in the shape of the data.

Account type as text with a CHECK, not a Postgres enum. A column constrained by `CHECK (type IN ('asset','liability', ...))` reads fine in plain SQL, and adding a type later is an ordinary migration instead of the awkward `ALTER TYPE` dance an enum forces. Low churn, so the enum's only real benefit does not apply.

No balance CHECK yet. The rule that a transaction's postings must sum to zero lives in the Go domain for now, in `Transaction.Validate()`. The database-level constraint that makes it impossible regardless of caller is deliberately Week 4 work, landing with the concurrency handling where it belongs. I am being honest about that gap rather than pretending the schema is already bulletproof.

5.5 Proving it works against a real Postgres

A schema you have not run is a guess. The Week 3 definition of done was a single integration test that exercises the whole happy path end to end, against a real Postgres, not a mock.

The test uses testcontainers-go to spin up an actual `postgres:16-alpine` in a container, runs the migration with goose, then does the real thing: create two accounts, post a balanced two-leg transaction, read both balances back, and assert they are +10000 and -10000 and that they net to zero. That last

assertion is the whole point of the project, checked end to end:

```
cashBal, _ := repo.Balance(ctx, tenant, cash.ID)      // +10000
revBal, _  := repo.Balance(ctx, tenant, revenue.ID)   // -10000

if cashBal.Amount()+revBal.Amount() != 0 {
    t.Errorf("ledger does not net to zero")
}
```

There is a second test that creates an account under one tenant and tries to read it as another tenant, expecting “not found”. The isolation I designed into the foreign keys is not just a claim in this post, it is a test that fails if I ever break it.

One honest detail: the test skips, rather than fails, when no Docker daemon is reachable, so a teammate without Docker still gets a green `make test`. The risk is that a skip can hide a broken integration path, so CI runs Docker and exercises the real thing for real. On my own laptop that meant getting a container runtime going (I use colima, not Docker Desktop), watching the first run pull the Postgres image, and finally seeing three green `PASS` lines against a live database. That is the moment the schema stopped being a guess.

5.6 The AI companion, week 3

The running subplot of this series is that I build this with Claude Code as a pair, and I stay honest about the split. This week Claude wrote a lot of the mechanical surface: the migration SQL, the sqlc setup, the repository adapter, the integration test. But the decisions in this post, append-only over a balance column, tenant_id on day one, composite foreign keys for isolation, three tables not five, are mine, argued out before any code got written and recorded in **ADR-003**. The deal holds: Claude accelerates the typing, I own the why. A ledger I could not explain line by line would defeat the point.

5.7 Where this leaves us

The ledger now has a home on disk that cannot lie to itself. Balances are derived, postings are immutable, tenants are isolated by the database, and there is a test that runs the whole path against real Postgres and proves it.

What is missing is the hard part: what happens when two requests post to the same account at the same time. Right now the balance invariant is enforced in Go, optimistically, with nothing stopping two concurrent writers from racing. Next week is atomicity and concurrency: wrapping posting in a database transaction at SERIALIZABLE isolation, adding the database-level CHECK that makes an unbalanced transaction impossible, and then pointing 10,000 concurrent posts at the same accounts to see if the invariant holds. That is where ledgers get genuinely interesting.

The repo is open source at **`github.com/sohag-pro/go-ledger`**. Follow along or steal the schema.

CHAPTER 6

Week 3 Interview Q&A

Interview-style questions for the Week 3 work: the `0001_initial` migration, the sqlc query layer, the `domain.Repository` port and its `internal/postgres` adapter, the testcontainers integration test, and ADR-003. Every question is answerable from this repo's code, ADRs, and tests.

6.1 Schema design

6.1.1 1. Walk me through the schema you landed on for Week 3. What are the three tables and how do they relate?

What a strong answer covers:

- `accounts` (identity, type, currency, tenant), `transactions` (a single-currency grouping, tenant), `postings` (signed `bigint amount` linking a transaction to an account).
- A transaction has two or more postings; balances are not stored, they are `SUM(amount)` over an account (carries ADR-001 into the DB).
- Postings are append-only: no `UPDATE` or `DELETE` query is written against them.
- Every table carries `tenant_id`; the whole ledger is scoped by tenant.

6.1.2 2. You chose append-only postings with derived balances over a mutable balance column. Why?

What a strong answer covers:

- A stored balance is a second copy of the truth that can drift from the postings, and once it drifts there is no way to tell which is right. That is exactly the bug double-entry exists to prevent.

- A mutable balance turns every posting into a read-modify-write that has to be serialized on the account row.
- Derived balance is always reconstructable from history; the posting log is the single source of truth.
- Tradeoff named: reads become an aggregate (`SUM`) instead of a column lookup. Fine at v1 scale; the escape hatch is a materialized rollup rebuilt from postings, never a mutable primary balance (recorded in ADR-003).

6.1.3 3. The build plan lists five tables but the migration creates three. Defend that.

What a strong answer covers:

- `idempotency_keys` and `audit_log` are Week 6 work; their columns are not known until the code that uses them exists.
- Designing a schema before its consumer invites an `ALTER` later anyway, so there is no real saving from doing it early.
- Migrations are cheap and ordered; adding tables in their own migration keeps each change reviewable and tied to a feature.
- YAGNI applied deliberately, not by accident: the decision is written down in ADR-003.

6.1.4 4. Why does currency live on the transaction and the account, but not on each posting?

What a strong answer covers:

- The domain already guarantees all postings of a transaction share one currency (`Transaction.Validate`), so storing it per posting would be redundant.
- A posting stays a single signed `bigint amount`; its `Money` is reconstructed from the transaction's currency on read (see `GetTransaction` in `repository.go`).
- The account also carries its own currency, which is what `Balance` uses to build the returned `Money`.
- Keeps the posting row minimal and the single-currency-per-transaction rule structurally visible.

6.2 Multi-tenancy and data integrity

6.2.1 5. There is no auth yet, so why is `tenant_id` on every table from day one?

What a strong answer covers:

- Retrofitting tenancy onto a populated ledger (adding the column, backfilling, rebuilding foreign keys) is a painful, risky migration nobody wants to write later.
- The column and the foreign-key discipline are cheap now and define the scoping seam the service and API will use.
- Repository methods are already tenant-scoped (`tenantID` is the first argument on every method), so the contract is set before any consumer depends on it.
- Auth and tenant resolution land later; the storage shape does not have to wait for them.

6.2.2 6. Explain the composite foreign keys on `postings`. What do they buy you that a plain FK on `id` would not?

What a strong answer covers:

- `accounts` and `transactions` have `UNIQUE (tenant_id, id)`; `postings` references `(tenant_id, account_id)` and `(tenant_id, transaction_id)`.
- This makes it impossible to insert a posting that points at an account or transaction in a different tenant: the database rejects it, not just application code.
- Tenant isolation becomes a data-integrity guarantee enforced by Postgres, not a convention you hope every query remembers.
- Demonstrated by `TestTenantIsolation`: a second tenant cannot read another tenant's account even with the correct id.

6.2.3 7. Account type is stored as `text` with a CHECK constraint instead of a Postgres enum. Why?

What a strong answer covers:

- `CHECK (type IN ('asset','liability','equity','income','expense'))` is readable in plain SQL and queryable without special casts.
- Adding or changing a type is an ordinary migration, versus the awkward `ALTER TYPE ... ADD VALUE` dance (which cannot run in a transaction in older Postgres and cannot remove values).
- The Go side maps the enum with `AccountType.String()` on write and `ParseAccountType` on read, so the text values and the domain enum stay in lockstep.
- Low-churn set, so the enum's main benefit (compact storage) is not worth its rigidity.

6.2.4 8. The schema has no CHECK that postings sum to zero. Isn't that the whole point of the ledger? Where is the invariant enforced?

What a strong answer covers:

- For Week 3 it is enforced in the domain: `CreateTransaction` calls `Transaction.Validate()` before writing anything.
- The database-level CHECK is deliberately Week 4 work, landing with the transaction-posting service and its concurrency handling where it actually belongs.
- Defense in depth is the goal: the domain check catches it early with a typed error; the DB constraint will make it impossible regardless of caller.
- Honest about the gap: until Week 4, a buggy caller that bypassed `CreateTransaction` could write an unbalanced set; that is why the constraint is scheduled, not skipped.

6.3 Repository and adapter design

6.3.1 9. Why is `Repository` an interface in `internal/domain` rather than just a struct in the postgres package?

What a strong answer covers:

- Ports and adapters: the domain owns the contract, the infrastructure implements it. Dependencies point inward.
- The domain stays free of pgx and uuid types; the port is a plain Go interface over domain types and `context`.
- The service layer (Week 4) can be tested against a fake repository without a database.
- A compile-time `var _ domain.Repository = (*Repository)(nil)` keeps the adapter honest about satisfying the port.

6.3.2 10. IDs are `uuid` in Postgres but `string` on the domain types. Walk through how that boundary is handled and why UUIDv7 generated in the app.

What a strong answer covers:

- The domain models ids as plain `string` so it carries no uuid dependency; the adapter parses to `uuid.UUID` at the boundary and stringifies on the way out.
- The app generates UUIDv7 before insert (`uuid.NewV7()`), writing it back to the passed-in pointer when the caller left the id empty.
- v7 is time-ordered, so it keeps b-tree index locality close to a serial key while staying globally unique and decoupled from the DB.
- Generating in Go (not `gen_random_uuid()`) means the caller knows the id without a round-trip and tests are deterministic.

6.3.3 11. How does `CreateTransaction` guarantee a transaction and its postings are written atomically?

What a strong answer covers:

- It opens a pgx transaction (`pool.Begin`), runs the transaction insert and every posting insert through `q.WithTx(tx)`, then commits.

- `defer tx.Rollback(ctx)` covers every early-return error path; rollback after a successful commit is a harmless no-op.
- `Transaction.Validate()` runs before any DB work, so an invalid transaction never opens a connection.
- Result: there is no window where a transaction row exists without its postings, or vice versa.

6.3.4 12. Why does `Balance` read the account first and then sum postings, rather than just running the SUM?

What a strong answer covers:

- The account is the source of the currency, needed to build the returned `Money`.
- It distinguishes "account exists but has no postings" (balance zero) from "no such account" (`ErrAccountNotFound`); a bare `SUM` over zero rows would return 0 and hide a missing account.
- The query uses `COALESCE(SUM(amount), 0)::bigint` so an account with no postings reads as 0, not NULL.
- Tradeoff: two round-trips instead of one. Acceptable for correctness; could become a single joined query if it ever mattered.

6.4 Testing

6.4.1 13. Describe the Week 3 integration test and how it proves the definition of done.

What a strong answer covers:

- `TestHappyPath` spins a real `postgres:16-alpine` via testcontainers-go, runs the goose migration, then creates two accounts, posts a balanced two-leg transaction, and reads both balances back.
- It asserts the specific balances (+10000 and -10000) and that they net to zero, which is the defining double-entry property end to end.

- It also round-trips the transaction through `GetTransaction` and re-runs `Validate()` on the reconstructed value.
- Backed by `TestGetAccountNotFound` and `TestTenantIsolation` for the error and isolation paths.

6.4.2 14. The test skips instead of fails when Docker is absent. Defend that choice and explain the risk.

What a strong answer covers:

- A developer without a running Docker daemon still gets a green `make test`; the suite does not punish them for missing infrastructure.
- The skip is explicit and logged with a clear reason, so it is visible, not silent.
- Risk: a skip can mask a genuinely broken integration path if CI does not run Docker, so CI must run it for real (it does).
- The skip key off the container failing to start, not a `testing.Short()` flag, so the real run needs no special invocation beyond Docker being up.

6.4.3 15. Migrations run through goose in the test but the app talks to Postgres through pgx. Why two database access paths, and how do they coexist?

What a strong answer covers:

- goose is built on `database/sql`, so the test opens a `sql.DB` with the pgx stdlib driver (`_ "github.com/jackc/pgx/v5/stdlib"`) just to run migrations, then closes it.
- The repository itself uses a `pgxpool.Pool` (the native pgx path) for all real queries.
- Migrations are embedded with `go:embed` (`postgres.Migrations`) so goose runs them without a path on disk, the same FS the server can use at startup later.
- Clean separation: schema management and query execution are different concerns and do not have to share a driver.

6.5 Trade-offs and stack defense

6.5.1 16. Defend pgx + sqlc over an ORM like GORM. What do you give up?

What a strong answer covers:

- sqlc generates type-safe Go from hand-written SQL, so the queries are explicit, reviewable, and exactly what runs; no hidden N+1 or surprise queries from an ORM's lazy loading.
- pgx gives full access to Postgres features and a fast native protocol path, plus first-class transaction control used directly in `CreateTransaction`.
- For a ledger, predictable SQL and tight control over transactions and isolation matter more than the CRUD convenience an ORM optimizes for.
- Given up: less automatic scaffolding, you write the SQL and the migrations yourself. Accepted as a feature, not a cost, for this domain.

6.5.2 17. Someone argues UUIDv7 primary keys are over-engineering and a `bigserial` would be simpler and faster. How do you respond?

What a strong answer covers:

- `bigserial` couples id assignment to the database, forcing a round-trip to learn the id and making cross-tenant uniqueness and external references messier.
- UUIDv7 keeps most of serial's index locality (time-ordered) while being globally unique and client-generatable, which keeps the domain and tests independent of the DB.
- For a multi-tenant system that may later shard or expose ids externally, opaque non-sequential keys avoid leaking volume and ordering and avoid id collisions across tenants.
- Honest cost: 16 bytes versus 8, and slightly larger indexes. Acceptable for the isolation and decoupling gained; documented in ADR-003.

WEEK 4

CHAPTER 7

Atomic Transaction Posting in Go

Imagine an account with 100 taka in it, and two withdrawals of 100 taka arriving at the exact same instant. Both read the balance, both see 100, both say “yep, enough money,” and both go through. The account is now at minus 100. Nobody wrote a bug, exactly. Each withdrawal was correct on its own. They were just allowed to happen at the same time, and the system let them both believe they were alone.

That is the whole problem of week 4. Last week I put the ledger on disk and proved a single transaction posts correctly. This week the question is harder: what happens when a hundred of them land on the same accounts at once, and how do I make absolutely sure the books still balance? This is post 4 of 12.

7.1 The thing that must never happen

The ledger’s one rule, from day one, is that every transaction’s postings sum to zero. Money moves, it never appears or vanishes. Under concurrency the failure mode is subtle. It is not that one transaction is wrong. It is that two correct transactions, interleaved, produce a state neither of them intended.

Here is my dinner-debt example from earlier in the series, now with a twist. Say two different transactions both touch my wallet at the same moment: I pay back a friend 600 taka, and at the same time a refund of 200 lands.

```
   Transaction A: pay friend        Transaction B: refund arrives
   sohag:wallet  -600               sohag:wallet  +200
   friend:wallet +600               shop:wallet   -200

   Both read sohag:wallet's history at the same instant.
   Both append their postings based on what they saw.
   If the database lets them interleave wrong...
   ...the final balance can reflect one, but not the other.
```

Both transactions are individually balanced. The danger is in the interleaving. So the real work this week was not writing the posting logic. It was choosing

how the database serializes conflicting transactions, and then proving it holds.

7.2 Two ways to be safe

There are two classic answers, and they are opposites.

The **pessimistic** one: lock the rows. Before touching an account, grab an exclusive lock on it (`SELECT ... FOR UPDATE`), do the work, release. Anyone else who wants that account waits their turn. It works, but correctness now depends on me remembering to lock the right rows, in the right order, on every single code path, forever. The day someone adds a new way to write a posting and forgets the lock, the race is back, silently.

The **optimistic** one: don't lock, just let the database watch. Run every transaction at `SERIALIZABLE` isolation, the strictest level Postgres offers. Postgres tracks the reads and writes of concurrent transactions and, if committing them all would produce a result that no serial (one-at-a-time) ordering could, it aborts one of them with a specific error. Nothing blocks. Conflicts surface as a retry-this error.

I went optimistic. The reason is in the failure mode: with locking, a forgotten lock fails silently and corrupts money. With SERIALIZABLE, the database refuses to let an unsafe interleaving commit, no matter what the application code does or forgets. Correctness stops being something I have to remember and becomes something the database guarantees. I wrote this up in **ADR-004**.

The catch is that "abort with a retry-this error" means I have to actually retry.

7.3 Retrying without making it worse

When Postgres aborts a transaction for a serialization conflict, it returns SQLSTATE 40001. The fix is to roll back and run the whole thing again, because the second time around the conflicting transaction has usually finished and yours can now commit cleanly.

One subtlety bit me immediately: the conflict often does not show up when you run your statements. It shows up at `COMMIT`. So the retry logic has to watch both the work and the commit:

```
attempt:
  BEGIN SERIALIZABLE
  insert transaction + postings   --> maybe 40001 here
  COMMIT                          --> or maybe 40001 here
  if 40001 anywhere: roll back, wait a bit, try again
```

The “wait a bit” matters more than it looks. If a hundred transactions all conflict and all retry after exactly the same delay, they just collide again, in lockstep, forever. So the backoff is exponential with **full jitter**: each retry waits a random amount up to a growing cap. The randomness is the point. It scatters the retriers so they stop landing on top of each other. Bounded retries, and if they ever run out, the caller gets a typed “conflict, try again” error that the API will map to a 503, not a confusing 500.

7.4 Belt and suspenders: enforce it in the database too

The domain already checks the sum-to-zero rule in Go before anything touches the database. But I wanted the database itself to refuse an unbalanced transaction, so that a bad migration, a direct `psql` session, or some future second service physically cannot write one.

The trouble is that the rule spans many rows. A normal `CHECK` constraint only sees one row at a time, and a single posting of plus 600 is not “wrong,” it is half of a pair. The sum only makes sense once all the postings are in.

The answer is a **deferred constraint trigger**. It is queued up and fires once, at `COMMIT`, after every posting is written, and it checks that each transaction’s postings sum to zero:

```
BEGIN
  insert posting +600   (trigger queued, not checked yet)
  insert posting -600   (trigger queued, not checked yet)
COMMIT
  --> now the trigger runs: SUM(postings) for this txn = 0 ? yes ->
      commit
                                                          no  ->
                                                              abort
```

Now the invariant is guaranteed in two independent places: the Go domain catches it early with a clean error, and the database guarantees it absolutely.

There is a test that bypasses all my Go code, inserts a single lopsided posting with raw SQL, and confirms the commit is rejected.

7.5 The afternoon one in six payments failed

Then I wrote the stress test the week was really about: 100 goroutines, 10,000 balanced transactions, all hammering a shared pool of accounts. The bar was zero invariant violations.

The first run was ugly. Roughly **one in six posts failed**, all with the same serialization error, even after retries. Nearly 2,000 transactions gave up. My first instinct was that my retry logic was weak, so I improved the backoff. It got better, not fixed. Still over a hundred failures, and the run took a full minute.

The real cause was somewhere I never would have guessed: a missing index.

My balance trigger reads postings filtered by their transaction id. There was no index on that column, so Postgres did a sequential scan to find them. And here is the part that is easy to miss: under `SERIALIZABLE`, a scan does not just read rows, it takes a broad **predicate lock** over the range it scanned. Two transactions writing completely unrelated postings would both scan, both lock wide swaths of the table, and Postgres would see overlapping read/write footprints and abort one of them as a conflict. The transactions were not actually fighting over the same data. The unindexed read was inventing the conflict.

I added one index on `postings(transaction_id)`. The failures dropped to **zero**, and the run went from about 62 seconds to 5.

```
before: trigger reads postings by transaction_id
        no index -> sequential scan -> wide predicate lock
        -> phantom serialization conflicts -> ~1 in 6 posts fail

after:  same read, now an index lookup -> narrow lock
        -> real conflicts only -> 10,000 / 10,000 commit, zero failures
```

That is the lesson I will keep from this week. Under SERIALIZABLE, an unindexed read is not merely slow. It actively manufactures conflicts that have nothing to do with your actual data. Indexes are not a performance nicety here, they are load-bearing for correctness-under-load.

One honest footnote on speed: my local numbers were p50 around 35ms, p99 around 68ms, which is above the sub-10ms target I had in mind. The cause is that I run Postgres in a Linux VM on my Mac, and every commit waits for the disk to fsync the write-ahead log through that virtualization layer. I am reporting the real numbers rather than turning off durability to get a prettier graph. For a ledger, the durable write is the entire point.

7.6 What a fresh pair of eyes caught

Before merging, I had my AI pair do a senior-level review of everything through week 4, specifically from a fintech angle. It caught a genuinely good one. Currency lived on accounts and on transactions, but nothing in the database tied them together, so a USD transaction could technically post into a EUR account. Invisible today because everything is USD, but exactly the kind of gap that becomes a real bug the day multi-currency arrives. That became a second small trigger and its own short decision record. The review also pushed the metrics endpoint off the public port (it leaks volumes and latencies) and turned a few raw database errors into typed ones the API can map cleanly.

That is the deal I keep describing in this series: the AI accelerates the work and pressure-tests my thinking, and I own every decision and can explain why each line exists. The currency catch is a good example. It suggested the gap, I decided how to close it and wrote down why.

7.7 Where this leaves us

The ledger now posts transactions atomically, stays correct under heavy concurrency, enforces its core invariant in two independent layers, and has the metrics to watch contention. The stress test runs 10,000 concurrent posts and the books balance to the cent, every time.

What is still missing is a way to actually call any of this over the network. Everything so far is exercised by tests. Next week is the REST API: chi, real endpoints to create accounts and post transactions, proper validation, and RFC 7807 error responses, all sitting on top of the service this week built. That is when the ledger stops being a test suite and starts being something you can curl.

The repo is open source at **github.com/sohag-pro/go-ledger**. Follow along, or go read your own database's docs on predicate locks before they teach you the hard way.

CHAPTER 8

Week 4 Interview Q&A

Interview-style questions for the Week 4 work: the `TransactionService`, the `RunInTx` SERIALIZABLE retry loop, the deferred balance trigger and the per-account currency trigger, the Prometheus metrics, the 10,000-post stress test, and ADR-004 and ADR-005. Every question is answerable from this repo's code, ADRs, and tests.

8.1 Concurrency model

8.1.1 1. Walk me through what happens when two requests post transactions that touch the same account at the same time.

What a strong answer covers:

- Each `Post` runs in its own SERIALIZABLE transaction via `RunInTx`; Postgres's SSI tracks read/write dependencies between them.
- If committing both would violate serial-equivalent ordering, Postgres aborts one with SQLSTATE 40001, often only at COMMIT time.
- The adapter catches that, rolls back, waits a jittered backoff, and replays the whole unit of work; the loser eventually commits once the winner is done.
- The ledger is never left unbalanced: a conflicting transaction either commits in full or not at all.

8.1.2 2. Why SERIALIZABLE plus retry instead of READ COMMITTED with SELECT FOR UPDATE?

What a strong answer covers:

- SERIALIZABLE makes correctness the database's job; the application cannot forget to lock the right rows in the right order, which is the failure mode of pessimistic locking (ADR-004).
- Pessimistic locking serializes access to hot accounts by blocking; optimistic lets non-conflicting work proceed and pays only for actual conflicts.
- The trade is that conflicts cost retries rather than blocked connections, and `fn` must be safe to run more than once.
- Defends against the "what if someone adds a new write path" risk: a future code path can't silently skip a lock it never knew about.

8.1.3 3. Serialization conflicts often surface at COMMIT, not mid-transaction. How does `RunInTx` account for that?

What a strong answer covers:

- The retry loop watches both the result of `fn` and the result of `tx.Commit`; either returning 40001/40P01 triggers a replay.
- A naive loop that only checks `fn`'s error would miss the common case and leak the failure to the caller.
- `isSerializationFailure` classifies via `pgconn.PgError` codes (40001 serialization failure, 40P01 deadlock), the two worth retrying.
- On a non-retryable error it returns immediately; there is a test (`TestRunInTxNonRetryablePropagates`) that pins this.

8.1.4 4. Your retry backoff is exponential with full jitter. Why the jitter specifically?

What a strong answer covers:

- Without jitter, a crowd of transactions that conflicted will all back off the same amount and retry in lockstep, colliding again, a thundering herd.
- Full jitter (`retryBackoff` returns a random duration in `[0, exp]`, capped) de-synchronizes retriers so they spread out and clear.
- Backoff is bounded (25 attempts, cap 100ms) and respects context cancellation while waiting.
- Exhaustion returns a typed `domain.ErrConflict`, not a raw pg error, so transport can map it to 503.

8.2 Enforcing the invariant in the database

8.2.1 5. The balance invariant is already in the Go domain. Why also enforce it in Postgres, and how?

What a strong answer covers:

- Defense in depth: the domain gives a fast, clean error, but the database guarantees the invariant holds against a bad migration, a direct psql write, or a second service (ADR-004).
- A row-level CHECK can't express it because the sum spans many posting rows.
- Migration 0002 uses a `DEFERRABLE INITIALLY DEFERRED` constraint trigger that fires at COMMIT, after all postings are in, and rejects any transaction whose postings don't sum to zero.
- Proven by `TestUnbalancedRejectedByTrigger`, which inserts raw unbalanced rows and asserts COMMIT fails.

8.2.2 6. Why DEFERRABLE INITIALLY DEFERRED rather than a trigger that fires on each insert?

What a strong answer covers:

- A transaction is built from multiple posting inserts; after the first insert the set is not yet balanced, so an immediate per-row check would reject every valid transaction.
- Deferring to COMMIT means the check sees the complete set, regardless of how many statements inserted the rows.
- It is independent of insert order and of which client wrote the rows.
- The trade: it is FOR EACH ROW (constraint triggers must be), so it runs N times for an N-leg transaction; N is tiny, so the repeated SUM is negligible, and the migration comments say so to stop a wrong "optimization."

8.2.3 7. (Hard) The stress test went from roughly 1 in 6 posts failing to zero failures after one change. What was it, and why did it matter so much?

What a strong answer covers:

- The balance trigger reads postings by `transaction_id`, and there was no index on that column, so the read was a scan.
- Under SSI, a scan takes broad predicate locks, which manufacture false-positive serialization conflicts between transactions that don't actually conflict.
- Adding `postings_transaction_idx` (migration 0002) made the trigger's read a narrow index lookup, collapsing the conflict rate to zero and cutting runtime from about 62s to 5s.
- The lesson: under SERIALIZABLE, an unindexed read is not just slow, it actively creates conflicts. Indexes are load-bearing for concurrency, not just performance.

8.2.4 8. You added a second trigger for currency. What gap did it close?

What a strong answer covers:

- Currency lives on accounts and on transactions, but postings carry only an amount, so nothing tied a transaction's currency to each account's currency (ADR-005).
- Without it, a USD transaction could post into a EUR account; invisible at single-currency v1, but a latent integrity hole.
- Migration 0003 adds an immediate AFTER INSERT trigger that rejects a posting whose account currency differs from its transaction currency.
- It is immediate (not deferred) because the account and transaction already exist at insert time, and its reads are primary-key equalities, so the SSI footprint is negligible. Covered by `TestCurrencyMismatchRejectedByTrigger`.

8.3 Architecture and error handling

8.3.1 9. Where does the transaction boundary live, and why is the service thin?

What a strong answer covers:

- The domain owns a `Tx` unit-of-work interface and `Repository.RunInTx`; the adapter owns isolation and retry; the service composes the work inside `RunInTx`.
- `TransactionService.Post` validates up front, times the operation, records metrics, logs, and maps errors; it holds no SQL and no isolation knowledge.
- This keeps the SERIALIZABLE/retry detail in one place (the adapter, which knows pg error codes) and makes the service the single entry point REST and gRPC will both call.
- `RunInTx` is a general seam, so Week 6 can write the audit-log row inside the same transaction as the posting.

8.3.2 10. How do callers tell a transient conflict apart from a real failure, and why does that distinction matter for an API?

What a strong answer covers:

- Retry exhaustion returns `domain.ErrConflict` (transient), wrapping the SQLSTATE for logs; a duplicate id returns `domain.ErrDuplicateTransaction`.
- A transport layer maps `ErrConflict` to 503 (retry me) and a validation failure to 4xx, not a blanket 500.
- Returning a typed error instead of a raw `postgres: ...` string means callers match on `errors.Is`, not on text.
- An unbalanced transaction returns the domain error unchanged from `Post`, so it becomes a 4xx, not a server error.

8.3.3 11. You validate the transaction in the service, but not inside the retried unit of work. Walk through that decision.

What a strong answer covers:

- Validation happens once at the public entry points (`TransactionService.Post`

and the convenience `Repository.CreateTransaction`), before the transaction starts.
- `txRepo.CreateTransaction` trusts that `t` is already valid, so validation does not re-run on every retry.
- Fast-fail: an unbalanced transaction never opens a database connection.
- The cost is that `txRepo` is an internal type with a documented precondition rather than a defensively-validating public one.

8.4 Observability and operations

8.4.1 12. What did you instrument, and where is the metrics endpoint exposed?

What a strong answer covers:

- A histogram `transaction_post_duration_seconds` labeled by outcome, and a counter `transaction_post_serialization_retries_total` so contention is visible, plus standard go_/*process*_ collectors.
- The scrape endpoint runs on a separate server bound to loopback (`127.0.0.1:9090`), not the public API port, because metrics leak volumes, latencies, and retry rates (ADR-004).
- The public liveness check stays `GET /healthz`; in production `/metrics` returns 404 on the public interface.
- A scraper reaches metrics over the private network or an SSH tunnel.

8.4.2 13. (Hard) Your stress test runs 100 goroutines but you say the real concurrency is lower. Explain.

What a strong answer covers:

- The true ceiling on simultaneous transactions is the connection pool's `MaxConns`, not the goroutine count; extra goroutines block on pool acquisition.
- The test sets `MaxConns` to 25 explicitly and logs it, so "100 goroutines" means up to 25 transactions truly concurrent at Postgres (ADR-004).
- Quoting the goroutine count would overstate the contention; the honest number is the pool cap.

- `NewPool` also sets statement, lock, and idle-in-transaction timeouts so a stuck operation can't hold a connection indefinitely.

8.5 Testing

8.5.1 14. How do you test the retry logic deterministically, given real serialization conflicts are racy?

What a strong answer covers:

- `RunInTx` takes a function, so a test can pass an `fn` that returns a synthetic `pgconn.PgError{Code: "40001"}` on the first call and nil after, exercising retry-then-commit without a real conflict.
- `TestRunInTxRetriesThenCommits` asserts exactly two attempts and that the retries counter moved by one; `TestRunInTxExhaustionReturnsConflict` asserts a persistent conflict ends as `ErrConflict`.
- The full stress test still exercises real contention end to end; the unit tests pin the edges the stress test hits only incidentally.
- These lifted the adapter's coverage without depending on flaky timing.

8.5.2 15. (Hard) The integration tests passed locally but failed in CI with "connection reset by peer." What happened and how did you fix it?

What a strong answer covers:

- The container helper waited only for port 5432 to be listening, but Postgres opens that port during initdb and then restarts it, so the port-open signal precedes real readiness.
- Locally on a slow, serial container runtime it never raced; in CI, parallel container startup hit the gap and goose got connection resets.
- Fix: wait on the log line "database system is ready to accept connections" with `WithOccurrence(2)` for the initdb-then-server double start.
- Broader point: container readiness is about the service accepting queries, not a TCP port being open; the test's correctness depended on the wait strategy.

8.6 Defending the stack

8.6.1 16. Someone argues SERIALIZABLE is too slow and you should drop to READ COMMITTED for throughput. How do you respond?

What a strong answer covers:

- For a ledger, correctness under concurrency is the product; READ COMMITTED would let anomalies (lost updates, write skew) through that double-entry exists to prevent.
- The measured cost is retries under contention, and with the right index plus jittered backoff the test commits all 10,000 posts with zero failures.
- If a specific hot path ever needs it, the optimization is a narrower one (better indexing, batching, or targeted FOR UPDATE), not weakening isolation globally.
- The latency baselines (p50 about 35ms, p99 about 68ms on a VM) are dominated by WAL fsync, not isolation, and durability is non-negotiable for money.

8.6.2 17. Defend pgx over database/sql here, given Week 4 leans on transaction control.

What a strong answer covers:

- `RunInTx` needs explicit `BeginTx` with `pgx.Serializable`, typed error codes via `pgconn.PgError`, and a pool it can tune (`NewPool` sets MaxConns and timeouts); pgx exposes all of this cleanly.
- The sqlc-generated queries bind to the pgx transaction via `WithTx`, so the same typed queries run inside or outside a transaction.
- `database/sql` would abstract away the Postgres-specific error codes and some pool controls the retry logic relies on.
- goose still uses `database/sql` for migrations via the pgx stdlib driver, so each tool is used where it fits.

WEEK 5

CHAPTER 9

Designing a REST API for a Payment Ledger

Open your banking app and scroll the statement. Every row has a little running balance on the right: after this coffee you had 4,210, after that salary you had 54,210, and so on down the list. It feels like each of those numbers was saved next to the transaction when it happened. It was not. That number does not exist anywhere until you ask for it, and producing it correctly, in pages, fast, turned out to be the most interesting thing I built this week.

For four weeks go-ledger has been a thing that only its own tests could talk to. This week I gave it a REST API and put it online. As of now you can actually curl it at `go.sohag.pro`. This is post 5 of 12.

9.1 Resources and verbs, not remote procedures

A ledger has a small, sharp set of nouns: accounts and transactions. So the API is boring in the good way, just those resources and the standard verbs:

```
POST /v1/accounts                    create an account
GET  /v1/accounts/{id}               read one
GET  /v1/accounts/{id}/balance       its current balance
GET  /v1/accounts/{id}/statement     its postings, with running balance
POST /v1/transactions                post a balanced transaction
GET  /v1/transactions/{id}           read one
```

People sometimes ask why not GraphQL, since it is the fashionable default for a new API. GraphQL earns its keep when clients need to shape wildly different views over a big, interconnected graph, and when over-fetching is a real problem. A ledger is the opposite of that. The resource set is tiny and well defined, the relationships are obvious, and the queries that matter (“what is this account’s balance”, “show me this account’s statement”) are known up front. GraphQL would buy me a flexible query language I do not need and hand me N+1 queries and caching headaches I would rather not have. REST over JSON maps cleanly onto these resources, it is curlable and cacheable, and everyone already understands it. gRPC is still coming later for internal

service-to-service calls, sharing the same core, but the public front door is REST.

9.2 Money on the wire

Here is the first decision that matters for a payment API: how does an amount look in JSON? go-ledger sends money as a signed integer count of the smallest unit, plus a currency code:

```
{ "amount": 10000, "currency": "USD" }
```

That is 100.00 dollars, as 10,000 cents. No decimal point, no floating point, no string parsing at the edge. It maps one to one onto the `int64` the ledger uses internally, so the API is exactly as precise as the ledger is. This is the same choice Stripe and most payment APIs make, and for the same reason: the moment you let `100.00` become a float somewhere, you have invited rounding error into a place where money lives. A debit posting is a positive amount, a credit is negative, and a transaction's postings still have to sum to zero. Posting that coffee sale looks like this:

```
POST /v1/transactions
{
  "currency": "USD",
  "postings": [
    { "account_id": "...cash...",    "amount": 450,  "description": "
        latte" },
    { "account_id": "...revenue...", "amount": -450, "description": "
        latte" }
  ]
}
```

Two legs, plus 450 and minus 450, summing to zero. The API hands the request to the same domain code and the same database triggers from the earlier weeks, so all the invariants still hold. The HTTP layer is thin on purpose: it translates JSON to domain types, calls a service, and maps any error to a clean status code. An unbalanced transaction comes back as a 422, a missing account as a 404, a write conflict as a 503, all as RFC 7807 `problem+json`.

9.3 The statement, and the number that is not stored

Back to that running balance. A statement is a list of the postings that touched one account, newest first, and next to each one, the balance of the account as of that posting. Remember the rule from week 3: balances are never stored, they are summed from the postings. So the running balance on row N is the sum of every posting up to and including row N.

```
  postings for "Cash", oldest to newest        running balance
  +-------------------------------+            (cumulative sum)
  | +5000  opening float          |  ----->      5000
  | -450   latte                  |  ----->      4550
  | +1200  pastry sale            |  ----->      5750
  | -450   latte                  |  ----->      5300
  +-------------------------------+
        the statement shows these newest first,
        each carrying its own running total
```

In SQL, that running total is a window function: `SUM(amount)OVER (ORDER BY created_at, id)`. It sums the postings in order, carrying the total forward row by row. Easy enough, until you remember the statement is paginated. A page shows maybe 50 rows out of an account's thousands. If I compute the window over just the 50 rows on the page, the running balances are wrong, they restart from that page instead of from the beginning of the account's history. The window has to see the whole history even when I only return a slice of it. So the query computes the running balance over all of the account's postings first, then slices out one page.

That is honestly expensive: each page scans the account's full history to rebuild the running totals, because I refuse to store a balance anywhere (storing one is exactly the drift bug double-entry exists to prevent). At this project's scale it is a non-issue, and the index from week 4 makes the scan a tight range. If it ever becomes a problem, the fix is a cached rollup rebuilt from the postings, never a mutable stored balance. The history stays the only truth.

9.4 Paging without lying to the user

The other half of statements is how you ask for the next page. The obvious way is "limit 50, offset 50, offset 100", and it is also subtly broken. If someone

posts a new transaction while you are paging, every offset shifts by one, and you either see a row twice or miss one entirely. For a bank statement, silently skipping a line is not a great look.

So go-ledger uses keyset pagination. Instead of "skip 50 rows", each page hands back an opaque cursor that points at the exact last row it showed, and the next page asks for "everything older than this row":

```
page 1:  newest ... down to row X   ->  cursor = position of row X
page 2:  everything strictly older than row X   ->  cursor = row Y
page 3:  ...older than row Y...      ->  no more rows, cursor = null
```

The cursor is the row's `(created_at, id)` pair, base64-encoded so the client treats it as opaque. Because it anchors on a real row rather than a count, new inserts at the top do not shift anything underneath you. Pages stay stable. When a page comes back shorter than the limit, that is the end, and the response says `next_cursor: null`.

9.5 When the query generator pushed back

A small war story. I write SQL by hand and use sqlc to generate type-safe Go from it. My first version of the statement query put the window function in a subquery and used a tuple comparison, `(created_at, id)< (cursor)`, for the keyset. Postgres is perfectly happy with that. sqlc was not: its analyzer could not resolve the subquery alias and refused to generate. I tried expanding the comparison out longhand. Still no. What finally worked was moving the window into a CTE (a `WITH` block) with plain unqualified columns.

The lesson stuck with me: sqlc has its own SQL analyzer, separate from Postgres, and a query the database accepts can still need restructuring for the generator to understand it. That is the tax for type-safe queries with no ORM in the middle, and I will pay it, but it is worth knowing the tax exists. I cover the actual behavior with an integration test against a real Postgres, because sqlc only proves it can parse and type the query, not that it returns the right rows.

9.6 The docs cannot drift

One thing I am quietly proud of: I never write the OpenAPI spec by hand. The API is built with huma, where each endpoint is a typed Go function, and the spec is generated from those same functions. There is a test that fails the build if the committed spec does not match what the handlers produce. So the interactive docs at `/playground` and the spec at `/openapi.json` are always exactly the running API, not a hopeful description of it that rots over time. Adding an endpoint and forgetting to document it is not possible here, the test catches it.

9.7 It is real now

The quiet milestone this week is that go-ledger stopped being a test suite. For four weeks the server ran with no database at all, deliberately, because nothing it served needed one. This week the endpoints need real persistence, so the server now connects to Postgres, runs its own migrations on startup, and refuses to boot without a database, because a ledger API with no ledger behind it should not pretend to work.

That also meant standing up Postgres on the actual server for the first time: a native install, a dedicated database and user listening only on localhost, tuned down to fit a 1 GB box, with a nightly backup. Then I pushed, the pipeline built and deployed, and a few seconds later I created an account and posted a transaction against the live API and watched the balance come back correct. The thing works in the world now, not just on my laptop.

9.8 The AI companion, week 5

Same arrangement as every week: I build this with Claude as a pair and stay honest about the split. This week it wrote a lot of the mechanical surface, the handler structs, the error mapping, the fake repository for the HTTP tests, and it was genuinely useful when sqlc started rejecting my query, working through the CTE rewrite with me. The shape of the API, the choice to avoid GraphQL, money as integer minor units, keyset over offset, the decision to keep balances derived even when it makes statements more expensive, those were mine,

argued out and written down in **ADR-006**. The deal holds: it accelerates, I decide and understand.

9.9 Where this leaves us

The ledger has a front door now. You can create accounts, post transactions, read balances, and pull a paginated statement with a running balance, all over a clean REST API with generated docs, live on the internet.

What it does not have yet is safety against the thing that haunts every payment API: the retry. A client posts a payment, the network drops the response, the client retries, and now you have charged someone twice. Next week is idempotency keys and an immutable audit log, the two patterns that make a payments API boringly safe to call. That is where this gets properly fintech.

The repo is open source at **github.com/sohag-pro/go-ledger**. Go curl it, or go read why your favorite API uses keyset pagination and never told you.

CHAPTER 10

Week 5 Interview Q&A

Interview-style questions for the Week 5 work: the v1 REST API built with huma on a chi router, the account statement endpoint (running balance + keyset paging), per-posting descriptions, money on the wire, the default-tenant seam, error mapping, the server's database wiring with startup migrations, and ADR-006. Every question is answerable from this repo's code, ADRs, and tests.

10.1 API design

10.1.1 1. Walk me through the v1 API surface and how a request flows from chi to the database.

What a strong answer covers:

- chi is the router with a middleware chain (request id, recovery, slog request logging); huma rides on it via the humachi adapter and owns operations, validation, and error formatting.
- A request hits a typed huma operation (input struct validated from tags), the handler calls an application service (`AccountService`/`TransactionService`), which calls the repository port, which runs sqlc queries on pgx.
- Six endpoints: create/get account, balance, statement, create/get transaction.
- The transport is thin: handlers translate between JSON and domain types and map errors; no SQL or business rules live there.

10.1.2 2. huma generates the OpenAPI spec from the handlers. Why did you choose that over hand-writing the spec or using go-playground/validator plus a separate generator?

What a strong answer covers:

- One source of truth: validation, problem+json errors, and the spec all come from the same typed handler, so the published docs cannot drift from the running API (there is a drift test that fails CI if `api/openapi.yaml` is stale).
- The build plan's literal wording (go-playground/validator, separate generator) predates the huma decision; huma's tag validation supersedes it and removes three moving parts.
- chi stays the router, so the locked decision holds; huma is a layer on top, not a replacement.
- Cost named: huma owns request decode/encode and uses its own validation tag dialect, so very custom responses mean working within its model.

10.1.3 3. How is money represented on the wire, and why?

What a strong answer covers:

- A signed integer count of minor units plus a currency code (`{"amount": 10000, "currency": "USD"}`), mapping 1:1 to the internal `int64`.
- No floating point and no decimal string parsing at the boundary, so the API is as exact as the ledger.
- This is the common payment-API convention (Stripe and others).
- The alternative (decimal strings) was rejected: it adds parsing and per-currency scale validation for no gain (ADR-006).

10.1.4 4. There is no auth yet, but the schema is multi-tenant. How does a request get a tenant?

What a strong answer covers:

- A single default tenant (`DEFAULT_TENANT_ID`, with a fixed fallback) is injected by the API layer into every service call; tenancy is not in URLs or a client header.

- When auth lands, it populates the same tenant value from a token instead, with no change to any resource URL.
- Rejected alternatives: a trusted `X-Tenant-ID` header (unauthenticated, spoofable) or tenant-in-path (bakes tenancy into the resource hierarchy).
- The repository methods are already tenant-scoped, so the seam is in place.

10.2 Account statement

10.2.1 5. Explain the statement endpoint: what it returns and how it paginates.

What a strong answer covers:

- `GET /v1/accounts/{id}/statement` returns the account's postings, newest first, each with the running balance as of that posting, plus an optional `next_cursor`.
- Paging is keyset (cursor) on `(created_at, id)`: the cursor is an opaque base64 token of the last entry's position, and the next page returns entries strictly older than it.
- A full page hands back a cursor; a short page returns `next_cursor: null`, signaling the end.
- Keyset paging is stable under concurrent inserts in a way offset paging is not.

10.2.2 6. Why keyset pagination instead of limit/offset?

What a strong answer covers:

- Offset can skip or duplicate rows when new postings are inserted mid-paging; keyset anchors on a real row position, so pages stay stable.
- Keyset is also cheaper at depth: no scanning past N rows to reach the offset.
- The cursor encodes `(created_at, id)`; `id` (the posting id) is the unique tiebreaker, so two postings sharing a timestamp still page deterministically.
- The trade is a slightly more complex cursor and that you page forward in one direction.

10.2.3 7. (Hard) The running balance needs every prior posting, but a page returns only N rows. How do you compute it correctly, and what does that cost?

What a strong answer covers:

- The query is a CTE that computes `SUM(amount)OVER (ORDER BY created_at, id)` over the account's full posting history, then a keyset filter and limit select one page from it.
- A window function applied only to the page's rows would compute a per-page sum, not the absolute running balance; the window must see the whole history.
- Because balances are derived and never stored (ADR-003), this scans the account's history per page, served by the `(tenant_id, account_id)` index.
- It is O(history) per page, fine at v1 scale; the future optimization if it ever bites is a materialized running balance rebuilt from postings, never a mutable primary balance.

10.2.4 8. (Hard) sqlc rejected your first statement query. What was the issue and how did you resolve it?

What a strong answer covers:

- The first version used a windowed derived subquery aliased `s` and a row-value tuple comparison `(s.created_at, s.id)< (...)` in the WHERE; sqlc's analyzer could not resolve the alias there.
- Rewriting the keyset as an expanded comparison did not help; moving the window into a CTE with unqualified columns did.
- The lesson: sqlc uses its own analyzer, so some valid Postgres (windowed derived tables, row-value comparisons) needs restructuring to generate.
- The behavior is verified by an integration test against real Postgres, since sqlc only checks that it can parse and type the query.

10.3 Data and error handling

10.3.1 9. You added a per-posting description. Where is it validated and bounded?

What a strong answer covers:

- `domain.Posting` carries an optional `Description`, bounded at `MaxPostingDescriptionLen` (256) and checked in `Posting.Validate`.
- Migration 0004 adds the column with a matching `CHECK (char_length(description)<= 256)`, so the database enforces the same bound.
- It appears on every posting in requests and responses, including statement entries.
- Defense in depth, consistent with how the other invariants are enforced in both the domain and the database.

10.3.2 10. How do domain errors become HTTP responses?

What a strong answer covers:

- A single `toHumaErr` mapper turns domain errors into huma status errors, which render as RFC 7807 `application/problem+json`.
- Not-found to 404, duplicate id to 409, an exhausted serialization conflict (`ErrConflict`) to 503, validation and invariant failures (unbalanced, currency mismatch, too few postings, bad description) to 422, and anything unrecognized to a 500 that does not leak internals.
- Returning the domain error unchanged from the service lets the handler map it; an unbalanced transaction becomes a 422, not a server error.
- This keeps error semantics in one place rather than scattered across handlers.

10.3.3 11. The plan listed five endpoints; you shipped six. Why, and is that scope creep?

What a strong answer covers:

- The sixth is the account statement, a core ledger feature (listing a single account's entries with a running balance); a ledger without one is hard to defend.
- It reuses the existing schema and the derived-balance model, so it is a natural extension, not a new subsystem.

- It is bounded: keyset paging, a max page size, and the same tenant and currency handling as the rest.
- Knowing when an addition is in the spirit of the system versus genuine creep is the judgment being shown.

10.4 Server and operations

10.4.1 12. The server now depends on Postgres. How are migrations applied, and why that way?

What a strong answer covers:

- `cmd/server` runs the embedded goose migrations on startup, before serving, using a short-lived `database/sql` handle over the pgx stdlib driver.
- On a single instance this is the simplest correct option: the binary that needs a column also creates it, atomically with the deploy.
- It does not scale to multiple instances (they would race); the upgrade path is moving the same `goose.Up` into a CI step.
- `DATABASE_URL` is required and the server fails fast without it: a ledger API with no database must not serve.

10.4.2 13. Walk me through how the server is wired now: router, services, and the two HTTP servers.

What a strong answer covers:

- `run` loads config, applies migrations, opens the tuned pool (`NewPool`, bounded conns + timeouts), builds the account and transaction services, and constructs the chi router.
- The router mounts the landing page, the Scalar playground, and the huma API (`/v1/*`, `/healthz`, `/openapi.*`); metrics stay on a separate loopback server (ADR-004).
- Middleware: request id, recovery, and a structured slog logger that records method, path, status, size, duration, and request id.
- `RealIP` is intentionally omitted because it trusts spoofable forwarding headers without a trusted-proxy allowlist.

10.4.3 14. (Hard) How did you test the API without standing up a database for every test?

What a strong answer covers:

- Handler tests run against an in-memory `fakeRepo` that implements the `domain.Repository` port, so the full HTTP path (routing, validation, error mapping, serialization) is exercised with no database.
- The fake covers the happy path and every 4xx: bad enum/pattern to 422, missing resource to 404, unbalanced to 422, plus statement pagination with running balances.
- The real database paths (the statement window query, the balance trigger, concurrency) are covered separately by testcontainers integration tests.
- This keeps the handler tests fast and deterministic while still proving the SQL against real Postgres elsewhere.

10.5 Defending the stack

10.5.1 15. Why a REST API at all, and not gRPC or GraphQL for the public surface?

What a strong answer covers:

- REST over JSON is the lowest-friction public surface: curlable, cacheable, universally understood, and it maps cleanly to the ledger's resources (accounts, transactions) and verbs.
- gRPC is still coming for internal/service-to-service use (a later week), sharing the same domain core; the two are not exclusive.
- GraphQL's flexible query graph buys little for a small, well-defined resource set and adds N+1 and caching complexity; the API avoided it deliberately (the blog title says as much).
- The OpenAPI spec and playground give REST much of the discoverability people reach to GraphQL for.

10.5.2 16. Defend pgx + sqlc again now that there is a window function and keyset pagination in play.

What a strong answer covers:

- sqlc generates type-safe Go from hand-written SQL, so the window/CTE statement query is explicit and reviewable, and its parameters and row type are checked at generate time.
- When sqlc could not analyze the first query shape, the SQL was restructured rather than hidden behind an ORM that might have generated something slower or surprising.
- pgx gives the typed error codes and pool tuning the rest of the system relies on (serialization retries, timeouts).
- An ORM would abstract the exact SQL away, which is the opposite of what a ledger wants for its read and write paths.

WEEK 6

CHAPTER 11

Idempotency Keys and the Immutable Audit Log

You are paying for something on your phone, you tap Pay, and the spinner just sits there. No confirmation, no error, nothing. Did it go through? Your thumb hovers over the button. Tap it again and maybe you pay twice. Do nothing and maybe you did not pay at all. We have all been in that half second of doubt, and the reason most of us tap again without really panicking is that somewhere a backend engineer already solved this for us. The second tap does not charge you twice. That property has a name, idempotency, and this week I built it into go-ledger, along with its close cousin: an audit log that can record what happened and never let anyone quietly rewrite it.

This is post 6 of 12. Last week the ledger got a REST API and went live at `go.sohag.pro`. This week is about making that API safe to retry and easy to trust.

11.1 The retry problem is not rare, it is the default

Here is the uncomfortable truth about networks: a request that times out did not necessarily fail. Your `POST /v1/transactions` left the phone, reached the server, posted the transaction, and then the response got lost on the way back. From the client's point of view that looks identical to a request that never arrived. So the client retries, because retrying is the correct thing to do when you are not sure. And now the server sees the same payment twice.

Without protection, that is two transactions, two movements of money, one angry customer. The fix is an idempotency key: the client generates a unique string once per intended action and sends it on every retry of that action.

```
POST /v1/transactions
Idempotency-Key: 7f3a9c2e-pay-dinner-share
{ "currency": "BDT", "postings": [ ... -600 ... +600 ... ] }
```

The contract is simple to state and surprisingly interesting to build:

- First time the server sees this key: do the work, remember the result.
- Same key again, same request: do not do the work again, return the original result.
- Same key, but a different request body: refuse, because the client is confused and reusing a key for two different things is a bug worth shouting about (HTTP 409).

11.2 The naive version has a hole in it

The obvious implementation is: look up the key in a table, if it is there return the saved response, if not then post the transaction and save the response. Read, then decide, then write.

That works fine until two copies of the same request arrive at nearly the same instant, which is exactly what a double tap or an aggressive retry produces. Both requests look up the key. Neither finds it, because neither has finished yet. Both decide "new key, go ahead." Both post the transaction. You are back to charging twice, except now it only happens under load, which is the worst kind of bug because your tests never see it.

The check and the write have to be one atomic step, and the database is the only thing in this picture that can make that guarantee. So instead of checking first, I let the database's uniqueness constraint be the referee. The idempotency key row goes into a table whose primary key is the tenant plus the key. Inserting it inside the same database transaction that posts the ledger transaction means the whole thing commits together or not at all:

```
                  100 identical requests, same Idempotency-Key
                                  |
       +--------+--------+--------+ ... +--------+--------+
       |        |        |              |        |
       v        v        v              v        v
    [ each opens a SERIALIZABLE transaction and tries: ]
       insert transaction + postings
       insert idempotency_key (tenant, key)   <-- unique, this is the
           referee
       insert audit_log row
       commit
                                  |
            the FIRST to commit the key wins ---+---> commits everything
                                  |
            the other 99 hit a duplicate key ---+---> whole transaction
                                                      rolls back,
```

```
                                                          nothing
16                                                      they wrote
                                                          survives
17                                    |
18                                    v
19              losers look up the winner's row and REPLAY its response
20                                    |
21                                    v
22               result: 1 transaction, 1 audit row, 99 replays
```

The key insight is that the losing requests do not just get rejected. When a request loses the race, its entire transaction rolls back, so the transaction and postings it optimistically wrote vanish too. Then it reads back the row the winner committed, loads that original transaction, and returns it as if it had done the work. From the client's side, all 100 retries got a clean, identical answer. Under the hood, money moved exactly once.

To make the "same key, different body" case work, the stored row also holds a fingerprint of the original request: a hash of the currency and every posting. On a repeat, if the incoming request hashes to the same fingerprint it is a genuine retry and gets the replay. If it hashes differently, someone reused a key for a different payment, and the ledger returns 409 rather than silently doing the wrong thing. I spent a little care making that fingerprint framing collision proof, so that two different transactions can never accidentally hash the same, but that is a detail for the repo.

11.3 The part I got wrong on the first pass

My first version of the fingerprint glued the fields together with a separator byte, which is the kind of thing that looks fine and passes every test you think to write. Then I pictured a hostile input: what if a field itself contains that separator byte? You could, in principle, craft two different transactions that produce the same glued-together string, collide their fingerprints, and slip a different payment through on a reused key. The fix was to length-prefix every field so the boundaries can never be faked. Nobody would hit this by accident, but "nobody would hit this by accident" is not a sentence you want anywhere near the code that decides whether a payment is a duplicate. I only caught it because I went looking for it, which is the whole argument for building these things yourself.

11.4 The audit log: write it once, never touch it again

The second pattern this week is the audit log. A ledger already keeps immutable postings, so why a separate log? Because the postings tell you *what the balances are*, and the audit log tells you *what happened and who did it*. When an auditor or an incident review asks "show me every action that touched this account, in order, with the full before and after," you want a single, honest, tamper-evident answer.

Two design decisions make it trustworthy. First, the audit row is written inside the same database transaction as the posting it describes. There is no separate "log this later" step that could fail on its own and leave you with a transaction nobody recorded. If the posting commits, its audit row committed with it. If anything fails, both roll back. They are one fact.

```
 one database transaction (all of it, or none of it)
+-----------------------------------------------------+
|  insert transaction + postings                      |
|  insert idempotency key (if one was supplied)       |
|  insert audit_log row: action, actor, snapshot      |
+-----------------------------------------------------+
                          |
                  commit succeeds
                          |
 posting and its audit record are now the same
 indivisible piece of history
```

Second, "append-only" is not just a promise in the application code, it is enforced by the database. A Postgres trigger rejects any attempt to UPDATE or DELETE a row in the audit log. Even a direct `DELETE FROM audit_log` from a psql prompt bounces off it. The only sanctioned exception is the demo seeder that resets `go.sohag.pro` every few hours, and it has to explicitly flip a per-transaction flag to be allowed through. The application's normal path can never set that flag, so in production the log is genuinely immutable, not immutable-until-someone-writes-the-wrong-migration.

You can read the log back by transaction or by account:

```
GET /v1/transactions/{id}/audit     what happened to this transaction
GET /v1/accounts/{id}/audit         every audited action touching this
    account
```

The by-account view is paginated with the same keyset cursor as last week's statement endpoint, because "list everything that ever happened to a busy account" is exactly the kind of query that needs a ceiling before it faces a real production account.

11.5 Proving it, not hoping it

The definition of done for this week was a specific, slightly mean test: fire 100 requests with the same idempotency key at the same time, then check the database. Not "it returned 200 a hundred times," but the thing that actually matters: exactly one transaction row exists, exactly one audit row exists, and 99 of the responses were replays of the first. That test runs against a real Postgres in a container, with the race actually happening, not mocked away. Watching it come back green, with the counts landing on exactly one, is a different feeling than reading that unique constraints work. That gap, between knowing a thing and having watched it hold under fire, is the entire reason I am building this in the open.

11.6 My AI companion this week

Same deal as always: I own the decisions, Claude does a lot of the typing. This week the collaboration was interesting because the work was mostly judgment, not code volume. I decided the key had to be inserted inside the posting transaction rather than checked beforehand, and Claude implemented that cleanly and wrote the 100-goroutine hammer test. But the fingerprint collision I described above surfaced during review: a reviewer pass (also Claude, wearing a different hat) flagged the separator-byte weakness, I agreed it was real, and we hardened it before it ever shipped. That is roughly the rhythm I have settled into. I set the invariants and the definition of done, the machine moves fast inside those rails, and the interesting moments are the disagreements and the "wait, what about this input" catches. A payment duplicate check is precisely the code you do not hand over without understanding every line, and I do.

11.7 Where this leaves us

go-ledger now survives retries the way a real payment API has to, and keeps a record of itself that nobody can quietly edit. The API got no bigger in surface area, just harder to misuse, which is the good kind of boring. Next week the ledger grows a second front door: a gRPC layer sitting on the exact same domain services, so REST and gRPC are two protocols over one source of truth rather than two implementations drifting apart.

The repo is open source at **`github.com/sohag-pro/go-ledger`**, and the live API is at `go.sohag.pro`. Go tap Pay twice on something and trust the ledger to have your back.

CHAPTER 12

Week 6 Interview Q&A

Interview-style questions for the Week 6 work: the optional `Idempotency-Key` header on `POST /v1/transactions`, the exactly-once design that inserts the key row inside the posting's SERIALIZABLE transaction, the request fingerprint and 409-on-mismatch contract, the append-only `audit_log` written in the same transaction and enforced by a Postgres trigger, the seeder's gated purge, the keyset-paginated audit reads, and the 100x concurrent test. Every question is answerable from this repo's code, migrations (`0006`), and tests.

12.1 Idempotency design

12.1.1 1. Walk me through what happens when a client sends `POST /v1/transactions` twice with the same `Idempotency-Key`.

What a strong answer covers:

- The key is optional: when the header is present the handler builds a `*domain.Idempotency` and passes it into `TransactionService.Post`; when absent it passes nil and the post proceeds as before.
- First request: inside one SERIALIZABLE transaction it inserts the transaction and postings, inserts the `(tenant_id, idempotency_key)` row, and writes the audit row, then commits.
- Second request with the same key: the key insert hits the primary-key collision, the whole transaction rolls back, the service looks up the stored row, loads the original transaction, and returns it with `Idempotent-Replayed: true`.
- Same key with a different body: returns 409, because the stored fingerprint does not match.

12.1.2 2. Why insert the idempotency key inside the posting transaction instead of checking a table first and then posting?

What a strong answer covers:

- Check-then-act has a race: two concurrent copies of the same request both read "key absent," both decide to post, and both create a transaction. The bug only appears under concurrency, which is where retries actually happen.
- Making the check and the write one atomic step needs the database as the referee; the unique primary key `(tenant_id, idempotency_key)` is that referee.
- Because the key insert shares the transaction with the postings, a losing request's postings roll back too, so nothing partial survives.
- The result is exactly-once without an application-level lock or a distributed mutex.

12.1.3 3. How does the losing request end up returning the right response after it rolls back?

What a strong answer covers:

- `InsertIdempotencyKey` maps the PK collision (constraint `idempotency_keys_pkey`) to the internal sentinel `domain.ErrDuplicateIdempotencyKey`.
- `Post` catches that specific error (only when `idem != nil`) and calls a `replay` helper.
- Postgres blocks the second insert on the same key until the first transaction resolves, so by the time the loser sees the duplicate error the winner has already committed; the loser's follow-up read (outside the transaction) is guaranteed to see the committed row.
- `replay` loads the original transaction via the stored `transaction_id`, overwrites the caller's `*t`, increments `IdempotencyReplays`, and reports `replayed = true`.

12.1.4 4. What is the request fingerprint, and why not just compare the raw request bodies?

What a strong answer covers:

- `Transaction.Fingerprint()` is a SHA-256 over the transaction's semantic content: the currency and each posting's account, signed amount, and description, in order. The transaction id is deliberately excluded.
- Comparing raw bytes would be fragile: whitespace, key order, or an echoed id would make two logically identical requests look different and defeat the replay.
- Comparing the semantic content instead means a genuine retry matches even if serialized slightly differently.
- The stored fingerprint is what distinguishes a true retry (replay) from key reuse with a different body (409 via `ErrIdempotencyConflict`).

12.1.5 5. You length-prefix each field when computing the fingerprint. What attack does that prevent?

What a strong answer covers:

- The first version joined fields with a separator byte; if a field can itself contain that byte, an attacker could shift the boundaries so two different transactions produce the same hash.
- A fingerprint collision on a reused key would let a different payment slip through as a "replay," which is a correctness and security hole in the duplicate check.
- The fix writes each field's length as a fixed-width prefix before its bytes, so field boundaries cannot be forged regardless of content.
- There is a targeted test with adversarial pairs (embedded NUL, boundary-shift) proving the framed fingerprints stay distinct.

12.2 Data integrity and the audit log

12.2.1 6. Why a separate audit log when postings are already immutable?

What a strong answer covers:

- Postings answer "what are the balances"; the audit log answers "what happened, when, and who did it."
- For an incident review or an auditor, a single ordered record of actions with

a snapshot is more useful than reconstructing intent from raw postings.
- In v1 it is create-only: one row per posted transaction, `action = transaction.created`, `before` NULL, `after` a JSON snapshot, `actor` the tenant id until auth lands.
- It is deliberately not a generic before/after diff framework yet, because nothing mutates in place (YAGNI).

12.2.2 7. How do you guarantee that every posted transaction has an audit row, and no transaction exists without one?

What a strong answer covers:

- The audit insert runs inside the same `RunInTx` closure as the transaction and postings, so they commit together or roll back together.
- There is no "log it afterwards" step that could fail independently and leave a gap.
- On the replay path no audit row is written, because that transaction rolled back and no new posting occurred; the audit log records only real creations.
- The `after` column is `NOT NULL` and the service always marshals a snapshot, so a row can never be half-written.

12.2.3 8. "Append-only" is easy to say. How is it actually enforced?

What a strong answer covers:

- A Postgres trigger on `audit_log` rejects UPDATE and DELETE (`BEFORE UPDATE OR DELETE ... FOR EACH ROW`), raising an exception; migration 0006 defines it.
- Enforcement is at the database, not just a convention in the app, so even a direct `DELETE FROM audit_log` in psql is refused.
- This mirrors the balance and currency triggers from earlier weeks: defense in depth against a buggy caller or a bad migration.
- The trigger raises with a named constraint so the failure is identifiable rather than an opaque error.

12.2.4 9. The demo seeder wipes the tenant every few hours, but the audit log rejects deletes. How do you reconcile that?

What a strong answer covers:

- The trigger has one gated exception: it allows the mutation only when the transaction-local GUC `audit.allow_purge` is set to `on`.
- The seeder sets `SET LOCAL audit.allow_purge = 'on'` at the start of its reset transaction, before the deletes; `SET LOCAL` is transaction-scoped, so it cannot leak to other pooled connections.
- The application's normal path never sets the GUC, so in production the log stays immutable; `current_setting(..., true)` returns NULL when unset and the trigger falls through to the exception.
- The seeder's delete order puts `idempotency_keys` and `audit_log` before `transactions` to respect the foreign keys, and there is a test that reseeds over planted rows to prove the gated purge works.

12.3 Concurrency and correctness

12.3.1 10. Your definition of done was "hammer the same key 100 times, get exactly one transaction." What exactly does that test assert, and why that shape?

What a strong answer covers:

- It fires 100 concurrent `Post` calls with the same key against a real Postgres (testcontainers), then asserts exactly one transaction row, exactly one audit row, and that 99 responses were replays returning the same id.
- Asserting "returned 200 a hundred times" would pass even if it created 100 transactions; the counts are what prove exactly-once.
- It runs the real race rather than mocking it, so it exercises the unique-constraint referee and the rollback-then-replay path under genuine contention.
- Watching the counts land on one is the difference between believing unique constraints work and having seen it hold under fire.

12.3.2 11. `RunInTx` retries on serialization failures. Why does a duplicate idempotency key not get retried into an infinite loop?

What a strong answer covers:

- `RunInTx` only retries when the error is a serialization failure or deadlock (SQLSTATE 40001 / 40P01); it classifies via `isSerializationFailure`.
- A PK collision surfaces as `domain.ErrDuplicateIdempotencyKey`, which is not a `*pgconn.PgError` serialization code, so it is returned immediately, not retried.
- A genuine 40001 on the key insert is still wrapped such that it is detected and retried; on the next attempt it sees the committed key and gets a clean duplicate, which the service replays.
- So the two failure modes are cleanly separated: transient conflicts retry, a real duplicate replays.

12.3.3 12. Could a serialization retry cause a transaction to get two audit rows?

What a strong answer covers:

- No: `RunInTx` fully rolls back the transaction between attempts and re-runs the whole closure, so a retried attempt starts clean with nothing from the previous attempt persisted.
- The audit insert lives inside that closure, so it is replayed as part of the same all-or-nothing unit; only the attempt that finally commits leaves a row.
- The closure must be safe to run more than once, which it is: it does not mutate external state that survives a rollback.
- This is the same idempotent-closure contract the posting path already relied on from Week 4.

12.4 API and pagination

12.4.1 13. The audit log is queryable by transaction and by account. Why is the by-account endpoint paginated but the by-transaction one is not?

What a strong answer covers:

- A transaction has a fixed, tiny number of audit rows (one, in v1), so it is bounded by nature and a plain list is fine.
- An active account can accumulate audit rows for every transaction that ever touched it, so an unbounded query is a memory and latency risk on a real production account.
- The by-account endpoint reuses the same keyset cursor machinery as the Week 5 statement endpoint (opaque cursor, `next_cursor`, newest-first), rather than an offset.
- This was flagged in review as the one production concern before merge and closed by adding the cursor.

12.4.2 14. How does the `Idempotent-Replayed` header end up in the OpenAPI spec without hand-editing it?

What a strong answer covers:

- The create handler returns a `CreateTransactionOutput` with a `Replayed bool` field tagged as a response header; huma generates the spec from that typed output, so the header appears automatically.
- The `Idempotency-Key` request header is likewise a tagged input field, and the operation description documents the retry and 409 contract.
- `make openapi` regenerates the committed `api/openapi.yaml`, and a drift test fails if it is stale, so the published spec cannot diverge from the handlers.
- This keeps the single-source-of-truth property from Week 5 intact.

12.5 Trade-offs

12.5.1 15. Reviewers noted several deferred items: the trigger's gated branch would silently no-op an UPDATE, `InsertIdempotencyKey` checks the constraint name rather than a generic unique-violation, and there is no `PostDuration` bucket for replay latency. Why ship with those open?

What a strong answer covers:

- Each was judged safe to defer with a reason: the gate is only ever used for DELETE by the seeder, so a gated UPDATE cannot occur today; the constraint name only appears on a 23505, so the check is behaviorally equivalent to a unique-violation guard.
- The replay-latency gap is observability polish, not correctness; volume is already covered by the `IdempotencyReplays` and `IdempotencyConflicts` counters.
- The bar for shipping was the core invariants (exactly-once, atomicity, append-only, tenant isolation), all verified end to end by a whole-branch review, not zero cosmetic findings.
- Naming the deferrals explicitly, rather than silently leaving them, is the point: they are tracked follow-ups, not forgotten corners.

WEEK 7

CHAPTER 13

Adding gRPC to a Go Ledger Service

Picture two services inside a bank that need to talk to each other a few thousand times a second: a payments service telling the ledger "move this money." You could have them speak JSON over HTTP, the same way your phone talks to the ledger. It works. It is also a bit wasteful: text parsing, no shared schema, no generated client, every field name a string agreement that nobody checks until it breaks in production. For machine-to-machine traffic on the hot path, teams reach for gRPC instead: a binary protocol with a typed contract both sides generate their code from. This week I gave go-ledger a gRPC front door, right next to the REST one it already had.

This is post 7 of 12. Last week was idempotency and the audit log. This week is a second way to talk to the same ledger, and the whole trick is that it is the *same* ledger.

13.1 Why both, instead of picking one

REST is the friendly public front door: curlable, cacheable, understood by everyone, great for a phone app or a dashboard. gRPC is the service-to-service door: a strict `.proto` contract, generated clients in many languages, HTTP/2 under the hood, and no argument about field names because the schema is the source of truth. A real payment platform usually wants both, and go-ledger is meant to look like a real one.

The danger with two APIs is obvious the moment you say it out loud: two front doors, two chances to implement "post a transaction" slightly differently, and now one of them has a bug the other does not. That is how a system slowly grows two personalities.

The way out is to make sure neither front door contains any actual logic. Both are thin translators that speak to one brain.

```
   REST client                    gRPC client
(phone, browser)              (another service)
      |                               |
      v                               v
```

```
   +-----------+                        +-----------+
   | chi+huma  |                        | grpc-go   |   <- transport only
       :
   | handlers  |                        | handlers  |      decode,
      translate,
   +-----------+                        +-----------+      map errors
         \                                    /
          \                                  /
           v                                v
            +----------------------------+
            |   ledger domain services   |   <- ALL the rules live here:
            |  Post, Balance, Statement  |      validation, idempotency,
            +----------------------------+      the balance invariant
                          |
                          v
                     Postgres
```

The REST handler already worked this way from Week 5: it decodes JSON, calls `TransactionService.Post`, and turns the result back into JSON. So the gRPC handler is the same shape with a different translator: decode protobuf, call the *same* `TransactionService.Post`, turn the result into protobuf. The exactly-once idempotency, the atomic audit write, the postings-sum-to-zero rule: none of that is reimplemented, because it lives in the service both doors call. The best part of this whole week was a diff that added a new package and changed zero lines of domain code.

13.2 The contract comes first

With gRPC you write the schema before the code. Here is the shape of posting a transaction, in protobuf:

```
message Posting {
  string account_id = 1;
  int64  amount      = 2;   // minor units, signed: debit positive,
      credit negative
  string description = 3;
}

message PostTransactionRequest {
  string  currency = 1;
  repeated Posting postings = 2;
}

message PostTransactionResponse {
```

```
  Transaction transaction = 1;
  bool        replayed    = 2;   // true if an idempotency key matched an
      earlier post
}
```

Notice money is still a signed `int64` plus a currency string, exactly like the REST API. Same decision, same reason as post 5: no floats anywhere near money. The `.proto` is just a stricter way to write down the contract the REST API already had.

I used `buf` to manage and generate from the protos, which is the modern, boring choice: `buf lint` keeps the schema tidy, and `buf generate` produces the Go server and client stubs using remote plugins, so there is no `protoc` toolchain to install on every machine. One `make proto` and the generated code is regenerated, byte for byte identical, every time.

A quick honest note: the plan called for generating clients in Go, Node, and Python. I shipped Go only. Verifying three language toolchains actually work is a lot of surface for a week already marked "heavy," and the goal this week was one working client, not a polyglot demo. Node and Python are a config change I can add later. Cutting scope you said out loud is not failure, it is the plan working.

13.3 Translating errors, not inventing them

REST speaks HTTP status codes; gRPC speaks its own status codes. The domain, sensibly, speaks neither: it returns typed errors like "this transaction does not balance" or "that account does not exist." So each front door has exactly one small function that translates domain errors into its own dialect, and they translate the *same* errors the same way:

Domain error	REST	gRPC
account not found	404	NotFound
idempotency key reused with a different body	409	AlreadyExists
postings do not sum to zero	422	InvalidArgument
write conflict, retries exhausted	503	Unavailable

Domain error	REST	gRPC
anything unexpected	500	Internal

Because both mappers read from the same list of domain errors, the two protocols cannot quietly disagree about what "that is not allowed" means. And the unexpected case is deliberately vague on both sides: a client gets "internal error," never a stack trace or a database detail.

13.4 Idempotency, the gRPC way

Last week's `Idempotency-Key` header lets a client safely retry a payment. gRPC has no headers, but it has metadata, which is the same idea: key-value pairs that ride alongside the call. So the gRPC `PostTransaction` reads an `idempotency-key` from the request metadata and hands it to the very same `Post` method the REST handler uses. Retry the same call with the same key, and the second one comes back with `replayed = true` and the original transaction, because the exactly-once logic is not in the gRPC layer at all. It is in the service, and the service does not care which door you came through.

13.5 Proving it runs

The definition of done was concrete: post a transaction with `grpcurl`, and have a generated Go client work in an example. The server turns on gRPC reflection, which lets `grpcurl` discover the API with no `.proto` file in hand:

```
$ grpcurl -plaintext localhost:9091 list
ledger.v1.LedgerService

$ grpcurl -plaintext -d '{"currency":"USD","postings":[
    {"account_id":"...cash...","amount":10000},
    {"account_id":"...revenue...","amount":-10000}]}' \
    localhost:9091 ledger.v1.LedgerService/PostTransaction
```

And the automated proof: an integration test that stands up the real gRPC server over an in-memory connection, backed by a real Postgres in a container, drives it with the generated Go client (create two accounts, post between

them, read the balance back), and checks the ledger nets to zero. Same test I would write for REST, one door over. The gRPC server runs on its own port next to REST and the metrics endpoint, and shuts down gracefully with them on a stop signal.

13.6 My AI companion this week

The split was sharp this week. I owned the decisions that shape the system: standard grpc-go over the fancier one-port alternative, a separate port instead of multiplexing, Go-only clients to respect the scope warning, and the rule that no domain code may change. Claude did most of the typing inside those rails: the proto, the buf setup, the handlers, the interceptors, the bufconn test. The interesting moments were, as usual, in review. A reviewer pass caught that the gRPC endpoints had no upper bound on page size while REST capped them, a real divergence and a denial-of-service foot-gun, and that a panic inside a handler was being swallowed with no log at all. Both got fixed before merge. That is the pattern I keep coming back to: the machine is fast and tireless, and my job is to know what "correct" means and to keep looking until I am sure it holds.

13.7 Where this leaves us

go-ledger now has two front doors over one brain. A phone can curl it, a service can call it over gRPC, and both get the identical rules, the identical errors, and the identical idempotency guarantees, because there is only one implementation of any of that. Next week the ledger learns to explain itself under load: OpenTelemetry tracing, structured logs tied to a trace id, and one unbroken trace from an incoming call all the way down to the SQL query.

The repo is open source at **`github.com/sohag-pro/go-ledger`**, and the REST side is live at `go.sohag.pro`.

CHAPTER 14

Week 7 Interview Q&A

Interview-style questions for the Week 7 work: the gRPC front door built on standard grpc-go, generated from protobuf with buf, the thin `internal/grpcserver` adapter over the same `ledger` services REST uses, the `toStatus` error mapping, idempotency over metadata, the recovery/logging/tenant interceptors, the shared `internal/paging` cursor, server reflection and health, the bufconn integration test, and ADR-009. Every question is answerable from this repo's code and ADRs.

14.1 Architecture and the shared domain

14.1.1 1. You now have REST and gRPC. Walk me through how a PostTransaction request flows in each, and where they converge.

What a strong answer covers:

- REST: chi + huma decodes JSON into a typed input, the handler builds domain types and calls `TransactionService.Post`. gRPC: grpc-go decodes protobuf, the handler builds the same domain types and calls the same `Post`.
- They converge at the `ledger` services: all validation, idempotency, the audit write, and the balance invariant live there, not in either transport.
- Each transport only decodes, translates to and from domain types, and maps errors to its own status dialect.
- The gRPC layer (`internal/grpcserver`) added a package and changed zero domain or service code.

14.1.2 2. Why add gRPC at all when REST already works?

What a strong answer covers:

- REST is the public, curlable, cacheable front door for phones and dashboards; gRPC is the service-to-service door with a typed contract, generated multi-language clients, and HTTP/2.
- A payment platform typically wants both, and the project aims to look like a real one.
- gRPC's schema-first contract removes stringly-typed field agreements that only break at runtime.
- The cost is a second surface to run and secure, which the design accepts and scopes (ADR-009).

14.1.3 3. What is the risk of running two APIs, and how does the design neutralize it?

What a strong answer covers:

- The risk is drift: two front doors can implement the same operation differently, so a bug lives in one but not the other, and the system grows two personalities.
- The design makes both transports thin translators with no business logic, so there is only one implementation of any rule.
- This is possible because Week 5 already made the REST handlers thin; the gRPC handler is the same shape with a protobuf translator.
- Evidence: the gRPC handlers call `s.accounts`/`s.txns`/`s.audit`, the same service handles `api.Deps` holds.

14.2 Protobuf, buf, and codegen

14.2.1 4. Why buf with remote plugins instead of a local protoc toolchain?

What a strong answer covers:

- `buf generate` with remote plugins (`buf.build/protocolbuffers/go`, `buf.build/grpc/go`) needs no local protoc install, so generation is reproducible on any machine and in CI.
- `buf lint` keeps the schema consistent; `buf.yaml` sets lint STANDARD and breaking FILE.

- `make proto` regenerates byte-for-byte identical output, verified by a no-diff check.
- Generated Go lives under `internal/genproto/ledger/v1/` (package `ledgerv1`), which keeps it unimportable outside the module while the in-repo example can still use it.

14.2.2 5. How does money cross the wire in the proto, and why the same as REST?

What a strong answer covers:

- Signed `int64 amount` plus a `string currency`, identical to the REST contract and to the internal `int64`.
- No floating point near money, the same decision as ADR-002 and post 5.
- A `Posting` is `{account_id, amount, description}`; a debit is positive, a credit negative, and postings still sum to zero (enforced by the unchanged domain).
- The proto is just a stricter written form of the contract REST already had.

14.2.3 6. The plan asked for Go, Node, and Python clients. You shipped Go only. Defend that.

What a strong answer covers:

- The definition of done needs only grpcurl and a working Go client; verifying three language toolchains is a large surface for a week flagged heavy with scope-creep risk.
- Node and Python are a `buf.gen.yaml` change that can be added later without touching the server.
- Cutting a stated-out-loud scope item is the plan's scope discipline working, not a failure.
- Nothing about the server design blocks adding them later; the contract already exists.

14.3 Error handling and idempotency

14.3.1 7. How do domain errors become gRPC status codes, and how do you keep that consistent with REST?

What a strong answer covers:

- A single `toStatus` helper maps domain sentinels to gRPC codes, mirroring the REST `toHumaErr`: NotFound, AlreadyExists, InvalidArgument, Unavailable, Internal.
- Both mappers read the same domain error list, so the two protocols cannot disagree about what a given failure means.
- Uses `errors.Is` so wrapped errors still match.
- The default case returns a generic `codes.Internal` "internal error" with no stack or database detail, matching REST's no-leak 500.

14.3.2 8. gRPC has no headers. How did you carry the idempotency key, and why does exactly-once still hold?

What a strong answer covers:

- gRPC metadata carries an `idempotency-key`, the direct analog of the REST `Idempotency-Key` header; the handler reads it from the incoming metadata.
- It is threaded into the same `TransactionService.Post`, which owns the key-inside-the-transaction mutex from Week 6.
- The response carries a typed `replayed bool` (the equivalent of the REST `Idempotent-Replayed` header).
- Because the exactly-once logic lives in the service, not the gRPC layer, retrying over gRPC gets the same guarantee as over REST, proven by the bufconn replay test.

14.3.3 9. Metrics and the audit log: did the gRPC path get those for free or did you wire them again?

What a strong answer covers:

- For free: `PostDuration`, `IdempotencyReplays`, and the audit write all happen inside the service layer, so a gRPC post emits the same Prometheus metrics and writes the same audit row with no gRPC-side code.

- This is the direct payoff of keeping the transports thin.
- A gRPC post is indistinguishable from a REST post at the domain and persistence layers.
- The only gRPC-specific observability is the per-RPC logging interceptor.

14.4 Interceptors, lifecycle, and the tenant seam

14.4.1 10. Describe the interceptor chain and why the order matters.

What a strong answer covers:

- Chained unary interceptors: recovery (outermost), logging, then tenant, then the handler.
- Recovery turns a handler panic into `codes.Internal` and logs the method, the recovered value, and a stack, so a panic is observable rather than silent.
- Logging emits one structured slog line per RPC (method, code, duration), mirroring the REST request-logging middleware.
- Tenant injects the default tenant into the context; handlers read it via `tenantFrom`, the same seam REST uses, so real auth later replaces only this interceptor.

14.4.2 11. There is no auth yet. How does a gRPC handler know which tenant it is acting as, and is that safe?

What a strong answer covers:

- The tenant interceptor is installed unconditionally in `NewGRPCServer`, so every handler runs with a tenant in context; handlers call `tenantFrom(ctx)`.
- Today it injects a single configured default (`DEFAULT_TENANT_ID`), identical to REST's `deps.DefaultTenant`.
- When auth lands, it resolves the tenant from a token in that one interceptor, with no handler change.
- The repository methods are already tenant-scoped, so the seam is in place end to end.

14.4.3 12. The service runs three listeners now. How are they started and shut down?

What a strong answer covers:

- REST on `:8080`, metrics on `127.0.0.1:9090`, and gRPC on `GRPC_ADDR` (default `:9091`), each in its own goroutine feeding a shared error channel.
- On SIGTERM all three stop: the HTTP servers via `Shutdown` with a 10s deadline, gRPC via `GracefulStop`.
- `GracefulStop` is run in a goroutine with a fallback to `Stop()` if the shutdown deadline expires, so a stuck in-flight RPC cannot hang shutdown past the budget.
- The error channel is buffered for all three producers so a multi-failure cannot leak a blocked goroutine.

14.5 Testing and trade-offs

14.5.1 13. How did you prove the gRPC layer works, given grpcurl is an external binary?

What a strong answer covers:

- Automated: an integration test stands up the real `NewGRPCServer` over an in-process bufconn listener, backed by a real Postgres in a testcontainer, driven by the generated Go client (create two accounts, post, read balance, assert net zero).
- A second test sends the same idempotency-key metadata twice and asserts `replayed=true` and the same transaction id.
- grpcurl stays a documented manual smoke; server reflection is registered so it works with no proto files, satisfying the DoD.
- A Go client example under `examples/grpc-client` is the "generated Go client works in an example" deliverable.

14.5.2 14. You chose standard grpc-go over ConnectRPC. Defend the choice.

What a strong answer covers:

- Standard grpc-go matches the plan's "gRPC server with interceptors", works with grpcurl via reflection, and is the widely-understood, boring infrastructure choice.
- ConnectRPC would serve gRPC, gRPC-Web, and HTTP/JSON on one port, which is elegant, but REST already covers the HTTP/JSON case and it is a larger opinionated dependency.
- A separate port for gRPC was chosen over cmux/h2c multiplexing to keep the HTTP/1.1 and HTTP/2 split clean on a single VPS.
- These are recorded in ADR-009 so they are not relitigated later.

14.5.3 15. A reviewer found the gRPC read endpoints had no upper bound on page size while REST capped them. Why did that happen and why does it matter?

What a strong answer covers:

- REST enforces maxima through huma validation tags (500 for accounts, 200 for statement and audit); the gRPC handlers only defaulted a limit when it was non-positive, with no ceiling.
- A gRPC client could send a huge limit and trigger an unbounded scan and allocation, which is both a protocol divergence and a denial-of-service vector.
- The fix clamps the gRPC limits to the same maxima via named constants, with a comment noting they mirror the REST tags (which are string literals and cannot be shared as Go constants).
- The lesson: parity between two transports is not automatic for validation that lives in one transport's framework; it has to be re-established deliberately.

WEEK 8

CHAPTER 15

Distributed Tracing for a Go Service

A friend pings you: “hey, that payment felt slow just now.” You open the logs. What you get is a hundred requests worth of lines, all interleaved, all from the same few seconds. You can see that *something* took a while, but you can’t tell which lines belong to that one payment, or whether the time went to Postgres, to the domain logic, or to the network in between. You are staring at a haystack and someone is describing one specific piece of straw.

That is the exact pain distributed tracing solves. A trace ties every step of one request together under a single id, so you can pull that one thread out of the pile and see the whole thing: where it went, how long each hop took, and where it stalled. This week I wired it into go-ledger, and the goal was concrete: post one transaction, and get back one trace that runs from the HTTP handler, through the domain service, down to every SQL query, all linked. This is post 8 of 12.

15.1 What “one trace” actually looks like

Say you post the dinner-repayment transaction from Week 1: move 600 taka from my wallet to my friend’s. That single API call touches four layers. Tracing records each layer as a *span* (a timed unit of work with a name), and nests the child spans inside their parent. Lined up on a timeline, one post looks like this:

```
POST /v1/transactions                                  [============]
    14.2ms└─
   ledger.PostTransaction                            [==========]
       11.8ms├─
         BEGIN (serializable)                        [=]            0.9
             ms├─
         INSERT transaction                          [==]           2.1
             ms├─
         INSERT postings (x2)                        [===]          3.4
             ms├─
         INSERT idempotency_key                      [=]            1.2
             ms├─
         INSERT audit_log                            [==]           1.9
```

```
            ms└─
8       COMMIT                                              [==]
            1.8ms
```

Now the "it felt slow" conversation is a five-second look instead of a guessing game. If the COMMIT span is fat, you have write contention. If an INSERT is fat, maybe an index is missing. If the gap between the HTTP span and the first SQL span is fat, the time went somewhere in your own code, not the database. The shape of the waterfall tells you where to look before you have read a single line of code.

The tool that produces this is OpenTelemetry (OTel): a vendor-neutral standard for traces, metrics, and logs, with a Go SDK. You instrument your code once against the standard, and you can send the data to whatever backend you like. Locally I send it to Jaeger, which draws exactly the waterfall above in a browser. Remotely I send it to Honeycomb's free tier. The app code does not change between the two; only an environment variable does.

15.2 Instrument the edges, then the middle

Most of the tracing you want comes from the boundaries, and the boundaries already have libraries. I wrapped the HTTP router with `otelhttp` and added `otelgrpc` to the gRPC server (Week 7 left a clean seam for exactly this), so every inbound request opens a span automatically. For the database, `otelpgx` plugs into the pgx connection pool as a query tracer and emits one span per SQL statement, which is where those `INSERT` and `COMMIT` rows in the waterfall come from.

That covers the top and the bottom. The one span I wrote by hand is the middle one, `ledger.PostTransaction`, because that is my domain's unit of meaning and no library knows it exists:

```
ctx, span := s.tracer.Start(ctx, "ledger.PostTransaction",
    oteltrace.WithAttributes(
        attribute.String("tenant_id", tenantID),
        attribute.Int("transaction.posting_count", len(t.Postings)),
        attribute.Bool("idempotency.present", idem != nil),
    ),
)
defer span.End()
```

The `ctx` that comes back carries the span, and because every layer already

passes `context.Context` down the call chain, the SQL spans that happen later automatically nest under this one. That is the whole magic of context propagation: you thread one value through, and the tree assembles itself.

15.3 The part I'm quietly proud of: what is *not* in that span

Look again at those attributes. Tenant id, a count, a boolean. No amount. No account balance. No account id as a value. That restraint is deliberate.

Honeycomb is a US company's servers. The moment I ship a span there, I have handed a third party whatever I put in it. For a payment system, "600 taka from account A to account B" is exactly the kind of data you do not casually export to someone else's cloud because a default made it easy. So the rule I baked in, and wrote into the architecture decision record, is simple: **identifiers and counts, never money.** `otelpgx` is configured to leave query arguments off, so the actual values bound into a SQL statement never leave the process either. You can trace the *shape* of the work without leaking the *contents*.

This is the kind of thing that is trivial to get right when you decide it on purpose in week 8, and quietly awful to discover in an incident review two years later.

15.4 The two decisions I went back and forth on

Should everything go through OpenTelemetry? OTel does metrics too, not just traces. The tidy-looking move is to route *all* my metrics through it and have one instrumentation API. I didn't. I already had four hand-built Prometheus metrics from earlier weeks, including the transaction-latency histogram that any alert I ever write will be based on. The OTel Prometheus exporter renames things: it appends unit suffixes and adds its own labels. Renaming a metric is how you silently break an alert and only find out when the thing it was supposed to page you about happens in silence.

So I split it. The four existing metrics stay exactly as they are, same names. The new "how many requests, how many errors, how fast" numbers for HTTP

and gRPC (the classic Rate-Errors-Duration trio) come from the OTel side and get merged into the same `/metrics` endpoint. Two styles living together on purpose beats one clean style that breaks my alerting. Boring and stable won.

What happens when tracing isn't configured? My first instinct was: if there's no endpoint set, just print spans to stdout so you can still see them. I talked myself out of it. On the server, stdout is where my structured logs go. Dumping a flood of span JSON into that same stream, silently, because someone forgot to set a variable, is a great way to make a misconfiguration invisible. So the behavior is: an endpoint set means export to it; the explicit `OTEL_TRACES_EXPORTER=console` means print to stdout for local eyeballing; and *nothing* set means tracing is off with one clear warning line at startup. A missing config should be loud and boring, not a silent surprise.

15.5 The thing that makes logs and traces click together

Here is the payoff for the "which lines belong to my payment" problem I opened with. Every span has a `trace_id`. If I stamp that same id onto every log line emitted during the request, then finding a slow trace in Jaeger and finding its logs become the same lookup.

I did this with a tiny wrapper around Go's `slog` handler. On every log record, it checks the context for an active span and, if there is one, adds the trace and span ids:

```
func (h traceHandler) Handle(ctx context.Context, rec slog.Record) error {
    if sc := trace.SpanContextFromContext(ctx); sc.IsValid() {
        rec.AddAttrs(
            slog.String("trace_id", sc.TraceID().String()),
            slog.String("span_id", sc.SpanID().String()),
            slog.Bool("sampled", sc.IsSampled()),
        )
    }
    return h.inner.Handle(ctx, rec)
}
```

A dozen lines, no new dependency, and now a log line looks like `{"msg":"transaction posted","trace_id":"a1b2...","tenant_id":"..."}`. Copy the `trace_id`, paste it into Jaeger, and you are looking at the exact waterfall for

that exact request. Logs tell you *what happened*; the trace tells you *where the time went*; the shared id is the bridge.

15.6 One boring-on-purpose detail: don't trace everything

There is a demo seeder in go-ledger that resets and repopulates the sample data every few hours, writing a few hundred transactions each time. If I trace all of that at full volume, plus a span for every single SQL query, I will happily burn through Honeycomb's free tier on data nobody will ever look at. So sampling is configurable through the standard OTel environment variables (in production I send a quarter of traces, not all of them), and the health-check and metrics endpoints are excluded from tracing entirely. Otherwise ninety percent of your "traces" are just a load balancer asking if you're alive. Keep the signal, drop the noise.

15.7 Where this leaves the ledger

One transaction post now produces one trace across HTTP, the domain, and SQL, and its log lines carry the same id. It runs to Jaeger with `make jaeger` and a single environment variable, and to Honeycomb by changing that variable and adding an API key. No money leaves the process. The existing metrics and their future alerts are untouched.

The repo is open source at **`github.com/sohag-pro/go-ledger`**, and the full reasoning for the hybrid metrics, the loud no-op, and the no-money-in-spans rule lives in **`ADR-010`**. Next week: testing for real. Unit, integration, load, and pointing enough concurrency at the ledger to try to make it lie about someone's balance.

CHAPTER 16

Week 8 Interview Q&A

Interview-style questions for the Week 8 work: the `internal/observability` setup (Resource, TracerProvider, env-selected exporter, W3C propagator), the trace-correlating `slog` handler, the hybrid metrics decision (native Prometheus collectors plus an OTel meter for RED via one shared registry), spans across HTTP (`otelhttp`), gRPC (`otelgrpc`), the domain (`ledger.PostTransaction`), and SQL (`otelpgx`), the no-money-in-spans rule, sampling and the excluded noise paths, the deferred telemetry flush on shutdown, and ADR-010. Every question is answerable from this repo's code and ADRs.

16.1 Tracing architecture and propagation

16.1.1 1. Walk me through what happens, span by span, when a single transaction is posted over HTTP.

What a strong answer covers:

- `otelhttp` wraps the chi router and opens the root server span; the `otelRouteName` middleware renames it to the matched route pattern once routing resolves.
- The handler calls `TransactionService.Post`, which opens the `ledger.PostTransaction` span with `s.tracer.Start(ctx, ...)`; the returned `ctx` carries the span.
- `otelpgx` on the pgx pool emits a child span per SQL statement (BEGIN, INSERTs, COMMIT), nested because the same `ctx` threads down.
- The result is one trace: HTTP to domain to SQL, all linked, satisfying the ADR-010 definition of done.

16.1.2 2. Nothing explicitly passes a "current span" between these layers. How does the SQL span know it belongs under `ledger.PostTransaction`?

What a strong answer covers:

- Context propagation: `tracer.Start` returns a new `ctx` with the span embedded, and every layer already threads `context.Context` down the call chain.
- `otelpgx` reads the active span from the `ctx` it is handed and starts its spans as children of it.
- This is why the manual domain span reassigns `ctx` (`ctx, span := ...`) rather than discarding it.
- The same mechanism carries the trace across process boundaries via the W3C `traceparent` header, set up by the composite `TraceContext` propagator.

16.1.3 3. Why instrument at four layers instead of just wrapping the HTTP handler?

What a strong answer covers:

- A single top-level span tells you a request was slow but not where the time went; the value is in the waterfall.
- The edges (`otelhttp`, `otelgrpc`, `otelpgx`) come from libraries almost for free; only the domain span is hand-written because no library knows what `PostTransaction` means.
- Per-query DB spans localize latency to a specific statement (a fat COMMIT means write contention, a fat INSERT hints at an index).
- The one manual span is the unit of business meaning, and it carries the safe attributes the libraries cannot know to add.

16.2 The hybrid metrics decision (defend it)

16.2.1 4. OpenTelemetry can emit metrics too. Why did you keep four hand-built Prometheus collectors instead of routing everything through OTel for a single instrumentation API?

What a strong answer covers:

- The four existing metrics (including `transaction_post_duration_seconds`) are SLO metrics tied to earlier latency baselines and any future alert.
- The OTel Prometheus exporter renames output: unit suffixes, `otel_scope_*` labels, a `target_info` series. Renaming an alert-bearing metric silently breaks the alert.
- The single-API win does not pay for breaking working alerting in a payment service, so the domain metrics stay native with their exact names.
- RED for HTTP and gRPC still comes from the OTel side (`otelhttp`/`otelgrpc`) and merges into the same registry, so you get unified request metrics without the rename.

16.2.2 5. Then how do the native collectors and the OTel-produced metrics end up on one `/metrics` endpoint?

What a strong answer covers:

- `metrics.MeterProvider()` builds an OTel Prometheus exporter with `otelprom.WithRegisterer(registry)`, pointing it at the package's existing private registry.
- The same registry already holds the native collectors and the Go/process collectors, so one `Handler()` serves all of them.
- `WithoutScopeInfo()` and `WithoutTargetInfo()` drop the exporter's clutter.
- OTel runtime instrumentation is deliberately not enabled, so `go_*` and `process_*` stay single-sourced from the native collectors and cannot double-register.

16.2.3 6. A skeptic says “two metric systems in one service is a smell.” Defend the coexistence.

What a strong answer covers:

- It is a deliberate trade of API purity for operational stability: the invariant-critical metrics keep stable names, the convenient RED metrics come from instrumentation libraries.
- The seam is narrow: both write into one registry and one endpoint, so a scraper sees a single surface.
- The alternative (full unify) has a concrete, silent failure mode (broken alerts); the coexistence has only a readability cost.
- ADR-010 records the reasoning so the choice is defensible later, not mysterious.

16.3 Exporter selection and failure modes

16.3.1 7. Describe the three exporter modes and why a missing endpoint is a no-op instead of printing spans to stdout.

What a strong answer covers:

- Endpoint set (`OTEL_EXPORTER_OTLP_ENDPOINT`): OTLP/HTTP exporter, the only production mode.
- `OTEL_TRACES_EXPORTER=console`: stdout exporter, opt-in for local eyeballing.
- Neither: a no-op tracer plus one `warn` line at startup.
- On the server stdout carries the JSON logs; silently flooding it with span dumps because a variable was forgotten hides a misconfiguration. A missing config should be loud and boring, not a silent surprise.

16.3.2 8. Why OTLP over HTTP rather than gRPC, given you already run a gRPC server?

What a strong answer covers:

- OTLP/HTTP is the lower-ops, firewall-friendly choice and is Honeycomb-

native.
- Running gRPC elsewhere in the app does not make an OTLP/gRPC exporter the better fit; there is no payoff here.
- Both Jaeger and Honeycomb accept OTLP/HTTP on 4318, so the same code path serves local and remote.
- It keeps the exporter a single, well-understood dependency.

16.3.3 9. The sampler is not set in code. How is sampling controlled, and why leave it to the SDK default?

What a strong answer covers:

- `NewTracerProvider` is built without a `WithSampler` option, so the SDK honors `OTEL_TRACES_SAMPLER` and `OTEL_TRACES_SAMPLER_ARG`, defaulting to parent-based always-on.
- Production sets a ratio (documented at 0.25) because the demo seeder writes hundreds of transactions on an interval and each fans out to per-query DB spans, which would exhaust the Honeycomb free tier.
- Health and metrics paths are excluded from tracing regardless, so traces are real request work.
- Leaving it to env means operators tune cost without a rebuild.

16.4 Data integrity and what leaves the process

16.4.1 10. What span attributes does `PostTransaction` set, and what is deliberately excluded? Why does it matter here more than in most services?

What a strong answer covers:

- Included: `tenant_id`, `transaction.posting_count`, `idempotency.present`. Excluded: amounts, balances, account ids as values.
- `otelpgx` runs with query arguments off, so bind values (account ids, amounts, idempotency keys) never leave the process.
- Honeycomb is a third-party US SaaS; a span is data handed to someone else's cloud, and "600 from A to B" is exactly what a payment system should not casually export.

- The rule (identifiers and counts, never money) is written into ADR-010 so richer trace data requires a deliberate data-residency decision, not a default.

16.4.2 11. `otelpgx` can attach the SQL statement and its parameters. How did you configure it, and what is the residual exposure?

What a strong answer covers:

- Configured with `otelpgx.NewTracer(otelpgx.WithTrimSQLInSpanName())` and, crucially, without `WithIncludeQueryParameters`, so the v0.11.1 default of parameters-off holds.
- The statement text can still appear (table and column names), but not the bound values, so no account id or amount is emitted.
- Residual exposure to name is the request path on the HTTP span (which contains ids); those are identifiers, not money, and are inherent to the chosen `otelhttp`.
- The trade is documented: trace the shape of the work, not its contents.

16.5 Logs and traces together

16.5.1 12. How do log lines correlate to traces, and what is the one edge case your `slog` handler comment warns about?

What a strong answer covers:

- `NewTraceHandler` wraps the JSON handler; on each record it reads the span context from `ctx` and, if valid, adds `trace_id`, `span_id`, and `sampled`.
- Because handlers log with the request `ctx` (`InfoContext`, `ErrorContext`), the ids appear automatically; copy the `trace_id` into Jaeger to find the request.
- The edge case: if a caller used `WithGroup`, the inner handler would nest the correlation ids under that group instead of at the top level, defeating a flat lookup.
- The ledger's loggers log flat attributes, so it does not arise today, but the comment keeps it that way on purpose.

16.5.2 13. Why a hand-written 12-line handler instead of the `otelslog` bridge?

What a strong answer covers:

- The goal is correlation (trace ids on existing logs), not remote log export.
- `otelslog` turns log records into OTel log records shipped over OTLP, a third signal pipeline this week does not ship.
- The wrapper adds no dependency and keeps logs on stdout where the existing tooling reads them.
- It delegates `Enabled`, `WithAttrs`, and `WithGroup` to the inner handler, so grouped and pre-attributed loggers keep working.

16.6 Shutdown and failure modes (hard follow-ups)

16.6.1 14. A code review caught that telemetry was being shut down before the servers drained. Explain the bug and the fix.

What a strong answer covers:

- Original order flushed the trace and meter providers inside the graceful block, before `srv.Shutdown` and the gRPC graceful stop, so requests still draining produced spans and metrics against already-closed providers and lost them.
- Worse, all steps shared one 10s budget, so an unreachable OTLP endpoint could spend most of it flushing and starve the real request drain the budget exists to protect.
- The fix moves the flush into a `defer` with its own bounded context, so it runs after the servers have stopped accepting work (their shutdown runs earlier in `run`'s body) and also covers early-return error paths.
- Net effect: in-flight requests keep their spans, and a hung exporter cannot eat the drain budget.

16.6.2 15. Two requests hit the same hot account concurrently and one is retried after a serialization conflict. What does the trace show, and does tracing change the concurrency behavior?

What a strong answer covers:

- Each request is its own trace; the retried one shows the SERIALIZABLE transaction spans replayed (a second BEGIN-to-COMMIT cycle under the same `ledger.PostTransaction` span), and the native `transaction_post_serialization_retries_total` counter ticks.
- Tracing is observation only: spans and attributes do not touch the SERIALIZABLE isolation, the retry loop, or the balance invariant, so behavior is unchanged.
- The domain span records an error status only on a failed post or a validation rejection, not on a retried-then-committed post, so a successful retry is not a false error.
- The waterfall makes contention visible (a fat or repeated COMMIT), which is the practical payoff of the per-query spans.

WEEK 9

CHAPTER 17

Testing a Payment Ledger End to End

Here is a small program. It adds two amounts of money together.

```
a := MinInt64   // the most negative number an int64 can hold
b := MinInt64
sum, err := a.Add(b)
// sum is 0. err is nil.
```

Two enormous negative numbers, added together, and my ledger said the answer was zero and nothing went wrong. No error, no warning, just a clean, confident, completely wrong result. On a payment system, "we added two things and silently got zero" is the exact shape of a bug that makes money disappear.

I did not find this by staring at the code. I found it because this week I stopped asking "do my tests pass?" and started asking "what would it take to make this thing lie to me?" That is a different question, and it is the whole point of a testing week.

This is post 9 of 12. The ledger already has a domain model, a Postgres schema, concurrency control, REST and gRPC APIs, idempotency, and tracing. What it did not have was proof that any of it holds up when things go wrong. So this week is five kinds of test, each doing a job the others cannot.

17.1 Coverage is a liar (but a useful one)

The obvious place to start is coverage: what percentage of my lines does the test suite actually run? I set a gate in CI: the build fails if coverage on the core code drops below 80%.

But here is the trap. Coverage counts lines *touched*, not behavior *checked*. I can hit 80% by testing every getter and constructor while never once testing the branch that rejects an unbalanced transaction. On a ledger, the dangerous lines, the rollback path, the overflow check, the currency mismatch, are exactly the ones that are annoying to reach and easy to skip. A high number can mean "well tested" or it can mean "well padded," and the number itself will

not tell you which.

So I did two things. First, the gate is not one flat number. The money-critical packages carry higher floors than the plumbing:

```
package            floor   why
-----------------------------------------------------------
internal/domain     90%    the double-entry invariant lives here
internal/ledger     85%    posting, idempotency, rollback
internal/postgres    80%    the SQL and the DB constraints
everything else      80%    transport, wiring, plumbing
-----------------------------------------------------------
generated code     excluded  it tests the code generator, not me
```

Second, and more important, I treated the coverage number as a *floor to clear on the way to real tests*, not the goal itself. Hitting 90% on the domain package was worth doing only because getting there meant writing tests for the branches that actually matter. The percentage is a smoke alarm: useful for telling you a whole room is untested, useless for telling you the tests you do have are any good.

For that second question, you need a different tool.

17.2 Property tests, or how the ledger caught itself lying

Most tests are examples. You write down an input, you write down the answer you expect, you check they match. That is fine, but you only ever check the handful of cases you thought of. The bug that bites you is the one you did not think of.

Property-based testing flips this around. Instead of "for input 5, expect 10," you state a *property* that must hold for **all** inputs, and the library throws hundreds of random cases at it trying to find one that breaks it. For a ledger, the properties write themselves:

- Add any two amounts in the same currency, and either you get the exact mathematical sum, or you get a loud overflow error. Never a silent wrong answer.
- Apply any random sequence of balanced transactions, and every account's balance always equals the sum of its own postings, and the total across all

accounts is always zero.
- Adding two amounts in different currencies always fails.

I wrote the first one as: for any two int64 amounts, if their true sum fits in an int64, `Add` returns it exactly; if it does not, `Add` returns an overflow error. Then I let the library hunt.

It found `MinInt64 + MinInt64` in a few thousand tries.

The bug was in my overflow check. I was detecting overflow by looking at signs: two positives that produce a negative overflowed, two negatives that produce a positive overflowed. Reasonable, and wrong at one point. `MinInt64 + MinInt64` in two's complement wraps all the way around and lands on exactly zero. Zero is not positive, so my "two negatives became positive" check missed it, and the function returned zero with no error. The fix is the textbook signed-overflow test: adding a negative number should never make the result larger.

I want to be clear about what happened here, because it is the best argument for property testing I have. I wrote that money type weeks ago. I tested it. It passed. It shipped. It had a bug that could destroy money, sitting quietly in the one input I never thought to try, and no example-based test I would have written by hand was ever going to catch it. The randomness caught it in seconds.

17.3 Chaos: kill the database at the worst possible moment

Property tests check the logic. They do not check what happens when the infrastructure fails halfway through. For that, you have to actually break something, on purpose, at a precise moment.

The scenario I care about is the one that keeps ledger engineers up at night. A client sends a payment. The server commits it to Postgres. And then, before the "success" reply gets back to the client, the network drops. The commit *happened*. The money *moved*. But the client has no idea. All it knows is that its request died. So it does the natural thing: it retries.

If your system is not careful, that retry posts the payment a second time, and now your friend paid for dinner twice.

```
  Client                 Server              Postgres
    |                       |                    |
    |---- post payment ---->|                    |
    |                       |---- COMMIT ------->|
    |                       |                    | (committed, durable)
    |                       |<-- ok ------  X    |  <- reply lost here
    |         X  <----------|                    |
    |  (client never hears back: did it work??)  |
    |                       |                    |
    |---- retry, same key ->|                    |
    |                       |--- already done -->|
    |<--- original result --|                    |  <- no double post
```

I tested two versions of this. In the first, the connection dies *before* the commit reaches Postgres. The correct outcome is a clean rollback: nothing persists, not a single half-written posting, because a payment that only half-happened is not allowed to exist. In the second, the commit *lands* but the acknowledgement is lost, the diagram above, and the correct outcome is that the retry recognizes the idempotency key and returns the original transaction instead of creating a new one.

The tricky part is making "the connection dies at exactly the right moment" happen *reliably*. My first instinct was to just kill the Postgres container mid-request. That is a bad idea: it is a race. Sometimes you kill it before the commit, sometimes after, sometimes you miss the window entirely and test nothing, and occasionally it fails in CI for reasons nobody can reproduce. A flaky chaos test is worse than no chaos test, because it teaches you to ignore red builds.

So instead I put a programmable network proxy (a tool called toxiproxy) between the app and the database, and I cut the connection at a controlled point in the protocol, driven by the code, not by timing. For the rollback case, I sever the link the instant the app tries to send its COMMIT, so the commit never reaches the server. For the ambiguous case, I cut only the *reply* path: the COMMIT flows to Postgres and durably commits, but the acknowledgement coming back gets dropped. That second one is the money-losing scenario made real and repeatable, and the test proves the retry does not double-post. Same failure, every run, no luck involved.

17.4 Load: 500 requests a second, without cheating

The last question is boring but real: does it hold up under load? The target I set was 500 payments per second with a 99th-percentile latency under 100 milliseconds, run locally against the full stack in Docker.

Load testing a ledger is easy to fake, in two specific ways, and I was determined not to.

The first fake: idempotency keys. If every request in your load test reuses the same key, you are not testing the payment path at all. After the first one, every request is a cheap "oh, I've seen this key, here's the cached answer" lookup that never touches the real write logic. You would get a gorgeous throughput number that measures the wrong thing entirely. So every request in my test carries a unique key, with a deliberate small slice of duplicates mixed in to exercise the replay path honestly.

The second fake, and this one is a genuine temptation, is durability. That 100-millisecond target is dominated by one thing: how long Postgres takes to physically flush each commit to disk. There is a setting that makes commits return before they are safely on disk, and flipping it would make my latency numbers look fantastic. It would also mean that a power cut could lose a committed payment. On a ledger, that is not a performance tuning knob, it is a lie. Durability stays on. The number is what it is: 500 requests a second, p99 well under the target, with every commit actually on disk.

I also run two shapes of load, not one. A "fan-out" test spreads payments across many accounts to measure raw throughput. A "hot account" test slams a single account, because that is where real systems fall over: everyone contends for the same row lock, and the tail latency blows up. The system that looks fast when work is spread out is not the same system as the one under contention, and only testing the easy shape tells you nothing about the hard one.

17.5 The pyramid, and why each layer is load-bearing

Five kinds of test, each catching what the others structurally cannot:

```
               /\
              /  \      load: does it hold up at 500 RPS,
             / L  \     durability on, under contention?
            /------\
           /  chaos \   what happens when the DB dies
          /          \  mid-commit? rollback? double-post?
         /------------\
        / integration  \  does it work against a real
       /                \ Postgres, real constraints, real SQL?
      /------------------\
     /     property       \  does the invariant hold across
    /                       \ thousands of random inputs?
   /-------------------------\
  /          unit             \  do the individual pieces
 /                              \ do what I think they do?
/--------------------------------\
```

A unit test would never have found the ambiguous-commit double-post, that needs a real database and a real network failure. A chaos test would never have found the `MinInt64` overflow, that needs thousands of random inputs. A load test would never have told me the invariant holds, only that it is fast. Each layer sees a different failure. Skip one and you have a blind spot exactly the size of the thing it would have caught.

17.6 My AI companion

As with every week, I built this with **`Claude`** in my terminal, and the honest split is worth describing, because this was a week where the division of labor really mattered.

I owned the decisions and the skepticism. When Claude drafted the chaos test by literally killing the Postgres container, I pushed back: that is a race, use a proxy and cut the connection deterministically. When it wrote the ambiguous-commit case as a plain retry with no actual failure injected, an automated review I ran flagged it, and I sent it back to inject the real ack-loss, because the whole value of that test is the failure. I made the calls on the coverage floors, on keeping durability on, and on a coverage badge that uses no long-lived secrets.

Claude owned the execution, and it was genuinely good at it: writing the several thousand lines of test code across the packages, wiring up toxiproxy,

reconciling my plan against the actual code when my assumptions about environment variables and API shapes were slightly off, and running the whole thing task by task with a review after each one. And, satisfyingly, one of those reviews is what caught the weakened chaos test before it could ship. The overflow bug in my money type was found by the property test *it* wrote, running against the code *I* wrote weeks ago. Neither of us gets to feel too clever.

17.7 Where this leaves the ledger

The invariant that this whole project is built on, money never appears or disappears, is now something I have *watched* survive an overflow, a mid-commit database death, a lost acknowledgement, and 500 requests a second. That is a different kind of confidence than "the tests pass." It is closer to "I tried to break it in the specific ways it could break, and it did not."

The full strategy is written up in `ADR-011`, and the repo, with a coverage badge that is finally honest, is at `github.com/sohag-pro/go-ledger`.

Next week the project leaves the laptop for good: Docker and the real infrastructure that puts this thing on a server.

CHAPTER 18

Week 9 Interview Q&A

Interview-style questions for the Week 9 work: the coverage gate with money-path floors (`scripts/coverage.sh`, `make cover`), property tests with `rapid` (`internal/domain/money_prop_test.go`, `model_prop_test.go`) and the real overflow bug they exposed in `Money.Add`, the deterministic toxiproxy chaos test (`internal/ledger/chaos_test.go`) covering pre-commit rollback and ambiguous-commit idempotency, the k6 load test (`test/load/post_transactions.js`) with its fan-out and hot-account scenarios, the restructured CI pipeline, the self-hosted coverage badge, and ADR-011. Every question is answerable from this repo's code and ADRs.

18.1 Coverage strategy

18.1.1 1. Your CI fails the build if coverage drops below a threshold. Walk me through what your gate actually measures and why it is not just one number.

What a strong answer covers:

- One statement-weighted total from a single merged profile (`go test ./internal/... -coverprofile`), read from `go tool cover -func`'s `total:` line, not an average of per-package percentages (which would weight a 20-line package like a 2000-line one).
- Per-package floors, not a flat number: `domain` at 90%, `ledger` at 85%, `postgres` at 80%, everything else counts toward the global 80%.
- Generated code (`internal/genproto`, `internal/postgres/sqlc`) is stripped from the profile before the number is computed, because it tests the code generator, not the author.
- The higher floors sit exactly on the money path, where a missed branch costs real money.

18.1.2 2. Why put a higher floor on `internal/domain` than on `internal/grpcserver`? Isn't coverage coverage?

What a strong answer covers:

- Coverage counts lines touched, not invariants checked; the value of a covered line depends on what breaks if it is wrong.
- A missed branch in the double-entry invariant or the overflow check can create or destroy money; a missed branch in transport formatting shows a slightly wrong error string.
- Concentrating the pressure where a bug is catastrophic is a deliberate risk-weighting, not a uniform metric.
- The flat-floor alternative lets the money packages sit at the floor while transport code carries the average, which is the opposite of what you want.

18.1.3 3. A skeptic says "80% coverage is meaningless, you can hit it with tests that assert nothing." How do you respond, given your own gate is a percentage?

What a strong answer covers:

- Agreeing with the premise: the number is a floor to clear on the way to real tests, a smoke alarm for whole untested rooms, not a proof of test quality.
- Quality is enforced elsewhere: property tests for invariants, a review pass on every task that specifically flags assertions checking nothing or padding.
- The gate catches "a whole package is untested"; it cannot catch "these tests are weak," and pretending otherwise is the trap.
- Concrete evidence the tests are real: the property test found an actual overflow bug, which a padding test never would.

18.2 Property-based testing

18.2.1 4. You found a genuine bug in your money type this week. Tell me exactly what it was and how the test caught it.

What a strong answer covers:

- `Money.Add(MinInt64, MinInt64)` returned `Money{amount: 0}` with a `nil` error: a silent wrong answer on a value that should overflow.
- The old check detected overflow by sign (two negatives producing a positive); `MinInt64 + MinInt64` wraps in two's complement back to exactly `0`, which is not positive, so the check missed it.
- The property "for any two int64 amounts, `Add` returns the exact sum or an overflow error, never a silent wrong answer" is what `rapid` was fuzzing when it hit the case in a few thousand tries.
- The fix is the standard signed-overflow test: adding a positive must not decrease the value and adding a negative must not increase it (`sum < m.amount` / `sum > m.amount`), which catches the `MinInt64` wrap.

18.2.2 5. Why did property testing find this when your existing example-based unit tests, which passed, did not?

What a strong answer covers:

- Example tests only exercise the inputs the author thought of; the bug lived in the one boundary value nobody hand-picks.
- Property tests state a rule over the whole input space and let the library search for a counterexample, so they explore inputs a human would not write down.
- `MinInt64 + MinInt64` is a classic two's-complement corner that is easy to reason past and hard to guess.
- After the fix, a deterministic boundary test pins that exact pair so the regression is caught every run, not just probabilistically.

18.2.3 6. Your stateful property test applies a random sequence of transactions and checks the invariant after each step. What class of bug does that catch that a single-transaction test cannot?

What a strong answer covers:

- A bug where each individual operation is valid but the aggregate over a sequence drifts, for example an accumulation error that only shows once two transactions touch the same account.
- The model persists across steps (`ledgerModel` tracks running per-account balances), so an overwrite-instead-of-accumulate bug surfaces the moment two steps overlap.
- The checked invariants are the project's core: each balance equals the sum of its own postings, and the signed total across all accounts is zero.
- It mirrors how a real ledger accumulates state, rather than testing one operation in isolation.

18.2.4 7. Your concurrency property test runs under the race detector. Why is that combination necessary, and how did you make the test itself race-clean by construction?

What a strong answer covers:

- The property (no lost updates when many goroutines post concurrently) is about a concurrency invariant, so it must run where a data race would be detected.
- The test pre-generates all transactions before spawning goroutines, because `rapid.T` is not safe to call from multiple goroutines.
- The shared model is mutex-guarded, so removing the lock would be an unambiguous race the detector flags, which is what makes the test meaningful rather than lucky.
- Passing the loop value as a parameter avoids the closure-capture footgun.

18.3 Chaos testing (the hard part)

18.3.1 8. Describe the two database-failure scenarios your chaos test covers and why they have different correct outcomes.

What a strong answer covers:

- Pre-commit failure: the connection dies after `BEGIN` and the writes but before `COMMIT` reaches Postgres; correct outcome is a clean rollback, zero partial postings, because a half-written transaction is not allowed to exist.
- Ambiguous commit: the `COMMIT` lands and is durable, but the acknowledgement is lost, so the client does not know it succeeded; correct outcome is that a retry with the same idempotency key returns the original transaction, not a second one.
- The second is the money-losing case: the natural client reaction to an unknown outcome is to retry, and without idempotency that double-posts.
- The distinguishing assertion: case A asserts zero rows persisted, case B asserts exactly one row persisted despite the client-side error.

18.3.2 9. Why toxiproxy instead of just killing the Postgres container mid-request? Defend the added dependency.

What a strong answer covers:

- Killing a container is timing-racy: you often land before the commit or after it and test nothing, while occasionally going red in CI for no reproducible reason.
- A flaky chaos test is worse than none, because it trains you to ignore red builds.
- A programmable proxy lets you cut the connection at a controlled point in the protocol, driven by code, not wall-clock timing, so the failure is deterministic and repeatable.
- It is also faster (no image restart) and can inject direction-specific failures, which the ambiguous-commit case needs.

18.3.3 10. The ambiguous-commit case requires the COMMIT to reach Postgres while the acknowledgement is lost. How do you inject that deterministically rather than hoping the timing works out?

What a strong answer covers:

- A pgx `QueryTracer` fires on the `commit` statement before the bytes hit the wire; that hook installs the toxic, so it is causally ordered ahead of the commit, not racing it.
- The toxic is scoped to the downstream direction only (server to client), so the COMMIT flows upstream to Postgres unaffected and durably commits, while the reply is dropped.
- A downstream `reset_peer` can only fire on the first downstream byte, which is Postgres's reply, so by construction the commit already landed before anything is cut.
- The test proves the commit landed (it asserts one persisted transaction) and that a retry is idempotent, distinguishing it from the rollback case.

18.3.4 11. Your chaos test asserts the idempotency-key write and the transaction commit are atomic. Why does that specific property matter, and what regression would the test catch?

What a strong answer covers:

- If the key write and the transaction commit were in separate database transactions, a crash between them would either double-post (key not recorded, retry re-posts) or wedge the key (recorded but no transaction).
- The ambiguous-commit test is the guard: it only passes if both rows commit together or neither does.
- This is the invariant that makes idempotency actually safe under failure, not just under happy-path retries.
- Under no injected failure the property would pass trivially, which is why the earlier weak version of this test (a plain replay) was rejected in review.

18.4 Load testing

18.4.1 12. What is the single most common way to accidentally fake a ledger load test, and how does your k6 script avoid it?

What a strong answer covers:

- Reusing one idempotency key: after the first request, every other is a cheap replay lookup that never touches the write path, so you measure the wrong code.
- The script generates a unique key per request (from VU and iteration counters plus randomness, no remote dependency), with a deliberate small slice of duplicates to exercise replay honestly.
- The accounts are created for real in k6 `setup()` and their returned UUIDs are used, so transactions hit real rows, not fabricated ids that would fail validation.
- Postings are balanced two-leg transactions in a single currency, so nothing is rejected for the wrong reason.

18.4.2 13. You could hit your p99 latency target easily by turning off `synchronous_commit`. Why is that off the table?

What a strong answer covers:

- That setting lets a commit return before it is durably on disk, so a power cut could lose a committed payment.
- On a ledger that is not a performance knob, it is trading the durability guarantee for a benchmark number, which is a lie about what the system does.
- The load number is reported as a machine-dependent local measurement with durability intact, not a hard SLO gamed by relaxing safety.
- The honest number (500 RPS, p99 well under target, durability on) is worth more than a faster dishonest one.

18.4.3 14. Why run both a fan-out and a hot-account load scenario instead of one?

What a strong answer covers:

- Fan-out spreads payments across many accounts and measures raw throughput with minimal lock contention.
- Hot-account slams a single account, where everyone contends on the same row lock and tail latency blows up, which is where real systems fail.
- The fast, uncontended shape is not the same system as the contended one, so testing only the easy shape hides the failure mode that matters.
- The service already has a hot-account path from earlier weeks, so the load test exercises it on purpose.

18.5 CI and trade-offs

18.5.1 15. Your CI runs the whole test suite, including chaos and integration tests that need Docker, in one job rather than splitting unit from integration. Why, and what did you give up?

What a strong answer covers:

- The tests are Docker-gated by a skip, not separated by build tags, so integration and chaos tests run when Docker is present (the GitHub runner has it) and skip cleanly when it is not.
- Splitting unit from integration would mean either running the suite twice or inventing build tags the codebase does not use, so one honest `test` job is the better trade.
- Coverage is measured with Docker present, so the number reflects the real, integration-exercised code, not the low no-Docker figure.
- `cmd/` is compiled and smoke-run as a build check but kept out of the coverage profile, because wiring with no unit tests would drag the gate down for no signal.
- The cost is a slower single job and container contention risk, mitigated by capping test parallelism and waiting on the Postgres readiness log rather than a port.

18.5.2 16. Your coverage badge is self-hosted rather than using Codecov or a gist plus personal access token. Walk me through the design and why you avoided the alternatives.

What a strong answer covers:

- The badge job computes the same statement-weighted, generated-excluded number the gate uses and writes a shields.io endpoint JSON to a dedicated `badges` branch using the built-in `GITHUB_TOKEN`.
- No standing personal access token with gist scope sits on a public repo, and no external SaaS dependency or token is added, matching the project's self-hosted ethos.
- The badge must apply the same exclusion filter as the gate, or it would report a different (lower) number than the one enforced; a badge that disagrees with the gate is a defect.
- Codecov gives nicer PR annotations but adds an external account and token for a portfolio repo, which is not a trade worth making here.

SPECIAL EDITION

CHAPTER 19

I Audited My Own Payment Ledger Like It Held Real Money. The Front Door Was Wide Open.

There is a moment in every project where you stop admiring what you built and start trying to break it. I hit that moment with go-ledger last week. The tests were green, the invariant held under 10,000 concurrent transactions, the load test sustained 500 requests a second. I felt good.

So I did the thing that ruins a good mood on purpose. I sat down and reviewed my own code as if I were a hostile senior engineer being paid to find the way in, and as if real money, not demo data, flowed through it. Not "do the tests pass." A different question: **if someone could reach this API, what could they do?**

The answer took the good mood with it. This is a special edition, a bonus post between the numbered weeks, about what that audit found and how I fixed it.

19.1 The good news first

The accounting core was genuinely solid, and I want to say that plainly because it is the part that took months to get right. Money is stored as integer minor units, never floating point, so no rounding ghost ever creeps in. The double-entry invariant (every transaction's postings sum to zero) is enforced in two independent places: the Go domain and a Postgres constraint trigger, so even a direct `psql` write cannot create money from nothing. Balances are summed from immutable, append-only postings, never a mutable number someone could overwrite. Every write is atomic under `SERIALIZABLE`. Idempotency keys share the exact transaction that posts the money, so a retry cannot double-post.

None of that was the problem. The problem was everything in front of it.

19.2 The bad news: the front door had no lock

Here is the single line that summed up the whole risk. Every request in the service acted as one tenant, read from a config value:

```
deps.Transactions.Post(ctx, deps.DefaultTenant, txn, idem)
```

There was no authentication. None. Anyone who could send an HTTP request to `/v1/transactions` could move money between accounts, and anyone who could reach `/v1/accounts/{id}/balance` could read anyone's balance. The tenant was not derived from who you were, because there was no "who you were." It was a constant.

I had shipped a bank vault with a flawless internal mechanism and left the front door propped open with a rock. And I had done it on purpose, sort of: the comments even said "until an auth layer lands." That is a fine trade-off for a demo. It is a catastrophe the instant the words "real money" enter the sentence.

The audit found five things, ranked the way I would rank them if this were a real pentest report:

```
   #   severity    finding
   ------------------------------------------------------------------
   1   Critical    no authentication or authorization at all
   2   High        idempotency was opt-in, so a retry could double-post
   3   High        unbounded postings array and no request body limit
   4   Medium      no rate limiting, incomplete server timeouts
   5   Medium      audit log was guarded but not tamper-evident
   ------------------------------------------------------------------
```

I'll walk through the fixes, but two of them are worth slowing down for: the first, because it had a constraint that made it interesting, and the last, because fixing it broke something else.

19.3 Fix 1: a lock that does not lock out the demo

The obvious fix for "no auth" is "require auth." Add API keys, check them, done. But go-ledger has a public try-it console at go.sohag.pro/console that anyone can use with no login, and a live demo that resets every four hours. If I required authentication everywhere, I would break the one thing that makes

the project fun to show people.

The tension is real: authentication has to be mandatory for the thing to be safe, and optional for the demo to work. The trick is to notice those are not actually in conflict. I made authentication uniform (every `/v1` request needs a bearer API key, the tenant is derived only from that key, and a key for tenant A physically cannot touch tenant B's data), and then I gave the demo a real key that happens to be public:

```
  request ---> [ auth middleware ] ---> handler
                        |
                        | reads "Authorization: Bearer <key>"
                        | looks up the SHA-256 hash in api_keys
                        | puts the key's tenant in the request context
                        v
                tenant comes ONLY from the key, never from the request
                    body
```

The demo key is scoped to the demo tenant, rate-limited harder than a normal key, shipped right there in the console's HTML, and wiped-adjacent (the four-hour reset clears the tenant's data but never the keys, so the demo keeps working). Exposing it is safe on purpose, because it can only ever touch a tenant that gets erased every four hours. One code path, no special cases, and the demo survives.

The keys themselves are stored as SHA-256 hashes, never plaintext. A leaked database dump leaks no usable keys. And because the tenant is derived only from the key, the authorization question ("can this caller touch this account?") collapses into a question the database already answers for free: is the account in the key's tenant? The foreign keys from week three enforce that. I did not have to write a single access-control rule.

19.4 Fix 5: the tamper-evident audit log, and the trap it set

The audit log already rejected updates and deletes with a database trigger, so it was append-only. But append-only is not the same as tamper-evident. A privileged database user could still, in principle, rewrite a row's contents in place if they could get past the trigger. I wanted the log to be able to *prove* it had not been altered.

The standard tool for that is a hash chain, the same idea underneath a blockchain minus all the noise. Each audit row stores a hash of its own contents plus the hash of the previous row. Change any row and its hash no longer matches; change it and try to fix the hash, and now the *next* row's stored "previous hash" is wrong. The tampering cannot hide.

```
row 1            row 2            row 3
+----------+     +----------+     +----------+
| data     |     | data     |     | data     |
| prev: 00 |     | prev: h1 | <-- | prev: h2 |
| hash: h1 | --> | hash: h2 |     | hash: h3 |
+----------+     +----------+     +----------+

tamper with row 2's data -> recomputed hash != h2 -> chain breaks,
and row 3's "prev: h2" no longer points at anything valid.
```

I built it per tenant (so the four-hour demo wipe restarts one tenant's chain from scratch without touching anyone else's), added a `GET /v1/audit/verify` endpoint that walks the chain and reports the first break, and wrote tests that raw-edit a row and confirm the verifier catches it. Building it surfaced two genuinely nasty determinism bugs, the kind a hash chain is uniquely allergic to: Postgres stores timestamps at microsecond precision, so a nanosecond timestamp hashed on the way in did not match what came back out, and the `jsonb` type silently reformats your JSON, so the bytes you hashed were not the bytes you stored. Both would have made the chain report "tampered" on data nobody touched. Fixed by truncating the timestamp and storing raw `json`.

And then the load test went red.

19.5 The trap: correctness that quietly costs throughput

Here is the thing about a hash chain. To append row N, you must first read row N minus one, because you need its hash. That read-then-write, done inside a `SERIALIZABLE` transaction, means two transactions posting to the same tenant at the same time both read the same "latest" row and both try to append. Postgres notices the conflict and aborts one of them. Under 500 requests a second to a single tenant, that is not an occasional retry. It is a stampede. The retry budget exhausted and the ledger started returning `503`s.

I had traded a security win for a throughput regression, and the load test

caught it honestly. The audit chain, by its nature, serializes writes within a tenant. The question was how to make that serialization cheap instead of catastrophic.

My first instinct was a database advisory lock. It made things worse, and understanding why is the interesting part. Under `SERIALIZABLE`, a transaction's view of the database is frozen at its first query. If a second transaction takes a lock and waits, its frozen view was already taken before the first one committed, so when it finally runs it still reads a stale chain tail and still aborts. The lock delayed the collision without preventing it.

The fix that worked moves the serialization out of the database entirely and into the application. Since the service runs as a single instance, I put a tiny in-memory lock per tenant, held only while a post runs:

```
   posts to tenant A:  A1 --wait--> A2 --wait--> A3   (one at a time)
   posts to tenant B:  B1 --wait--> B2               (its own queue)

   A and B run fully in parallel. Only same-tenant posts queue,
   and they queue on a Go mutex, not on a held database connection.
```

The distinction that matters: transactions waiting for their turn block on an in-process lock, not on a checked-out database connection, so a busy tenant can never starve everyone else of connections. Different tenants run in parallel. Same-tenant posts serialize, which is exactly what a per-tenant hash chain requires anyway. And the `SERIALIZABLE` retry loop stays underneath as the correctness backstop, in case this ever runs as more than one instance.

Then I made the load test honest about all this by spreading its 500 requests a second across eight tenants, which is how a real ledger's traffic actually looks. The result: 38,417 requests, zero failures, 500 requests a second sustained at a 99th-percentile latency under 5 milliseconds, and not a single audit-chain conflict. The security feature and the throughput target both hold, at the same time, with everything switched on.

19.6 What the other three fixes were

The middle three are less dramatic but no less necessary. Idempotency went from optional to mandatory: a `POST /v1/transactions` with no `Idempotency-Key` header is now a `400`, because "retry safely" should be the default on an endpoint that moves money, not an opt-in a client can forget. The postings

array got a maximum and the request body got a size cap, so one request can no longer become a multi-million-row transaction that eats the server's memory. And every API key now gets its own rate limit, with the demo key throttled tighter than the rest.

19.7 The lesson I keep relearning

The gap between "my tests pass" and "this is safe for real money" is not a gap in the hard, interesting core. I had spent months making the accounting bulletproof. The gap was in the boring perimeter: the lock on the door, the cap on the input, the limit on the rate. The unglamorous stuff that does not show up in a demo and absolutely shows up in an incident.

If you are building something that moves money, or moves anything that matters, do the hostile read of your own code. Assume the attacker can reach every endpoint, because they can. Ask what they could do, not what your tests check. The list you get back is uncomfortable, and it is the most useful list you will write all quarter.

Every fix here is recorded in `ADR-012`, and the whole ledger, front door now firmly locked, is at `github.com/sohag-pro/go-ledger`.

The numbered series continues next week with Docker and real infrastructure. This one was a detour worth taking.

CHAPTER 20

Special Edition Q&A: Hardening a Payment Ledger for Real Money

Interview-style questions for the security-hardening work recorded in ADR-012: API-key authentication (tenant derived from the key), mandatory idempotency, input bounding, per-key rate limiting, and a per-tenant tamper-evident audit hash chain, plus the concurrency regression the audit chain caused and the in-process per-tenant mutex that fixed it. Every question is answerable from this repo's code and ADR-012.

20.1 The audit mindset

20.1.1 1. Your test suite was green and your load test hit 500 RPS. What made you audit the service anyway, and what did the audit assume that your tests did not?

What a strong answer covers:

- "Tests pass" answers "does it do what I designed"; a security audit answers "what can someone do who I did not design for."
- The audit assumed real money and a hostile caller who can reach every endpoint, not a cooperative client.
- The accounting core (integer money, double-entry enforced twice, derived balances, atomic posting, idempotency sharing the transaction) held up; the gap was the service perimeter.
- The five findings were all perimeter concerns (auth, idempotency-as-default, input bounds, rate limiting, tamper evidence), none in the core math.

20.2 Authentication and tenant isolation

20.2.1 2. Before the fix, how did the service decide which tenant a request acted as, and why is that a Critical finding under a "real money" premise?

What a strong answer covers:

- The tenant was a single config value (`DEFAULT_TENANT_ID`), passed to every handler; there was no authentication at all.
- Anyone who could reach `/v1/transactions` could move money; anyone reaching `/v1/accounts/{id}/balance` could read any balance.
- It was a deliberate demo-stage trade-off ("until an auth layer lands"), fine for a demo, catastrophic for real money.
- Severity is about impact under the threat model, not about whether it was intentional.

20.2.2 3. You chose API keys in a hashed database table over JWTs and over keys-in-config. Defend that.

What a strong answer covers:

- API keys fit a ledger with a handful of tenants: no token issuance or signing-key rotation story to build, lower ops than JWT.
- A database table (over config) makes keys revocable and provisionable without a redeploy, and supports per-key rate limits.
- Only the SHA-256 hash is stored, so a database dump leaks no usable key; the plaintext is shown once and never recoverable.
- A short in-memory cache keeps the per-request lookup off the database, at the cost of revocation lag bounded by the cache TTL.

20.2.3 4. Once auth landed, how much authorization logic did you have to write, and why so little?

What a strong answer covers:

- Almost none: the tenant is derived only from the key, and the composite foreign keys already make cross-tenant access impossible at the database.
- "Can this caller touch this account?" reduces to "is this account in the key's tenant?", which the schema enforces for free.

- Per-account ACLs were deliberately rejected as redundant with the tenant foreign keys.
- The isolation is verified against real Postgres (a key for tenant A cannot see tenant B's account), not just asserted.

20.2.4 5. The tenant comes only from the key. Why is it important that no request field can set or override it, and where is that enforced?

What a strong answer covers:

- If a request body or header could name a tenant, a caller could impersonate another tenant; the whole boundary would collapse.
- The middleware puts the resolved tenant in the request context and handlers read it from there; no handler reads a tenant from the request.
- gRPC uses the same resolver through an interceptor, so the boundary is uniform across both transports.
- Only `/v1` and non-health gRPC require a key; health, docs, static, and the console page stay open by design.

20.3 The demo constraint

20.3.1 6. Mandatory auth would break the public console. How did you keep the demo working without a second, unauthenticated code path?

What a strong answer covers:

- Authentication is uniform; the demo is reached with a real but public, low-privilege demo key scoped to the demo tenant.
- The key is rate-limited tighter than normal and the demo tenant is wiped every four hours, so exposing it in the console HTML is safe on purpose.
- The demo key survives the wipe because the seeder resets tenant data tables only, never `api_keys`.
- One code path, no special-casing an "open" tenant (which would be a permanent open write surface).

20.4 Idempotency and input

20.4.1 7. Idempotency was already implemented. Why change it from optional to mandatory, and why not just auto-generate a key when the client omits one?

What a strong answer covers:

- On a money-moving POST, safe retry should be the default, not something a client can forget and then double-post on a dropped response.
- A missing `Idempotency-Key` is now a 400 with a clear message.
- Auto-deriving a key from the request fingerprint was rejected: it would silently collapse two genuinely separate but identical payments into one, turning a real second payment into a false replay.
- Making the client name the key keeps "same request retried" distinct from "two payments that look alike."

20.4.2 8. What concrete resource-exhaustion vectors did the input hardening close?

What a strong answer covers:

- The postings array had a minimum but no maximum, so one request could become an enormous multi-row transaction; it now has a cap.
- There was no request body-size limit; an oversized body could exhaust memory before validation; it now returns 413.
- The server had only `ReadHeaderTimeout`; it now sets read, write, and idle timeouts so a slow client cannot hold a connection open.
- Per-key rate limiting bounds how fast any single key can hit the service.

20.5 The tamper-evident audit chain (and its cost)

20.5.1 9. The audit log already rejected updates and deletes. What does a hash chain add that an immutability trigger does not?

What a strong answer covers:

- The trigger prevents casual mutation; the chain detects a privileged rewrite that bypasses the trigger.
- Each row hashes its own content plus the previous row's hash, so any change breaks that row's hash and the next row's back-pointer.
- A `GET /v1/audit/verify` endpoint walks the chain and reports the first break, so tamper evidence is checkable, not just stored.
- It is defense in depth: prevention plus detection, not one instead of the other.

20.5.2 10. A hash chain is uniquely sensitive to any difference between what you hash and what you store. What two determinism bugs did building it surface, and how were they fixed?

What a strong answer covers:

- Postgres `timestamptz` stores microsecond precision, so a nanosecond `created_at` hashed on the way in did not match the value read back; fixed by truncating to microseconds before both hashing and storing.
- The `jsonb` type reformats JSON on read, so the stored bytes differed from the hashed bytes; fixed by storing raw `json`.
- The fix principle: apply any normalization before both the hash and the store, using the identical value for each.
- The chain also hashes the tenant id first, so a row moved to another tenant's chain is detectable.

20.5.3 11. Fixing the audit chain caused a throughput regression. Explain the mechanism precisely.

What a strong answer covers:

- To append a row the write must first read the previous row's hash; that read-then-write inside a `SERIALIZABLE` transaction conflicts when two posts hit the same tenant at once.
- Both transactions read the same chain tail and try to append; Postgres aborts one with a serialization failure (40001).
- Under 500 RPS to a single tenant the retry budget exhausts and the service returns 503, not an occasional retry but a stampede.
- The chain, by nature, serializes writes within a tenant; the problem was making that serialization cheap instead of an abort storm.

20.5.4 12. You tried a database advisory lock first and it made things worse. Why, and what actually fixed it?

What a strong answer covers:

- Under `SERIALIZABLE`, a transaction's snapshot is fixed at its first query; a waiter that blocks on a lock already took its snapshot before the holder committed, so it still reads a stale tail and still aborts.
- The advisory lock delayed the collision without preventing it, and it also risked holding a database connection during the wait.
- The fix moves serialization into the application: a per-tenant in-process mutex, held only while a post runs, on a single-instance deployment.
- Same-tenant posts queue on a Go mutex (not a held connection, so no pool starvation), different tenants run in parallel, and the SERIALIZABLE retry loop remains the multi-instance correctness backstop.

20.5.5 13. After the in-process mutex, single-tenant posts serialize. How did you still demonstrate 500 RPS, and why is that framing honest rather than a dodge?

What a strong answer covers:

- The load test spreads 500 RPS across eight tenants, so each tenant's serialized chain runs well under its sequential ceiling.
- That reflects how real ledger traffic actually looks (many tenants), rather than an artificial all-load-on-one-chain worst case.
- The result: 38,417 requests, zero failures, 500 RPS sustained, p99 under 5 ms, and zero audit-chain serialization errors, with auth and the chain fully

on.
- Honest because the single-tenant serialization is documented as an inherent property of a per-tenant hash chain, not hidden.

20.6 Trade-offs and consequences

20.6.1 14. What are the standing negative consequences of this hardening that a reviewer should know about?

What a strong answer covers:

- A per-request key lookup, kept off the database by a cache whose TTL bounds revocation lag.
- Same-tenant posts serialize in-process; a single very high-throughput tenant is bounded, and the mechanism assumes a single instance (multi-instance falls back to the retry loop).
- The demo key is public by design and only safe because it is tenant-scoped, rate-limited, and wiped; those three properties must stay true.
- Key management for real tenants is still manual (a row insert or a small CLI); there is no self-service issuance yet.

20.6.2 15. This work sits outside the numbered 12-week plan. Why do it as a special edition rather than fold it into a week, and what does that say about how you scope security work?

What a strong answer covers:

- The five findings came from a deliberate hostile audit, not the week's planned scope, so they earned their own record (ADR-012) and their own writeup.
- Security hardening is cross-cutting and event-driven (you do it when an audit finds something), not a tidy weekly feature.
- Doing it as a focused pass, with an ADR written before the code, keeps the decisions and their trade-offs on the record for the book.
- It models treating "make it safe for real money" as a distinct, first-class piece of work rather than a checkbox inside a feature week.

go-ledger

ARCHITECTURE DECISIONS

CHAPTER 21

ADR-001: Why Double-Entry Accounting

21.1 Status

Accepted: 2026-06-11

21.2 Context

go-ledger is a payment ledger service: it must record money movements between accounts and answer "what is the balance of account X?" correctly, every time, under concurrent writes.

The naive approach is a single `balances` table with one row per account that gets incremented and decremented in place. This is how many systems start, and it fails in predictable ways:

- **No audit trail.** A mutable balance tells you *what* the balance is, not *how* it got there. Reconstructing history requires separate logging that inevitably drifts from the source of truth.
- **Money appears and disappears.** A single-sided update ("add $10 to A") has no structural guarantee that the money came from anywhere. Bugs create or destroy money silently.
- **Race conditions corrupt state.** Read-modify-write on a balance row under concurrency loses updates unless every code path remembers to lock correctly. The schema itself offers no protection.

Double-entry bookkeeping, in use since the 15th century, solves all three structurally:

- Every transaction consists of two or more **postings** (ledger entries).
- Each posting debits one account and credits another.
- The sum of all postings in a transaction must equal zero.

21.3 Decision

go-ledger models all money movement as double-entry transactions:

- `Transaction`: an atomic, immutable unit of money movement.
- `Posting`: a single signed entry against one account; a transaction has two or more postings.
- **Invariant:** $\Sigma(\text{postings}) = 0$ for every transaction. Enforced in the domain types (`Transaction.Validate()`), and later at the database level with a CHECK constraint.
- Postings are append-only. Balances are *derived* (sum of postings), never stored as mutable primary state.

21.4 Consequences

21.4.1 Positive

- Money can never be created or destroyed by a write path bug; an unbalanced transaction is rejected before it persists.
- The ledger is its own audit trail: every balance is explainable as a sum of immutable postings.
- Append-only postings sidestep update races entirely; concurrency control reduces to serializing transaction inserts (Week 4).
- The model matches what accountants, auditors, and payment partners expect.

21.4.2 Negative

- Balance reads are aggregations, not single-row lookups. Acceptable for v1; if it becomes hot, a derived (and rebuildable) balance cache can be added; it stays a cache, never the source of truth.
- Slightly more conceptual overhead for API consumers: posting a payment means submitting balanced postings, not “set balance”.

21.5 Alternatives considered

- **Mutable balance table**: rejected. No audit trail, no structural integrity guarantee (see Context).
- **Event sourcing with a full event store**: rejected for v1. Double-entry postings already give us the append-only history we need without the operational complexity of projections and replay. Revisit only if product needs demand it (see scope discipline in the build plan).

CHAPTER 22

ADR-002: Money Representation

22.1 Status

Accepted: 2026-06-15

22.2 Context

The ledger moves money between accounts and must answer balance queries exactly. Floating point is unfit for money: 0.1 + 0.2 != 0.3 in IEEE-754, and rounding error accumulates across millions of postings. We need an exact, fast representation that maps cleanly to Postgres in Week 3.

Three candidates:

- **int64 minor units**: store the amount as a signed integer count of the currency's smallest unit (cents for USD). Exact, no allocation, maps to BIGINT. Must guard against overflow on arithmetic.
- **Arbitrary-precision decimal** (for example shopspring/decimal): exact and overflow-free, but every operation allocates a big.Int, and it maps to NUMERIC. More machinery than a single-currency v1 needs.
- **Custom decimal struct** (int64 coefficient + scale): full control, but we own the arithmetic, rounding, and a pile of tests for no v1 benefit.

22.3 Decision

go-ledger represents money as `int64` minor units in a `Money` value type that also carries its `Currency`. v1 is single-currency (USD), per the build plan's scope discipline, so there is no per-currency scaling to track yet.

- `Money` is an immutable value type. Operations return new `Money` values.
- Arithmetic (`Add`, `Sub`, `Neg`) is overflow-guarded and returns an error on overflow rather than panicking or wrapping.

- Cross-currency arithmetic returns `ErrCurrencyMismatch`. The check exists now so multi-currency (a future, out-of-scope item) cannot silently corrupt sums.
- Maps to `BIGINT` in Postgres (Week 3) with no conversion.

22.4 Consequences

22.4.1 Positive

- Exact arithmetic with zero allocation on the hot path.
- Clean, lossless mapping to BIGINT; no NUMERIC parsing.
- Overflow is a typed, testable error, not undefined behavior.

22.4.2 Negative

- int64 caps the representable amount near 9.2e16 minor units. Far beyond any realistic single-account balance, but arithmetic must still check, hence the overflow guard.
- Adding a second currency later means introducing per-currency exponents. Acceptable: multi-currency is explicitly out of scope for v1.

22.5 Alternatives considered

- **float64**: rejected. Not exact; unacceptable for money.
- **shopspring/decimal**: rejected for v1. Allocation and NUMERIC overhead with no benefit while we are single-currency. Revisit if multi-currency lands.
- **Custom decimal struct**: rejected. Owning rounding and arithmetic is cost without v1 value.

CHAPTER 23

ADR-003: Postgres Schema and Repository Layer

23.1 Status

Accepted: 2026-06-29

23.2 Context

Week 3 puts the domain model (ADR-001, ADR-002) onto disk. We need a schema for `accounts`, `transactions`, and `postings`, a repository layer over it, and an integration test that runs the full happy path against a real Postgres. Several choices had no obvious default and would be costly to reverse once data exists, so each is recorded here.

The hardest question is how balances are stored. Two shapes:

- **Mutable balance column**: each account row carries a `balance` updated in place on every posting. Reads are a single column lookup.
- **Append-only postings, derived balances**: postings are immutable rows; balance is `SUM(amount)` over an account. Nothing is updated in place.

A mutable balance is faster to read but reintroduces exactly the bug double-entry exists to prevent: the stored number and the sum of postings can drift, and once they drift there is no way to tell which is right. It also turns every posting into a read-modify-write that must be serialized.

23.3 Decision

Postings are append-only and balances are derived. This carries the core invariant from ADR-001 into the database. The schema, with the reasoning for each non-obvious choice:

- **Three tables now, not five.** The build plan lists five tables, but `idempotency_keys` (Week 6) and `audit_log` (Week 6) are designed and built in the weeks that use them, when their columns are actually known. `0001_initial` creates only `accounts`, `transactions`, and `postings`. Migrations are cheap; guessing a schema before its code exists is not.
- **Multi-tenant from day one.** Every table carries a `tenant_id`. There is no auth or tenant resolution yet (that comes later); the column and the foreign key discipline exist now because retrofitting tenancy onto a populated ledger is a migration no one wants to write. Repository methods are all scoped by tenant.
- **UUIDv7 primary keys, generated in the application.** Keys are `uuid` columns; the app generates a UUIDv7 before insert. v7 is time-ordered, so it keeps index locality close to a serial key while staying globally unique and decoupled from the database. Generating in Go (not `gen_random_uuid()`) means the caller knows the id without a round-trip and tests are deterministic.
- **Composite foreign keys enforce tenant integrity.** `accounts` and `transactions` carry a `UNIQUE (tenant_id, id)`. `postings` references both via `(tenant_id, account_id)` and `(tenant_id, transaction_id)`. A posting cannot point at an account or transaction in a different tenant: the database rejects it, not just application code.
- **Currency lives on the transaction, not the posting.** The domain already enforces that all postings of a transaction share one currency (`Transaction.Validate`). Storing it once on the transaction keeps a posting to a single signed `bigint amount`, and each posting's `Money` is reconstructed from the transaction's currency on read. Accounts also carry their own currency.
- **Account type is `text` with a CHECK, not a Postgres enum.** The five types are constrained by `CHECK (type IN (...))`. Readable in plain SQL, and adding or changing a type is an ordinary migration rather than the awkward `ALTER TYPE` dance an enum forces.
- **No balance CHECK constraint yet.** The sum-to-zero invariant is enforced in the domain (`Transaction.Validate`) for Week 3. The database-level CHECK lands in Week 4 alongside the transaction-posting service and its concurrency handling, where it belongs.

The repository follows a port-and-adapter split: `domain.Repository` is the port (the domain owns the contract); `internal/postgres` is the adapter, built on pgx and sqlc-generated queries. `CreateTransaction` validates the double-entry invariant, then writes the transaction and all its postings in one database transaction, so a half-written transaction can never exist. Tooling is goose for migrations, sqlc for type-safe queries, and testcontainers-go for the integration

test (all locked in the build plan).

23.4 Consequences

23.4.1 Positive

- The stored ledger cannot disagree with itself: there is no second copy of the balance to drift from the postings.
- Tenant isolation is enforced by the database, not just by careful queries.
- The domain stays free of storage and uuid types; the port is a plain Go interface, so the service layer (Week 4) can be tested against a fake.
- UUIDv7 keeps write-side index locality without coupling ids to the database.

23.4.2 Negative

- Balance reads are an aggregate (`SUM`) rather than a column read. Fine at v1 scale; if it ever bites, the answer is a materialized rollup that is rebuilt from postings, never a mutable primary balance.
- The integration test needs a Docker daemon. It skips (does not fail) when none is reachable, so local `make test` stays green; CI runs Docker and exercises the real path.

23.5 Alternatives considered

- **Mutable balance column**: rejected. Reintroduces drift between stored balance and posting history, the exact failure double-entry prevents, and serializes every posting behind a row update.
- **All five tables in `0001`**: rejected. Designs `idempotency_keys` and `audit_log` before the code that uses them; they get their own migrations in Week 6.
- **Single-tenant schema, add tenancy later**: rejected. Adding `tenant_id` and backfilling foreign keys on a live ledger is far more painful than carrying the column from the start.
- **DB-generated UUIDv4 (`gen_random_uuid()`)**: rejected. Random keys hurt index locality and force a read-back to learn the generated id.

- **Postgres enum for account type**: rejected. `ALTER TYPE` is more friction than a `text` column with a CHECK, for no real gain.

CHAPTER 24

ADR-004: Concurrency Control for Transaction Posting

24.1 Status

Accepted: 2026-06-29

24.2 Context

Posting a transaction writes a transaction row and two or more postings that must sum to zero, and it must do so correctly when many requests post to the same accounts at once. Two requests racing on the same account cannot be allowed to produce a ledger that does not balance, lose a posting, or read a stale balance.

The schema (ADR-003) already makes postings append-only and balances derived, so there is no mutable balance row to corrupt. What remains is making each posting atomic and keeping the double-entry invariant intact under concurrency.

There are two classic strategies:

- **Pessimistic: lock the rows.** Run at READ COMMITTED and `SELECT ... FOR UPDATE` the involved account rows, in a fixed order to avoid deadlock, before writing. Conflicting transactions block rather than fail, so there is nothing to retry. Correctness rides on remembering to lock the right rows, in the right order, every time.
- **Optimistic: let the database detect conflicts.** Run at SERIALIZABLE. Postgres's Serializable Snapshot Isolation (SSI) tracks read/write dependencies and aborts a transaction with SQLSTATE 40001 if committing it would break serial-equivalent ordering. No explicit locks; conflicts surface as errors the application retries.

24.3 Decision

go-ledger uses **SERIALIZABLE with automatic retry**. Correctness comes from the database guaranteeing serial-equivalent execution, not from the application remembering to take the right locks. A buggy or future code path cannot silently skip a lock it never knew to take.

The pieces:

- **Service owns the unit of work.** `ledger.TransactionService.Post` validates the transaction, then runs the write through the repository's `RunInTx`, which begins a SERIALIZABLE transaction, runs the work, and commits.
- **The adapter owns isolation and retry.** `RunInTx` (in `internal/postgres`) begins each attempt at `pgx.Serializable`. A serialization failure (40001) or deadlock (40P01) can surface either inside the work or, commonly, only at COMMIT, so both are watched and the whole unit of work is replayed. Retries are bounded (25 attempts) with exponential backoff and full jitter; the jitter is what stops a crowd of conflicting transactions from retrying in lockstep and colliding again.
- **The invariant is also enforced in the database.** A `DEFERRABLE INITIALLY DEFERRED` constraint trigger (`assert_txn_balanced`, migration 0002) runs at COMMIT and rejects any transaction whose postings do not sum to zero. This is defense in depth: the domain checks the invariant first (`Transaction.Validate`) for a fast, clean error, but the trigger guarantees it holds even against a bad migration, a direct psql write, or a second service.
- **The trigger's read is indexed.** The trigger reads postings by `transaction_id`. Without an index that read is a scan, and under SSI a scan takes broad predicate locks that turn into a storm of false-positive serialization conflicts. Adding `postings_transaction_idx` was the single change that took the stress test from roughly 1 in 6 posts failing to zero. This is the non-obvious lesson of the week: under SERIALIZABLE, an unindexed read is not just slow, it manufactures conflicts.

`SELECT FOR UPDATE` is not used. The two approaches were not combined: belt and suspenders here would mean taking explicit locks and running SERIALIZABLE and retrying, which is more moving parts and redundant guarantees for no gain.

24.4 Observability

Posting is instrumented with two Prometheus collectors: a histogram `transaction_post_duration_seconds`, labeled by outcome, for latency, and a counter `transaction_post_serialization_retries_total` so a climbing retry rate makes write contention visible. These sit alongside the standard Go runtime and process collectors.

The scrape endpoint is served on a separate server bound to loopback (`127.0.0.1:9090` by default), not on the public API port. A metrics endpoint leaks transaction volumes, latencies, and retry rates, so it is reached over the host's private network or an SSH tunnel, never the public interface. The public liveness check stays `GET /healthz`.

24.5 Measured baselines

Stress test: 100 goroutines posting 10,000 balanced transactions against a shared pool of 100 accounts. Result: all 10,000 commit, zero failures, and the sum of every account balance is exactly zero.

One honest caveat about the concurrency level: the real ceiling on how many posting transactions run at the database at once is the connection pool's `MaxConns`, not the goroutine count. The test sets `MaxConns` to 25 explicitly and logs it, so "100 goroutines" means up to 25 transactions truly concurrent at Postgres with the rest queued on pool acquisition. That is still meaningful contention on 100 shared accounts, and it is the number worth quoting rather than the goroutine count.

Latency on the local test machine (Postgres 16 in a colima VM): p50 about 35 ms, p99 about 68 ms. These are above the original p50 < 10 ms / p99 < 50 ms target. The dominant cost is WAL fsync on the VM's virtualized disk: every commit waits for a durable write, and a Lima VM's disk is slower than bare metal. The numbers are reported as measured rather than tuned (for example by disabling `synchronous_commit`), since durability is the point of a ledger. On bare-metal or a real server disk the same code should land much closer to the target. Latency is logged by the test, not asserted, so the suite does not flake on machine speed.

24.6 Consequences

24.6.1 Positive

- Correctness does not depend on the application taking the right locks; the database guarantees serial-equivalent execution.
- The double-entry invariant is enforced in two independent places: the domain and a deferred database trigger.
- Retry with jittered backoff absorbs contention transparently; under the test's sustained load every transaction eventually commits.
- `RunInTx` is a general unit-of-work seam, so Week 6 can write the audit-log entry inside the same transaction as the posting it records.

24.6.2 Negative

- Under contention some transactions do extra work (retries), and pathological contention could still exhaust the retry budget and surface an error to the caller. That is the optimistic trade: conflicts cost retries, not blocked connections.
- SSI is sensitive to read patterns. Indexes that look optional for correctness (like the one on `transaction_id`) are load-bearing for concurrency. Future reads inside a posting transaction must be considered for their SSI footprint.
- `fn` passed to `RunInTx` must be safe to run more than once, since it can be replayed.

24.7 Alternatives considered

- **READ COMMITTED + SELECT FOR UPDATE**: rejected. Pushes correctness onto the application to lock the right rows in the right order every time; a future code path that forgets reintroduces the race. Pessimistic locking also serializes access to hot accounts by blocking, rather than letting non-conflicting work proceed.
- **SERIALIZABLE + SELECT FOR UPDATE + retry**: rejected. Redundant guarantees and more moving parts than SERIALIZABLE alone needs.
- **Mutable balance with row locks**: rejected back in ADR-003; it reintroduces a second copy of the truth that can drift.

- **A non-deferred row CHECK**: impossible; the sum-to-zero invariant spans many posting rows and cannot be expressed in a single-row CHECK, hence the deferred constraint trigger.

CHAPTER 25

ADR-005: Per-Account Currency Integrity

25.1 Status

Accepted: 2026-06-29

25.2 Context

Currency lives in two places (ADR-003): every account has a currency, and every transaction has a currency (the single currency shared by all its postings). Postings themselves carry only a signed `amount`, not a currency, because the domain already guarantees all postings of a transaction share one currency.

That leaves a gap. Nothing tied a transaction's currency to the currency of each account it posts into. A USD transaction could post into an account whose currency is EUR, and the database would accept it. Today the ledger is single-currency (USD), so the gap is invisible, but it is exactly the kind of latent integrity hole that turns into a real one the moment multi-currency, a second writer, or a buggy import appears. A reviewer would not trust a ledger that lets a posting land in an account of a different currency.

The domain cannot close this on its own: `Transaction.Validate` sees the postings and their shared currency, but it does not know each referenced account's currency, which lives in the database.

25.3 Decision

Enforce it in the database. Migration 0003 adds an immediate `AFTER INSERT` trigger on `postings` (`assert_posting_currency`) that, for each inserted posting, looks up its account's currency and its transaction's currency and rejects the row if they differ.

- It is immediate, not deferred: the account and transaction both already

exist when a posting is inserted (the foreign keys require it), so there is nothing to wait for.
- The lookups are equality on primary keys, so under SERIALIZABLE they take only narrow predicate locks on rows that posting transactions read but never write. The stress test confirmed this adds no measurable serialization contention.

This is defense in depth, consistent with how the balance invariant is handled (ADR-004): the domain keeps postings in one currency, and the database guarantees that currency matches each account, even against a direct write.

25.4 Consequences

25.4.1 Positive

- A posting can never move money in a currency the account does not hold. The guarantee holds regardless of what client or code path wrote the row.
- The schema is ready for multi-currency: when it lands, this invariant is already enforced rather than retrofitted onto live data.

25.4.2 Negative

- A small per-posting read cost at insert time (two primary-key lookups). Negligible in practice and confirmed not to add serialization conflicts.
- The currency rule now lives in two places: the domain (postings share one currency) and the database (that currency matches the account). That redundancy is the point, but it is two places to keep in mind.

25.5 Alternatives considered

- **Denormalize currency onto postings and use a composite foreign key** `postings(account_id, currency) -> accounts(id, currency)`: rejected. It would put currency back on the posting row, undoing the ADR-003 decision to keep a posting to a single signed amount, for no extra safety beyond what the trigger gives.

- **Enforce only in application code** (check each account's currency in `CreateTransaction`): rejected. That is the very thing we want the database to guarantee; an application-only check is one forgotten code path away from a cross-currency posting.
- **Do nothing until multi-currency arrives**: rejected. Adding the constraint to a populated, multi-currency ledger later is far riskier than enforcing it now while the rule is trivially true.

CHAPTER 26

ADR-006: REST API Design

26.1 Status

Accepted: 2026-06-29

26.2 Context

Through Week 4 the ledger was exercised only by tests; the service had no network surface. Week 5 puts a REST API in front of it: create accounts, post transactions, and read accounts, balances, statements, and transactions. The router is chi (a locked decision), and the project already generates its OpenAPI spec from the handlers with huma, with a drift test and a self-hosted playground.

Several choices had no obvious default.

26.3 Decision

26.3.1 huma operations on chi

The endpoints are huma operations running on a chi router via the humachi adapter (swapped from humago). Each endpoint is a typed operation: huma derives request validation from struct tags, renders errors as RFC 7807 `application/problem+json`, and generates the OpenAPI spec, all from the same handler. chi owns routing and the middleware chain (request id, recovery, structured request logging).

This was chosen over plain chi handlers plus go-playground/validator plus hand-written problem+json (the build plan's literal wording). huma already solves validation, error formatting, and spec generation as one unit, and the existing drift test and playground depend on the spec being generated

from handlers. The plan predates the huma decision; huma's validation supersedes the separate validator cleanly. The cost is using huma's validation tags (`minLength`, `enum`, `pattern`) rather than go-playground's, which is minor.

26.3.2 Money as integer minor units

Amounts cross the wire as a signed integer count of minor units plus a currency code (`{"amount": 10000, "currency": "USD"}`), mapping 1:1 to the internal `int64`. No floating point and no decimal parsing at the boundary. This is what most payment APIs (for example Stripe) do, and it keeps the API as exact as the ledger.

26.3.3 Single default tenant for now

The schema is multi-tenant, but there is no auth yet. Rather than expose tenancy in URLs or trust a client header, every request acts as a single default tenant (`DEFAULT_TENANT_ID`, with a fixed fallback), injected by the API layer into each service call. When auth lands it will populate the same tenant value from a token instead, without changing any resource URL.

26.3.4 Account statement with running balance and keyset paging

Beyond the planned endpoints, `GET /v1/accounts/{id}/statement` lists the account's postings, newest first, each with the account's running balance as of that posting (a window `SUM` over the account's history), and pages with keyset (cursor) pagination on `(created_at, id)`. Keyset paging is stable under concurrent inserts in a way offset paging is not.

Because balances are derived, never stored (ADR-003), the running-balance window scans the account's full posting history per page (one index range on `(tenant_id, account_id)`). That is O(history) per page, fine at v1 scale; the future optimization, if it ever bites, is a materialized running balance rebuilt from postings, never a mutable primary balance.

26.3.5 Per-posting description

Each posting carries an optional free-text `description` (narration), bounded at 256 characters in both the domain and a database CHECK. It appears on every posting in requests and responses, including statement entries.

26.3.6 Server now owns its database, with startup migrations

`cmd/server` reads `DATABASE_URL` (required, fail-fast: a ledger API with no database must not serve), opens the tuned pool, and applies the embedded goose migrations on startup before serving. On a single instance this is the simplest correct option: the binary that needs a column also creates it, atomically with the deploy. If the project ever runs more than one instance, the same `goose.Up` moves into a dedicated CI step to avoid a migration race.

Production Postgres runs natively on the VPS (a dedicated database and user, listening on localhost, backed up with `pg_dump`), consistent with the native-binary deploy. This is provisioned out of band before the deploy; see the ops runbook.

26.3.7 Error mapping

Domain errors map to status codes in one helper: not-found to 404, duplicate id to 409, an exhausted serialization conflict to 503, validation and invariant failures to 422, and anything unrecognized to a 500 that does not leak internals.

26.4 Consequences

26.4.1 Positive

- One source of truth for handlers, validation, errors, and the spec; the published docs cannot drift from the running API.
- The API is as exact as the ledger (integer minor units), and statements give a real running balance with stable pagination.
- The transport stays thin: services hold the logic, handlers translate.

- Auth can be added later by changing only how the tenant is resolved.

26.4.2 Negative

- huma owns request decode/encode, so a very custom response shape means working within its model.
- Statement pages cost a scan of the account's posting history while balances stay derived. Accepted for v1; a rollup is the escape hatch.
- The server now hard-depends on Postgres, so the deploy requires the database to be provisioned and `DATABASE_URL` set first.

26.5 Alternatives considered

- **chi + go-playground/validator + hand-written problem+json**: rejected. Reintroduces three concerns huma already covers and risks spec drift.
- **Spec-first with oapi-codegen**: rejected. Drops the existing huma investment (playground, drift test, generator) and makes the spec a hand-maintained artifact, heavy for a solo build.
- **Decimal-string money in JSON**: rejected. Adds parsing and per-currency scale validation at the boundary for no gain over exact integer minor units.
- **Tenant in the URL or a trusted header now**: rejected. Bakes tenancy into the resource model or trusts an unauthenticated header; the default-tenant seam is cleaner to evolve into real auth.
- **Migrations as a separate step now**: deferred. Correct for multi-instance, but premature for a single VPS; startup migrations are simpler today.

CHAPTER 27

ADR-007: Idempotency Keys and the Immutable Audit Log

27.1 Status

Accepted: 2026-07-02

27.2 Context

Through Week 5 the ledger had a REST API but no protection against retries. A `POST /v1/transactions` that times out may have already committed: the request reached the server and posted, and only the response was lost. A correct client retries, and without protection that retry posts a second transaction, moving money twice. Week 6 adds an idempotency contract so a retry is safe, and an audit log so every posting leaves a tamper-evident record of what happened.

Both tables (`idempotency_keys`, `audit_log`) were designed in ADR-003 and left for this week; `RunInTx` was called out in ADR-004 as the seam the audit write would use. This ADR records the decisions those earlier ADRs deferred.

Several choices had no obvious default.

27.3 Decision

27.3.1 The idempotency key row is inserted inside the posting transaction

The client sends an optional `Idempotency-Key` header. When present, the key row is inserted into `idempotency_keys` inside the same SERIALIZABLE transaction (`RunInTx`) that inserts the transaction, its postings, and the audit row. The primary key `(tenant_id, idempotency_key)` is the exactly-once mutex: the

first request to commit the key wins, and any concurrent duplicate hits the unique violation, rolls back its whole transaction (so its optimistically written postings vanish too), and replays.

This was chosen over the obvious check-then-act approach (look the key up, and if absent, post). Check-then-act has a race: two concurrent copies of the same request both read "key absent," both post, and both create a transaction. The bug only appears under concurrency, which is exactly when retries happen. Making the check and the write one atomic step requires the database as the referee; the unique primary key is that referee, so no application lock or separate idempotency service is needed.

`InsertIdempotencyKey` maps the PK collision (constraint `idempotency_keys_pkey`) to an internal `ErrDuplicateIdempotencyKey`. `TransactionService.Post` catches it (only when a key was supplied), reads back the committed row, loads the original transaction, and returns it with `replayed = true`, which the API surfaces as the `Idempotent-Replayed` response header. Because Postgres blocks the second insert until the first transaction resolves, the loser's follow-up read always sees the committed winner; there is no visibility gap.

27.3.2 A request fingerprint distinguishes a retry from key reuse

The stored key row holds a fingerprint of the original request: `Transaction.Fingerprint()`, a SHA-256 over the currency and each posting's account, signed amount, and description, in order, with the transaction id excluded. On a repeat, a matching fingerprint is a genuine retry and gets the replay; a different fingerprint means the client reused a key for a different payment, and the service returns `ErrIdempotencyConflict`, mapped to HTTP 409.

The fingerprint length-prefixes each field rather than joining with a separator byte. A separator scheme is forgeable when a field can contain that byte: two different transactions could be crafted to collide, and a collision on a reused key would let a different payment through as a "replay." Fixed-width length prefixes make field boundaries impossible to fake regardless of content.

27.3.3 The idempotency key is optional, not required

An absent header posts normally, exactly as before Week 6. This keeps the change backward compatible (the console and seeder callers are unaffected) and matches common payment APIs where the key is recommended but not mandatory. A keyless post is still audited.

27.3.4 The audit log is written in the same transaction and is create-only

Each posted transaction writes one `audit_log` row inside the same `RunInTx` closure as the postings: `action = transaction.created`, `before` NULL, `after` a JSON snapshot of the transaction, `actor` the tenant id (until auth lands). Because the audit insert shares the transaction, a posting and its audit record commit together or roll back together; there is no separate "log it later" step that could fail on its own and leave a transaction nobody recorded. On the replay path the transaction rolled back and no new posting happened, so no audit row is written: the log records only real creations.

It is deliberately create-only. Postings are append-only, so there is no in-place update to diff; a generic before/after framework would be speculative (`before` stays NULL for now).

27.3.5 Append-only is enforced by a database trigger, with one gated exception

A trigger on `audit_log` rejects UPDATE and DELETE (`BEFORE UPDATE OR DELETE ... FOR EACH ROW`), so the log is immutable even against a direct `DELETE FROM audit_log` in psql, not just by application convention. This is the same defense-in-depth pattern as the balance and currency triggers (ADR-004, ADR-005).

The one sanctioned exception is the demo seeder, which resets the tenant every few hours and must clear the log. The trigger allows the mutation only when the transaction-local GUC `audit.allow_purge` is set to `on`; the seeder sets `SET LOCAL audit.allow_purge = 'on'` at the start of its reset transaction. `SET LOCAL` is transaction-scoped, so it cannot leak to other pooled connections, and the application's normal path never sets it, so production stays immutable.

27.3.6 Audit reads: by transaction unpaginated, by account keyset-paginated

`GET /v1/transactions/{id}/audit` returns a transaction's audit rows unpaginated, because a transaction has a fixed, tiny number of them. `GET /v1/accounts/{id}/audit` reuses the Week 5 keyset cursor on `(created_at, id)`, because an active account can accumulate a row for every transaction that ever touched it, and an unbounded query is a memory and latency risk on a real account. The by-account query joins `audit_log` to the account's postings and is tenant scoped on both the outer table and the subquery.

27.4 Consequences

27.4.1 Positive

- Retrying a payment is safe: 100 concurrent requests with the same key create exactly one transaction and one audit row, verified by an integration test that runs the real race against Postgres.
- Exactly-once needs no distributed lock or idempotency service; a unique constraint inside the existing transaction does it.
- Every posting leaves an immutable, transactionally-consistent audit record that the database itself refuses to alter.
- The change adds no API surface that can be misused: the key is optional and the audit reads are bounded.

27.4.2 Negative

- The idempotency key table grows without bound in v1 (no TTL or expiry); the demo seeder clears it on reset, and real retention is a later concern.
- The audit `after` snapshot duplicates data already in the postings; accepted, as the point of an audit log is a self-contained record.
- The immutability trigger's gated branch currently permits any mutation (not just DELETE) when the GUC is set; only the seeder's DELETE uses it today, so a gated UPDATE would silently no-op. Tracked as a follow-up (fail closed on UPDATE).
- Replay and conflict paths are counted (`IdempotencyReplays`, `IdempotencyConflicts`

) but not timed in the `PostDuration` histogram.

27.5 Alternatives considered

- **Check-then-act idempotency (look up, then post)**: rejected. Races under concurrent retries and creates duplicate transactions exactly when it matters.
- **A separate idempotency middleware with its own table and short transaction**: rejected. The key insert would not share the posting's transaction, reopening a window where the posting commits but the key does not (or two requests both proceed); keeping the mutex inside the posting transaction is simpler and strictly correct.
- **Required idempotency key**: rejected for v1. A breaking change to the existing contract and to the console and seeder callers, for a guarantee clients can opt into per request.
- **Fingerprint by hashing the raw request body**: rejected. Whitespace, key order, or an echoed id would make identical payments look different and break the replay; hashing the semantic content is stable.
- **Separator-byte fingerprint framing**: rejected. Forgeable when a field can contain the separator, allowing a collision on a reused key; length-prefixing is collision-safe.
- **Audit log immutability by convention only**: rejected. A trigger enforces it at the database, consistent with the ledger's other invariants, so a buggy caller or a bad migration cannot rewrite history.
- **Generic before/after audit framework now**: deferred. Nothing mutates in place, so create-only with a NULL `before` is enough; a diff model is speculative.
- **Unpaginated by-account audit read**: rejected. Fine for the demo, but an unbounded query is a production memory and latency risk; keyset paging matches the statement endpoint.

CHAPTER 28

ADR-008: Covering Index for Account History Reads

28.1 Status

Accepted: 2026-07-02

28.2 Context

ADR-006 documents that the account statement is O(history) per page: because balances are derived, never stored (ADR-003), the running-balance window scans the account's full posting history within a single index range on `(tenant_id, account_id)`, the two-column `postings_tenant_account_idx` from `0001_initial.sql`. ADR-007 added a second reader on the same shape: `GET /v1/accounts/{id}/audit` keyset-pages `(created_at, id)` through a join from `audit_log` to the account's postings.

Both queries filter on `(tenant_id, account_id)` and then need their rows ordered by `(created_at, id)`: the statement to return newest-first pages with a correct running balance, the audit-by-account read to keyset-page in the same order. The two-column index covers the filter but not the order, so Postgres has to sort every page's matched rows after the index range scan. On a busy account this sort runs on every single page request, competing for the one CPU core the production VPS has.

A coverage/capacity audit flagged this as the read-side gap most likely to show up first as the demo tenant's ~285 seeded transactions grow, and it is cheap to close without touching query text or the derived-balance model.

28.3 Decision

Migration `0007` replaces `postings_tenant_account_idx (tenant_id, account_id)` with `postings_tenant_account_created_idx (tenant_id, account_id, created_at, id)`.

The composite's leading two columns, `(tenant_id, account_id)`, still serve the balance `SUM` and any other equality-only lookup, exactly as the old index did. The trailing `(created_at, id)` means an index range scan on the tenant/account prefix already returns rows in the exact order the statement and audit-by-account queries need, so Postgres can walk the index directly into a page of results instead of scanning then sorting. No sqlc query text changes: the SQL was already selecting and ordering by these columns, it was just the index that did not match.

The old two-column index is dropped in the same migration. Every query that could use it can use the composite's leading prefix instead, so keeping both would only add write-side index maintenance for no read benefit.

This does not change what is derived versus stored. Balances are still computed by summing postings at read time; the index removes the per-page sort, it does not cache or materialize a balance. The O(history) scan cost ADR-006 already accepted is unchanged: this is a narrower fix for the sort that sat on top of it.

28.4 Consequences

28.4.1 Positive

- Statement and audit-by-account pages no longer sort after the index scan; the index range scan already returns rows in the order the query needs.
- No application or query-text change: `internal/postgres` and its sqlc queries are untouched, only the schema.
- One index does the work of two: the composite serves every read the old two-column index served, plus the ordered scans, at one index's write cost instead of two.

28.4.2 Negative

- The composite index is wider than the one it replaces (two extra columns per entry), so it costs slightly more to maintain per insert. On the production box's single core and 1GB of memory this is a small but real addition to write-side work; accepted, since posting throughput is not the bottleneck the audit flagged.
- Still O(history) per page in the sense that the scanned range grows with an account's full posting count, even though it is no longer sorted. A genuinely unbounded account still needs the materialized-rollup escape hatch ADR-006 already named, not another index.

28.5 Alternatives considered

- **Keep both indexes**: rejected. The two-column index becomes fully redundant once the composite exists (same leading prefix), so this would just pay index maintenance twice for the same set of queries.
- **Materialized running balance, rebuilt from postings**: deferred. This is the actual escape hatch ADR-003 and ADR-006 already named for when O(history) itself becomes the problem, not just the sort on top of it. Out of scope for this change: it would introduce a second, cached number that has to reconcile with the postings, which is the exact drift ADR-003 rejected a mutable balance column to avoid. Revisit only if index-scan cost itself, not the sort, becomes the bottleneck.
- **Do nothing**: rejected. Leaves an unbounded per-page sort competing for the single core on every statement and audit-by-account request, and the fix is a single, low-risk, additive migration.

CHAPTER 29

ADR-009: gRPC Layer over the Shared Domain

29.1 Status

Accepted: 2026-07-02

29.2 Context

Through Week 6 the ledger exposed one network surface: the REST API on chi + huma (ADR-006). Week 7 adds a gRPC front door. The point is not gRPC for its own sake: it is to prove that the domain services built in Weeks 2 to 6 are transport agnostic, so REST and gRPC are two thin adapters over one source of truth rather than two implementations that drift apart. This is only cheap because ADR-006 already made the handlers thin: they translate and delegate, and all logic lives in the `ledger` services.

The build plan flags Week 7 as heavy with scope-creep risk, and says to ship REST-only if it slips. Several choices had no obvious default.

29.3 Decision

29.3.1 Standard grpc-go, not ConnectRPC

The server uses `google.golang.org/grpc` with `protoc-gen-go` and `protoc-gen-go-grpc`, generated by buf. It is gRPC only (no HTTP/JSON on the same handler), works with grpcurl through server reflection, and takes unary interceptors, which matches the plan's wording ("gRPC server with interceptors").

ConnectRPC (connect-go) was the main alternative: it serves gRPC, gRPC-Web, and HTTP/JSON on one port and is curlable with plain curl. It is elegant and

buf-native, but it is a larger opinionated dependency and a different server model than the plan describes, and REST already covers the HTTP/JSON case. Standard grpc-go is the boring, widely-understood choice, which is what this project wants for infrastructure decisions.

29.3.2 buf for proto management, remote plugins, generated code under internal/

buf is a locked stack decision. `buf.gen.yaml` uses buf remote plugins (`buf.build/protocolbuffers/go`, `buf.build/grpc/go`) with managed mode, so `buf generate` needs no local `protoc` and is reproducible on any machine and in

CI. Proto sources live in `proto/ledger/v1/`; generated Go lands in `internal/genproto/ledger/v1/`. Placing it under `internal/` keeps the generated types unimportable outside the module while the in-repo Go client example, being in the same module, can still import them. `make proto` runs `buf lint` then `buf generate`, and regenerating must produce no diff.

29.3.3 Go server and Go client only for v1

The plan literally lists Go, Node, and Python clients. Only the Go server and Go client are generated now. The definition of done needs only grpcurl and a working Go client example; generating and actually verifying Node and Python toolchains triples the smoke-test surface in a week already flagged heavy. Node and Python are a later `buf.gen.yaml` change, documented, not built now. This is a direct application of the plan's own scope warning.

29.3.4 Full REST mirror of the RPC surface

`LedgerService` exposes the same operations as REST: CreateAccount, GetAccount, ListAccounts, GetBalance, GetStatement, PostTransaction, GetTransaction, plus the two audit reads. Both protocols cover the same domain, which is the point of the shared-domain refactor. The audit RPCs are the first cut if the week slips, then the health service; the seven core RPCs are the DoD. Messages mirror REST exactly: `int64 amount` plus `string currency`, signed postings that sum to zero, and the same opaque keyset cursor for statements.

29.3.5 Thin gRPC handlers over the unchanged ledger services

`internal/grpcserver` implements the generated `LedgerServiceServer` by translating proto to domain, calling the same `ledger` services the REST layer uses, and translating back. The domain and the services do not change. Domain errors map to gRPC status codes in one `toStatus` helper, mirroring `toHumaErr`: not-found to `codes.NotFound`, a duplicate or idempotency conflict to `codes.AlreadyExists`, validation and invariant failures to `codes.InvalidArgument`, an exhausted serialization conflict to `codes.Unavailable`, and anything unrecognized to `codes.Internal` without leaking internals.

29.3.6 A separate gRPC port with the same graceful-shutdown treatment

gRPC listens on its own port (`GRPC_ADDR`, default `:9091`), separate from REST (`:8080`) and metrics (`:9090`), added as a third serving goroutine with its own `GracefulStop` under the existing SIGTERM handling. Single-port multiplexing (cmux or h2c) was rejected: it adds connection-muxing complexity and muddies the clean split between HTTP/1.1 (chi) and HTTP/2 (gRPC) for no benefit on a single VPS.

29.3.7 Interceptors now, tracing deferred to Week 8

Chained unary interceptors: recovery (panic to `codes.Internal`), slog request logging (mirrors the REST middleware), and a tenant interceptor that injects the default tenant into the context (the same seam as REST; real auth later replaces only this). Server reflection is registered so grpcurl works, and the standard gRPC health service reports SERVING. A tracing interceptor is deliberately deferred: OpenTelemetry is Week 8's scope. A clean interceptor seam is left, with no tracing code now.

29.3.8 Idempotency parity through metadata

`PostTransaction` reads an `idempotency-key` from the incoming gRPC metadata, mirroring the REST `Idempotency-Key` header, and threads it into the same

`TransactionService.Post`. The response carries a typed `bool replayed`, the equivalent of the REST `Idempotent-Replayed` header. This is cheap parity because the service already supports idempotency.

29.4 Consequences

29.4.1 Positive

- One domain, two transports: REST and gRPC call the identical services, so they cannot drift, and the shared-domain claim is demonstrated, not asserted.
- grpcurl and a generated Go client both exercise the running service, satisfying the DoD, with reflection and a health check for easy operation.
- The write path keeps every guarantee it already had (exactly-once idempotency, atomic audit, SERIALIZABLE posting) because gRPC reuses the same service.
- Adding a second serving goroutine on its own port keeps the deploy and shutdown model unchanged.

29.4.2 Negative

- A second network surface to run, secure, and eventually observe. The tracing gap is explicit and closes in Week 8.
- Generated code is a new build input; `make proto` and a no-diff check must stay green, and the buf remote-plugin fetch needs network at generate time (not at build or run time).
- gRPC is unauthenticated for now, same as REST, guarded only by the default tenant and the VPS network posture.

29.5 Alternatives considered

- **ConnectRPC (connect-go)**: rejected for v1. One-port gRPC plus HTTP/JSON is attractive, but it is a larger opinionated dependency and a different model than the plan, and REST already serves the HTTP/JSON case.

- **Node and Python clients now**: deferred. The DoD needs only grpcurl and a Go client; three toolchains is the scope creep the plan warns about. A later `buf.gen.yaml` change adds them.
- **Single-port multiplexing (cmux / h2c)**: rejected. Adds complexity and blurs the HTTP/1.1 vs HTTP/2 split for no gain on one box; a separate port is simpler.
- **Local protoc instead of buf remote plugins**: rejected. Remote plugins keep `buf generate` hermetic and machine-independent, consistent with the locked buf decision.
- **Idempotency key as a request field instead of metadata**: rejected. Metadata mirrors the REST header semantics (a cross-cutting request attribute, not part of the transaction body), keeping the two protocols' contracts aligned.
- **A tracing interceptor now**: deferred to Week 8, which owns OpenTelemetry; a seam is left so it drops in without reshaping the chain.

CHAPTER 30

ADR-010: Observability with OpenTelemetry

30.1 Status

Accepted: 2026-07-07

30.2 Context

Through Week 7 the ledger had two network surfaces (REST on chi, gRPC) and two observability signals that did not talk to each other: native Prometheus metrics on a loopback `/metrics` server (ADR-004), and structured slog lines with a chi request id but no trace context. ADR-009 deliberately left a clean interceptor seam and deferred tracing to this week.

Week 8 closes that gap. The definition of done: a single transaction post produces one distributed trace spanning HTTP or gRPC, the domain service, and the SQL round-trips, with every log line for that request carrying the same `trace_id`. Traces run to Jaeger locally and to Honeycomb's free tier remotely.

The build plan flags this as a heavy week with an OpenTelemetry-config rabbit hole risk, and says to time-box to eight hours and ship slog-correlation plus Prometheus RED if the remote export fights back. Several choices had no obvious default, and one early instinct (route every metric through OpenTelemetry) turned out to be the wrong call for a service that already has metrics tied to alerting baselines.

30.3 Decision

30.3.1 Hybrid metrics: native domain collectors, OTel meter for RED

The four existing domain collectors stay exactly as they are, native `client_golang` on the dedicated registry, with their current names: `transaction_post_duration_seconds`, `transaction_post_serialization_retries_total`, `transaction_idempotency_replays_total`, `transaction_idempotency_conflicts_total`. These are SLO metrics tied to the Week 4 latency baselines. Renaming them is how you silently break an alert and find out during an incident, so they do not move.

The Rate, Errors, Duration signals for the two transports come through the OpenTelemetry metric API instead: `otelhttp` emits `http.server.request.duration` and `otelgrpc` emits `rpc.server.duration`, both with status attributes that give rate and errors for free. Those flow through an OTel `MeterProvider` whose Prometheus exporter is registered into the same registry the native collectors use, so `/metrics` stays a single loopback endpoint. Runtime instrumentation is not enabled on the OTel side: the existing `NewGoCollector` and `NewProcessCollector` remain the only source of `go_*` and `process_*`, avoiding a double registration.

The rejected alternative was routing all metrics through OpenTelemetry. It buys a single metric API, but the OTel Prometheus exporter mangles output (unit suffixes, `otel_scope_*` labels, a `target_info` series) and would have renamed the alert-bearing metrics for no operational gain. The hybrid keeps the invariant-critical metrics stable and still gets unified RED. A one-API purity win was not worth breaking working alerts in a payment service.

30.3.2 Tracing exporter selected by environment, loud when misconfigured

One tracer setup, three modes chosen at startup:

- `OTEL_EXPORTER_OTLP_ENDPOINT` set: an OTLP/HTTP exporter (`otlptracehttp`), which reads the standard OTLP environment variables for endpoint and headers. This is the only mode used in production and against Jaeger.
- Endpoint unset but `OTEL_TRACES_EXPORTER=console`: a stdout exporter, for eyeballing spans in local development without running a collector.
- Neither set: a no-op tracer and one `warn` log line at startup.

The stdout exporter is opt-in, not the silent default. On the VPS a missing or misconfigured endpoint must be a loud no-op, not a flood of span JSON dumped into the same stdout stream the logs are correlated on. OTLP/HTTP over gRPC because it is the lower-ops choice, firewall-friendly, and Honeycomb-native; the app already runs a gRPC server, so an OTLP/gRPC exporter would not be exotic, but there is no payoff over HTTP here.

30.3.3 Head sampling, and the noise paths excluded

The sampler is `ParentBased` with the root sampler read from the standard `OTEL_TRACES_SAMPLER` and `OTEL_TRACES_SAMPLER_ARG` variables, defaulting to always-on for local development. Production sets a ratio (documented as `parentbased_traceidratio` at `0.25`) because the demo seeder writes about 285 transactions every four hours, each fanning out to per-query DB spans under SERIALIZABLE retries, and Honeycomb's free tier is twenty million events a month. Always-on plus that write volume burns the budget on demo noise.

Two classes of span are dropped regardless of sampler: health and metrics requests are filtered out of HTTP tracing (`otelhttp WithFilter` on `/healthz`, and the loopback metrics server is not wrapped at all), so traces are real request work and not a wall of health checks.

30.3.4 Spans at four layers, no money or PII in attributes

- HTTP: the chi router is wrapped with `otelhttp`; a small middleware upgrades the span name to the matched chi route pattern (not the raw path) so URL cardinality cannot explode the trace backend.
- gRPC: `otelgrpc.NewServerHandler` is added as a `StatsHandler` alongside the existing interceptor chain from ADR-009, which stays unchanged.
- Domain: `TransactionService.Post` and the key `AccountService` reads open manual spans. Failure paths call `RecordError` and set an error status, and the double-entry invariant rejection from `Transaction.Validate()` is recorded as a span error, because that is the event a reviewer actually wants to find.
- Database: `otelpgx` is the pool query tracer, one span per SQL statement.

Span attributes are restricted to safe identifiers: `tenant_id`, `transaction_id`, posting count, and whether an idempotency key was present. No amounts, no balances, no account identifiers as values, and `otelpgx` runs with query arguments off so bind values never leave the process. Honeycomb is a US

SaaS processor, and shipping customer financial metadata to a third party is a decision that needs to be made on purpose, not leaked through a default. For a demo and portfolio service the rule is simply: identifiers and counts, never money.

30.3.5 Logs correlate through a wrapping slog handler

A thin `slog.Handler` wraps the existing JSON handler. On each record it reads the active span context and, when valid, injects `trace_id`, `span_id`, and a `sampled` flag, so every line logged with a request context correlates to its trace. It delegates `WithAttrs` and `WithGroup` to the inner handler and rewraps, so grouped and pre-attributed loggers keep working. This is a dozen lines and no new dependency.

The rejected alternative was the `otelslog` bridge, which turns log records into OTel log records shipped over OTLP. That is remote log export, a signal this week does not ship and the plan does not scope. The goal is correlation, not a third pipeline.

30.3.6 Service version from build info, not a hand-set string

The trace `Resource` carries `service.name`, `service.version`, and `deployment.environment`. The version is read from `runtime/debug.BuildInfo` (the VCS revision the binary was built from) rather than a hand-maintained constant, so every trace can be attributed to a release without a manual bump. The CI pipeline builds from a checked-out commit, so the revision is populated.

30.4 Consequences

30.4.1 Positive

- One transaction post yields one linked trace across transport, service, and SQL, and its log lines carry the same `trace_id`: the definition of done, verifiable in Jaeger locally and Honeycomb remotely.
- The alert-bearing domain metrics keep their exact names, so nothing built on the Week 4 baselines breaks, while RED for both transports arrives for

free through the instrumentation libraries.
- Tracing is off by default with no endpoint, so local development and the VPS run clean without a collector, and a misconfiguration is a loud no-op rather than a silent log flood.
- No money, balances, or bind values leave the process, so enabling a third-party backend does not quietly export financial data.

30.4.2 Negative

- Two metric APIs now coexist: native `client_golang` for the domain metrics and the OTel meter for RED. That is a deliberate trade of purity for stability, and it means a contributor sees both styles in the codebase.
- OpenTelemetry adds several dependencies (`otel/sdk`, the OTLP/HTTP and stdout trace exporters, the Prometheus metric exporter, `otelgrpc`, `otelpgx`) and a startup path that must be shut down cleanly within the existing graceful-shutdown budget so a batch exporter flush cannot hang termination.
- Span attributes are deliberately thin. Anyone wanting richer trace data must first decide what is safe to send to a third party, which is friction by design.

30.5 Alternatives considered

- **All metrics through OpenTelemetry**: rejected. It renames alert-bearing SLO metrics and adds `target_info` and `otel_scope_*` noise for a single-API win that does not pay for breaking working alerts. Hybrid keeps the domain metrics native and still unifies RED.
- **stdout as the silent no-endpoint default**: rejected. A misconfigured endpoint in production would dump span JSON into the correlated log stream. Console export is opt-in; the real default is a loud no-op.
- **OTLP over gRPC (port 4317)**: not chosen. HTTP/protobuf is the lower-ops, firewall-friendly path and Honeycomb-native; running gRPC elsewhere does not make it the better exporter here.
- **Always-on sampling**: rejected for production. The seeder plus per-query DB spans would exhaust the Honeycomb free tier on demo traffic; a parent-based ratio plus dropping health and metrics paths keeps the signal and the budget.

- **`otelslog` log bridge**: deferred. It ships logs over OTLP, a signal this week does not export. A wrapping handler gives the correlation the DoD asks for without a new pipeline.
- **`otelchi` for route-named HTTP spans**: not adopted. A few lines of chi middleware over the already-present `otelhttp` set the route pattern without another dependency.

CHAPTER 31

ADR-011: Testing strategy: unit, property, integration, chaos, load

31.1 Status

Accepted: 2026-07-07

31.2 Context

Through Week 8 the ledger accumulated tests package by package as each feature landed: table-driven unit tests in `internal/domain`, testcontainers-backed integration tests against real Postgres in `internal/postgres` and `internal/ledger`, a stress test that throws concurrent transactions at the service, and one property test on the balance invariant. What it did not have was a strategy: a stated coverage bar, a way to enforce it, a load profile with a number attached, or any test that asks what happens when Postgres dies in the middle of a commit.

Week 9 is the testing week. The definition of done from the build plan is CI green on a clean push and a load test that sustains 500 RPS at p99 under 100ms locally. That is the visible target. The harder and more valuable goal is that the test suite actually protects the money invariant under the conditions that break payment systems in production: concurrency, partial failure, and retries after an ambiguous outcome.

Coverage percentage is a weak proxy for that. Eighty percent line coverage on a ledger can miss every branch that matters (rollback, overflow, currency mismatch, double-post on retry) while padding the number with getters. So the decisions here are less about hitting a percentage and more about which invariants get tested and how the failure injection stays deterministic instead of flaky. A flaky chaos test or a load test that secretly measures the wrong path is worse than none: both ship a false “tested” signal, which is the last thing a payment service wants.

31.3 Decision

31.3.1 A five-layer test pyramid, each layer with a distinct job

- **Unit** (fast, no Docker): domain types, money arithmetic, API and gRPC error mapping, paging cursors. Table-driven. These already dominate the suite and stay the default.
- **Property** (fast, no Docker, `pgregory.net/rapid`): invariants that must hold across a large input space, not just hand-picked cases. See below.
- **Integration** (Docker via testcontainers): the service and repository against real Postgres, so the schema, triggers, `CHECK` constraints, and SERIALIZABLE retry behaviour are exercised, not mocked.
- **Chaos** (Docker via testcontainers plus toxiproxy): partial-failure behaviour when the database connection dies mid-transaction. See below.
- **Load** (k6 against a Compose stack): throughput and tail latency under sustained traffic, in two contention profiles.

31.3.2 Coverage: an 80% global floor, higher on the money path, generated code excluded

The gate is enforced by a `make cover` target and run in CI. It fails the build below the floor rather than only reporting a badge, because a coverage number no one blocks on is a number that silently rots.

The global floor on `internal/` is 80%. The money-critical packages carry higher per-package floors, because that is where a missed branch costs real money: `internal/domain` at 90%, `internal/ledger` at 85%, `internal/postgres` at 80%. Transport and plumbing packages (`grpcserver`, `api`, `metrics`) count toward the global floor but do not carry their own higher bar.

Generated code is excluded from the measurement: `internal/genproto` (protoc output) and `internal/postgres/sqlc` (sqlc output). Testing generated code measures the generator, not our work. The mechanics matter and are easy to get wrong: the target emits a single merged coverage profile from one `go test ./internal/... -coverprofile` run, strips the generated paths out of the profile file, then reads the statement-weighted total from `go tool cover -func`. Averaging per-package percentages would be a lie, because it weights a 20-line package the same as a 2000-line one.

31.3.3 Property tests target invariants, not just `Validate()`

The existing property test checks that `Transaction.Validate()` accepts a balanced set and rejects an unbalanced one. That is the easy part. Three more properties carry the real weight:

- **Money round-trip and bounds**: minor-unit `int64` amounts survive string-to-value-to-string round-trips, addition does not overflow silently, and zero and negative amounts are rejected where the domain forbids them. Money representation bugs are the classic ledger failure, so they get direct randomized coverage.
- **Stateful model check** (rapid state machine): a random sequence of transactions is applied, and after every step two invariants hold: each account balance equals the sum of its own postings, and the signed total across all accounts is zero. This catches an error that unit tests structurally cannot: a sequence of individually valid operations that corrupts the aggregate.
- **Concurrency invariant**: many goroutines post concurrently and the sum-of-postings invariant still holds with no lost updates, run under `-race`.

31.3.4 Chaos uses toxiproxy for deterministic injection, and covers the ambiguous-commit case

Killing the Postgres container with `docker kill` at the right instant is timing racy and slow: you usually land after the commit or before the write is in flight, and then the test proves nothing while still occasionally going red on

CI. Instead a toxiproxy container sits between the app and Postgres, and the test cuts the connection at a controlled point. That is repeatable, fast, and needs no image restart.

Two distinct failures are tested, because they have different correct outcomes:

1. **Connection lost after `BEGIN`, before `COMMIT`**: the post must fail and roll back cleanly, leaving zero partial postings. The append-only invariant means a half-written transaction is not allowed to exist.
2. **Connection lost after `COMMIT` is sent but the acknowledgement is lost**: the client cannot tell whether the transaction committed. This is the scenario that loses money, because the natural client reaction is to retry. The test replays the same idempotency key and asserts exactly one transaction exists, not two.

This also pins down an invariant the integration tests take for granted: the idempotency-key write and the transaction commit are one atomic database transaction. If they were separate, a crash between them would either double-post or wedge the key, and the ambiguous-commit test is what would catch a regression that split them.

31.3.5 Load: two contention profiles, unique keys, durability left on

The k6 script runs against a Compose stack (Postgres plus the app plus the existing Jaeger) under a dedicated `load-test` profile. Two scenarios, because they measure different ceilings:

- **Fan-out**: transactions spread across many accounts. This measures raw write throughput with minimal row-lock contention, the number the 500 RPS target is about.
- **Hot account**: many transactions against one account. This serializes on the per-account row lock and is where p99 blows up. The service already has a hot-account path; the load test exercises it on purpose rather than pretending contention does not happen.

Two rules keep the load test honest:

- **Every request carries a unique idempotency key**, with a small deliberate slice of duplicates mixed in. Without this the test would hammer the idempotency short-circuit (a cheap replay lookup) and report a throughput number for the wrong code path.
- **Durability stays on.** p99 on a laptop is dominated by fsync latency, and the tempting way to hit 100ms is to turn off `synchronous_commit`. On a ledger that is cheating: it trades the durability guarantee for a benchmark number. The target is reported as machine-dependent, measured with durability intact, not treated as a hard SLO.

The `load-test` Compose profile sets `SEED_ENABLED=false`. The demo seeder resets the demo tenant every four hours and would collide with a load run, so the load test uses its own tenant against a stack with the seeder off.

31.3.6 CI: parallel test jobs, a non-gating smoke-load, deploy unchanged in intent

The existing `deploy.yml` runs one test-and-lint job before deploy. Week 9 splits the pre-deploy work into jobs that run in parallel rather than chained, so a PR is not paying for artificial serialization:

- `lint`: golangci-lint.
- `test`: the race-enabled suite plus the `make cover` gate. This one job runs the unit, property, integration, and chaos tests together, because the suite is not separated by build tags: the integration and chaos tests skip themselves when no Docker is present and run for real when it is. The GitHub runner has Docker, so the full suite runs and coverage reflects the real numbers. The testcontainers tests wait on the Postgres readiness log (`"ready to accept connections"` seen twice), not on the port opening, because port-open races with the initdb restart and produces connection resets on CI. Splitting unit from integration would mean either running the suite twice or inventing build tags the codebase does not use, so one honest `test` job is the better trade. The coverage profile is scoped to `internal/` only: `cmd/main.go` is wiring with no unit tests, and folding its zero into the number would drag the gate down for no signal, so `cmd/` is compiled and smoke-run as a build check but kept out of the coverage measurement.
- `smoke-load`: boots the Compose stack and runs a short low-RPS k6.

`deploy` needs `lint` and `test`, and still only runs on a push to `main`. PR test runs get their own cancelable concurrency group so a force-push cancels the stale run; the deploy concurrency group keeps `cancel-in-progress: false` so a release is never interrupted mid-flight.

The smoke-load is deliberately not gated on the 500 RPS or 100ms figures, which are laptop numbers and would flake on a shared runner. It is gated on the regression signals that are stable across machines: a zero error rate and a loose p99 ceiling (around 500ms). That catches a server that fell over or a ten-times latency regression without going red because a CI runner was slow.

31.3.7 Badges without a standing credential

The coverage badge is produced by writing a shields.io endpoint JSON to a dedicated `badges` branch using the built-in `GITHUB_TOKEN`, and pointing shields at the raw file. This avoids a long-lived personal access token with gist scope

sitting on a public repository: the built-in token is scoped to the run and needs no manual secret. The Go Report Card badge is a plain URL to `goreportcard.com`, which generates on first request and needs no setup.

31.4 Consequences

31.4.1 Positive

- Coverage is enforced, not decorative, with a higher bar where money moves and generated code excluded so the number reflects hand-written work.
- The invariant that defines the whole service is checked across a large random input space, including the aggregate-over-a-sequence case that example tests miss, and under concurrency with the race detector.
- The two failure modes that actually lose money in a payment system, a rollback after mid-transaction failure and a double-post after an ambiguous commit, have explicit deterministic tests rather than a hope that it works.
- The load test measures the real write path in both the throughput and the contention regime, with durability intact, so the number means something.
- CI stages run in parallel with a stable, non-flaky load gate, and the coverage badge needs no third-party account or standing token.

31.4.2 Negative

- Two more infrastructure dependencies enter the test path: toxiproxy for chaos injection and a fuller Compose stack plus k6 for load. Both are Docker-gated and skip cleanly when Docker is absent, but they lengthen a full local run.
- Per-package coverage floors add friction: a change that drops `internal/ledger` below 85% fails CI even if the global number is fine. That is the intended pressure, but it is pressure.
- The load numbers are machine-dependent and not a portable SLO, so the DoD figure is a local measurement, not a guarantee that holds on a different box.
- toxiproxy is one more container image to pull and pin, and the ambiguous-commit test depends on injecting failure at a precise protocol point, which is more intricate than a plain integration test.

31.5 Alternatives considered

- `docker kill` **for chaos injection**: rejected. Timing-racy and slow, it usually fires before the write or after the commit and proves nothing, while still flaking on CI. toxiproxy cuts the connection at a controlled point, repeatably and without an image restart.
- **Testing only the pre-commit failure**: rejected as incomplete. The ambiguous-commit case (commit sent, acknowledgement lost, client retries) is the one that double-posts, so it is the case worth writing.
- **A flat 80% coverage floor with no per-package bar**: rejected. It lets the money-critical packages sit at the floor while transport code carries the average. Higher floors on `domain`, `ledger`, and `postgres` put the pressure where a missed branch costs money.
- **Averaging per-package coverage percentages**: rejected as arithmetically wrong. It weights a tiny package equally with a large one. A single merged, statement-weighted profile is the honest number.
- **Load testing with a fixed idempotency key or a single account**: rejected. Fixed keys measure the replay short-circuit, and a single account measures only the contended path. Unique keys plus both a fan-out and a hot-account scenario measure the real write path and the real tail.
- **Relaxing `synchronous_commit` to hit the latency target**: rejected outright. It trades durability for a benchmark number, which is not a trade a ledger makes.
- **Gating the smoke-load on the 500 RPS / 100ms figures**: rejected. They are laptop numbers and would flake on shared CI. Gating on zero errors and a loose p99 ceiling catches real regressions without machine-dependent flakiness.
- **A gist-plus-PAT coverage badge (`schneegans/dynamic-badges-action`)**: rejected. It puts a long-lived gist-scoped personal access token on a public repo. Writing the badge JSON to a `badges` branch with the built-in `GITHUB_TOKEN` needs no standing secret.
- **Codecov for coverage**: not adopted. It adds an external SaaS dependency and a token for a portfolio repo whose whole ethos is self-hosted and dependency-light. A self-computed badge fits better.

CHAPTER 32

ADR-012: API authentication, mandatory idempotency, input hardening, and a tamper-evident audit chain

32.1 Status

Accepted: 2026-07-08

32.2 Context

A security and correctness audit of the service (performed against the premise "assume real money flows through this") found the accounting core sound but the service perimeter wide open. The double-entry invariant is enforced twice, money is integer minor units with no float, balances are derived from immutable postings, and posting is atomic under SERIALIZABLE with a correctly-scoped idempotency mutex. None of that is the problem. The problem is everything in front of it.

The audit's top five findings:

1. **No authentication or authorization.** Every request acted as a single tenant read from config (`DEFAULT_TENANT_ID`). Anyone who could reach `/v1` could move money and read every balance.
2. **Idempotency was opt-in.** A client that omitted the `Idempotency-Key` header and retried after a dropped response would double-post.
3. **Unbounded input.** The `postings` array had a minimum but no maximum, and there was no request body-size limit, so one request could become an enormous transaction.
4. **No rate limiting, incomplete server timeouts.** Only `ReadHeaderTimeout` was set.
5. **The audit log was guarded but not tamper-evident.** An immutability trigger rejected UPDATE and DELETE, but the rows were not cryptographically chained, so a sufficiently privileged database role could rewrite history.

The hard constraint on any fix: the public try-it console at `/console` and the Scalar playground must keep working with no login, and the demo tenant must keep resetting every four hours (see the demo seeder). A naive "require auth everywhere" breaks the demo, which is the point of the public deployment.

This ADR records the decisions that close all five findings while keeping the demo intact.

32.3 Decision

32.3.1 API-key authentication, keys stored hashed in the database

Authentication is a bearer API key: `Authorization: Bearer glk_<random>`. A new `api_keys` table holds one row per key: `id`, `tenant_id`, `name`, `key_hash` (unique), an optional `rate_limit_rpm`, `created_at`, and a nullable `revoked_at`. Only the SHA-256 hash of the key is stored; the plaintext is shown once at creation and is never recoverable from the database. A leaked database dump does not leak usable keys.

An authentication middleware reads the bearer token, hashes it, looks up an unrevoked row, and puts the resolved `tenant_id` (and the key's rate limit) into the request context. A missing or unknown or revoked key is a `401`. To keep the hot path off the database on every request, resolved keys are cached in memory by hash with a short TTL; revocation is therefore effective within that TTL, which is an acceptable trade for a service at this scale.

The tenant is derived **only** from the key. No request field, header, or body can set or override it. That is what makes tenant scoping the authorization boundary: a key for tenant A can only ever act on tenant A, and the composite foreign keys from Week 3 already make a posting into another tenant's account impossible at the database. So "can this principal touch this account?" reduces to "is this account in the key's tenant?", which the schema enforces for free. We deliberately did not add per-account access-control lists: they would be redundant with the tenant foreign keys and add a whole authorization surface for no gain at this scope.

Only `/v1/*` requires a key. Liveness (`/healthz`), the landing page (`/`), the console, the playground, static assets, the OpenAPI documents, and the loopback

metrics endpoint stay unauthenticated: they are either public by design or already off the public interface. The gRPC surface moves the same money, so it gets the same key check through its existing interceptor chain (ADR-009), reading the token from request metadata. Leaving gRPC open would make the whole exercise a REST-only speed bump.

32.3.2 A public demo key keeps the console open, with one auth path

Rather than special-case the demo tenant as unauthenticated (two code paths, and a permanently open write surface), authentication is uniform and the demo is reached with a real, low-privilege **demo key**. It is provisioned at startup from `DEMO_API_KEY` (a known, public value with a safe default), scoped to the demo tenant, and carries a tighter rate limit than a normal key. The console and the playground ship it. Exposing it is fine on purpose: it can only touch the demo tenant, it is rate limited, and that tenant is wiped every four hours.

The demo key survives the wipe because the seeder resets tenant **data** (accounts, transactions, postings, audit rows, idempotency keys) and never touches the `api_keys` table. So after each four-hour reset the console keeps working with the same key against a fresh ledger, exactly as before.

32.3.3 Idempotency is mandatory on money-moving POSTs

`POST /v1/transactions` now requires an `Idempotency-Key`; its absence is a `400` with a clear message. We rejected auto-deriving a key from the request fingerprint when the header is missing: that would silently collapse two genuinely separate but identical payments (same accounts, same amount) into one, turning a real second payment into a false replay. Making the client name the key keeps "these are the same request, retried" distinct from "these are two payments that happen to look alike." The console generates a fresh UUID per post.

32.3.4 Input hardening: bounded arrays, bounded bodies, complete timeouts

The `postings` array gets a maximum (100 legs), and the HTTP handler is wrapped with a request body-size limit, so one request can no longer become an arbitrarily large transaction or exhaust memory before validation. The HTTP server gains the timeouts it was missing: `ReadTimeout`, `WriteTimeout`, and `IdleTimeout`, alongside the existing `ReadHeaderTimeout`, so a slow client can no longer hold a connection open indefinitely.

32.3.5 Per-key rate limiting

Each key is rate limited independently: an in-memory token bucket per key hash, with the limit taken from the key's `rate_limit_rpm` (a default applies when the column is null). Over the limit is a `429` with `Retry-After`. The demo key is set lower; a normal key uses the default; and the local load-test stack provisions a high-limit key so the 500 RPS load test is exercising the ledger, not the limiter. The default is configurable so production and the load stack can differ without code changes.

32.3.6 A per-tenant, tamper-evident audit chain

The audit log gains two columns, `prev_hash` and `row_hash`. Each row's hash is `SHA-256` over its own content (tenant, action, transaction id, actor, before, after, created_at) plus the previous row's hash, with every field length-prefixed so no field's bytes can be mistaken for a boundary (the same framing the idempotency fingerprint already uses). `prev_hash` is the previous audit row's `row_hash` **for that tenant**, or a fixed genesis constant for a tenant's first row. The chain is extended inside the same database transaction that posts the ledger transaction, so a committed transaction always leaves the chain consistent, and `created_at` is set by the application (not the database default) so the hash is deterministic and verifiable.

The chain is per tenant, not global, for two reasons: it keeps tenants decoupled, and it means the four-hour demo wipe restarts only the demo tenant's chain from genesis without touching or invalidating any other tenant's history.

A new `GET /v1/audit/verify` endpoint walks the caller's tenant chain oldest

first, recomputes every row hash, and returns `{valid, checked, first_break_id}`, so tamper-evidence is not just stored but checkable. This sits on top of, not instead of, the existing immutability trigger: the trigger prevents casual mutation, and the chain detects a privileged rewrite that bypasses it.

32.3.7 Same-tenant posts serialize with an in-process mutex

The audit chain's read-then-insert above is a genuine same-tenant conflict. Every post reads the tenant's latest `row_hash` and then inserts the next row, in the same SERIALIZABLE transaction as the ledger posting, so two concurrent same-tenant posts reading the same chain tail is a real read-write antidependency. PostgreSQL aborts the loser with a serialization failure (SQLSTATE 40001), and under sustained same-tenant concurrency the repeated abort can exhaust the retry budget (25 attempts) and surface to the caller as a `503`.

We first tried closing this with a per-tenant PostgreSQL session advisory lock (`pg_advisory_lock`), taken on a connection checked out from the pool before the SERIALIZABLE transaction began, so a blocked waiter's snapshot was not fixed until the lock holder had committed. A review found this had two problems specific to this service's own configuration, serious enough to replace rather than patch:

1. The connection pool sets `lock_timeout` to 3 seconds. `pg_advisory_lock` waits are subject to `lock_timeout`, so a waiter under sustained contention could be aborted by Postgres with SQLSTATE 55P03. That code is not a serialization failure, was not retried, and was not mapped to `domain.ErrConflict`, so it surfaced as an opaque `500` instead of the intended `503`.
2. The lock was acquired on a connection checked out from the pool before the wait began, and held for the whole call, including every retry's backoff. A burst of same-tenant posts could check out and hold most or all of the pool's connections while merely waiting on the lock, starving every other tenant's posts of a connection. That defeats the entire point of scoping the lock per tenant: different tenants are supposed to stay fully parallel.

The fix, since go-ledger runs as a single VPS instance rather than a fleet, is an in-process keyed mutex in the repository adapter (`internal/postgres/repository.go`): one `sync.Mutex` per tenant id, held for the whole `RunInTx` call but never around a checked-out database connection. A post for a tenant that already has one in flight blocks on that Go mutex, holding no connection while it waits; once unblocked it begins its own attempt exactly as before

the advisory lock existed, acquiring a connection from the pool only when it actually runs, and releasing it between attempts (including during backoff). Different tenants use different mutexes and never wait on each other. The map of per-tenant mutexes grows with the number of distinct tenants ever seen and is never evicted; at this service's tenant count that stays small and cheap, so eviction was not built.

This closes both problems above: there is no database lock wait, so `lock_timeout` and SQLSTATE 55P03 do not apply, and a waiting goroutine never holds a pool connection, so a hot tenant's backlog cannot starve other tenants of connections. The SERIALIZABLE retry loop is unchanged and remains the correctness backstop, not the primary serialization mechanism for same-tenant posts: on today's single instance it now runs only after the in-process mutex has already made same-tenant execution one at a time, so it should rarely see a same-tenant conflict at all. If go-ledger is ever run as more than one instance, the in-process mutex only serializes posts within one instance; the retry loop is what still guarantees correctness across instances, since a same-tenant race between two different instances remains possible (rare) and would retry rather than corrupt the chain.

This supersedes ADR-004's negative consequence "conflicts cost retries, not blocked connections," specifically for same-tenant posting. That trade still holds for the SERIALIZABLE mechanism itself and for any cross-instance conflict, but for same-tenant posts on today's single instance the practical cost is now waiting on an in-process mutex, not a retry, and not a blocked database connection.

32.4 Consequences

32.4.1 Positive

- `/v1` and gRPC both require a key, the tenant comes only from the key, and cross-tenant access is impossible by construction. The perimeter is real.
- Retrying a payment is safe by default: a missing idempotency key is rejected up front rather than silently double-posting.
- One request can no longer exhaust memory or hold a connection open, and a single key cannot flood the service.
- The audit log is now cryptographically tamper-evident and verifiable through the API, not merely trigger-guarded.

- The public demo and its four-hour reset are unchanged: one uniform auth path, a public demo key that survives the wipe, no special-casing.

32.4.2 Negative

- Every `/v1` request now does a key lookup; the in-memory cache keeps that off the database on the hot path but means revocation lags by up to the cache TTL.
- The audit chain requires same-tenant posts to serialize: one at a time, in the order they arrive. On today's single-instance deployment this is enforced by an in-process per-tenant mutex (see the Decision section above), not by a database lock, so different tenants stay fully parallel and a hot tenant's queued posts hold no database connection while they wait. It is still real coupling that a very high-throughput single tenant would feel as added latency, just not as blocked connections or a database lock timeout.
- Key management for real tenants is still manual this pass (insert a row, or a small CLI): there is no self-service key issuance UI, which is out of scope.
- The demo key is public by design. That is safe only because it is tenant-scoped, rate limited, and wiped, and those three properties must stay true.

32.5 Alternatives considered

- **API keys in config instead of a table**: rejected. Config keys are not revocable or provisionable without a redeploy, and a hashed table is barely more code while supporting rotation and per-key rate limits.
- **JWT bearer tokens**: rejected for v1. Stateless tokens need an issuance path and a signing-key rotation story that a ledger with a handful of tenants does not need yet. API keys are the lower-ops fit.
- **Leaving the demo tenant unauthenticated**: rejected. It means two code paths and a permanently open write surface. A uniform auth path with a weak public key is simpler and safer.
- **Auto-deriving an idempotency key from the fingerprint**: rejected. It rejects two legitimately-identical separate payments as replays, which on a payment system is a correctness bug, not a safety feature.
- **Per-account access-control lists**: rejected as redundant. The tenant foreign keys already make cross-tenant access impossible, and the key already resolves to exactly one tenant.

- **A single global audit chain**: rejected. It couples all tenants into one hash sequence and makes the per-tenant four-hour wipe impossible without breaking the chain. Per-tenant chains keep the demo reset clean.
- **Cryptographic signing of each audit row (HMAC or asymmetric) instead of a hash chain**: deferred. A hash chain detects reordering, insertion, and deletion, which is the property the audit needed. Signing adds authenticity of the writer on top, which matters once there is more than one writer identity; that is a later concern.
- **A per-tenant PostgreSQL session advisory lock, to serialize same-tenant posts**: tried and reverted. It closed the audit-chain serialization storm but introduced two problems of its own: the lock wait was subject to the pool's `lock_timeout` and could abort with an unmapped SQLSTATE 55P03 (an opaque `500` instead of a `503`), and the lock was held on a connection checked out before the wait began, so a same-tenant burst could hold most or all of the pool's connections purely waiting, starving other tenants. An in-process per-tenant mutex has neither problem, since it never touches a database connection while waiting.

≡ go-ledger

Built in public, one week at a time.

Read, clone, follow along

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> go.sohag.pro
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